

Trigger change! Innovations in Sustainable Urban Land Restoration

Course manual



United Nations
Convention to Combat
Desertification



G20 GLOBAL
LAND INITIATIVE



Trigger change!
**Innovations in Sustainable
Urban Land Restoration**

Course manual

Disclaimer

The designations employed and the presentation of materials in this information product do not imply the expression of any opinion whatsoever on the part of the United Nations Convention to Combat Desertification (UNCCD) concerning the legal or development status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

The mention of specific companies or products of manufacturers does not imply that these have been endorsed or recommended by the UNCCD in preference to others of similar nature that are not mentioned. The views expressed in this information product are those of the authors or contributors and do not necessarily reflect the views or policies of the UNCCD.

For more Information on the UNCCD and GLI products, please visit:

unccd.int

G20land.org

Citation: United Nations Convention to Combat Desertification/ G20 Global Land Initiative. (2025). Course Manual. *Trigger Change! Innovations in Sustainable Urban Land Restoration*. Bonn: UNCCD/G20 GLI.

Cover photo: iStock

Developed by:



UNU
EHS

Technology
Arts Sciences
TH Köln



FAO
GREEN
CITIES
INITIATIVE

UCUENCA
TERRITORIUM



IFLA
INTERNATIONAL FEDERATION
OF LANDSCAPE ARCHITECTS

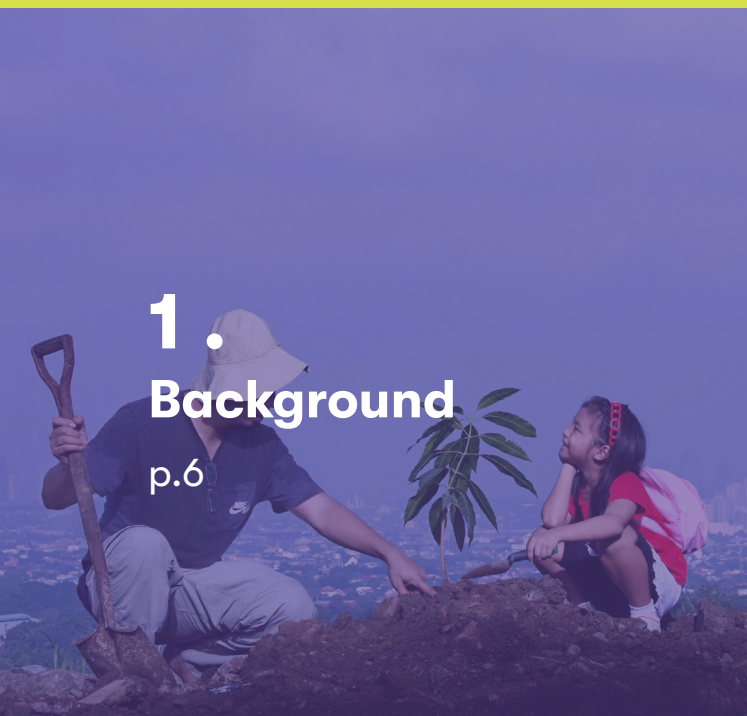
SALAR International

Acronyms

ASU	Ain Shams University, Egypt
BMA	Barcelona Metropolitan Area
CBA	Cost-Benefit Analysis
CSBL	Case Study-Based Learning
CSR	Corporate Social Responsibility
CU	University of Cuenca, Ecuador
DLDD	Desertification/Land Degradation and Drought
ECTS	European Credit Transfer System
EIA	Environmental Impact Assessment
FAO	Food and Agriculture Organisation of the United Nations
GDPR	General Data Protection Regulation
GIS	Geographic Information System
GLADA	Global Land Degradation Assessment
GLASOD	Global Assessment of Soil Degradation
GLI	Global Land Initiative
IFLA	International Federation of Landscape Architects
ILUP	Integrated Land Use Planning
IoT	Internet of Things
IPBES	Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services
IPCC	Intergovernmental Panel on Climate Change
IUCN	International Union for Conservation of Nature
KU	Kenyatta University, Kenya
LADA	Land Degradation Assessment in Drylands
LDA	Land Degradation Assessment
LDN	Land Degradation Neutrality
MCDM	Multi-Criteria Decision Making

MCH	Mitigation Conservation Hierarchy
MEA	Millenium Ecosystem Assessment
NbS	Nature-based Solutions
NDVI	Normalized Difference Vegetation Index
PIA	Participatory Impact Assessment
PPP	Public Private Partnership
ROAM	Restoration Opportunities Assessment Methodology
SALAR	SALAR International, Sweden
SCBA	Social Cost-Benefit Analysis
SDGs	Sustainable Development Goals
SFDRR	Sendai Framework for Disaster Risk Reduction
SLM	Sustainable Land Management
SROI	Social Return on Investment
SULR	Sustainable Urban Land Restoration
THK	TH Köln
tOD	The Other Dada
UNCCD	United Nations Convention to Combat Desertification
UNSSD	United Nations System Staff College
UGM	Universitas Gadjah Mada, Indonesia
ULD	Urban Land Degradation
ULR	Urban Land Restoration
UNEA	United Nations Environmental Assembly
UNEP	United Nations Environmental Programme
UNEP-CCC	United Nations Environment Programme Copenhagen Climate Centre
UNU-EHS	United Nations University, Institute for Environment and Human Security
USD	United States Dollar

Table of Content



1 .
Background
p.6



2 .
From urban land degradation to Sustainable Urban Land Restoration through innovative practices
p.15



3 .
Innovative didactics and competencies for “Triggering Change!”
p.24



4 .
Course module descriptions
p.33



5 .
Case studies descriptions
p.120

List of figures and tables

Figure 1. Course modules and corresponding number of “contact” or teaching hours	8	Figure 16. Course overview	34
Figure 2. G20 Global Land Initiative/UNCCD four pillars of action	10	Figure 17. Besòs River in Barcelona Metropolitan Area, Catalonia, Spain	122
Figure 3. The doughnut of planetary boundaries and social foundation	17	Figure 18. Left: Before Miyawaki forest restoration (2019); Right: Postrestoration (2021), Beirut, Lebanon	123
Figure 4. SULR is the intersection in the management of ecological and social processes in urban areas, more specifically between land	18	Figure 19. Left: Before urban renewal project; Right: After urban renewal project implemented by the community	124
Figure 5. ULD results in short-term socio-economic gains, which eventually plummet as there are few ecosystem services to support them	19	Figure 20. Left: Work in progress: Urban greening and planting of restored land in Al-Fustat site	125
Figure 6. Our course conceptual framework defines urban land as the interconnections between land	20	Figure 21. Left: Shinshin creek before renovation; Right: Shinshin creek, Cuenca, Ecuador postrenovation rendition	126
Figure 7. SULR conceptual framework with four types of actions	21	Figure 22. Left: Mangrove Ecosystem in North Jakarta, Indonesia. Right: Unprotected Mangrove Areas in North Jakarta, Indonesia	127
Figure 8. SULR scenario: “reduce” land degradation increases both ecosystem services and socio-economic benefits	21	Figure 23. Public-private-partnership kitchen garden programs at Kathmandu schools, Nepal	128
Figure 9. Cross-scale and multilevel considerations	22	Figure 24. Left: Michuki Memorial Conservation Park, Nairobi, Kenya, before restoration; Right: Post-restoration	129
Figure 10. The range of scales and types of land restoration	22	Figure 25. Left: Maori carving NZ. Right: Wellington, NZ.	130
Figure 11. NbS actions contribute to achieving SULR	23	Figure 26. Renaturation of the Emscher river system	131
Figure 12. Trigger Change! Innovations in Sustainable Urban Land Restoration	23	Figure 27. Emscher Park, Germany	131
Figure 13. Revised Bloom’s taxonomy	25	Table 01. SULR course conceptual framework based on the Land Degradation Neutrality and Mitigation Conservation Hierarchy frameworks	19
Figure 14. Constructive alignment in teaching	26		
Figure 15. The “sandwich” 20-minute interactive learning approach for more sustainable learning outcomes	27		

1. Background



1.1 Rationale

Urbanization is the most transformative process shaping the modern world. It has brought innovation, employment but also pollution, heat, land degradation and a human health crisis. People under 25 years old currently constitute 60 per cent of the population in many developing countries (UNCCD/G20 GLI, 2023), with millions of young people living in urban centers. While urban growth has contributed to economic development, social progress and innovation, it has also led to profound environmental degradation with costly consequences to human health (UNCCD/G20 GLI, 2023). The challenge is how to revitalize degraded urban areas, while prioritizing ecological, social, and economic sustainability. Tackling these challenges requires innovative and transdisciplinary approaches to land restoration that are not adequately covered in most urban planning /landscape architecture university courses, hence the genesis of our course: “*Trigger Change! Innovations in Sustainable Urban Land Restoration (SULR)*”.

Land in urban areas is often overlooked for the many ecosystem services they provide, most of which are public goods (e.g. urban trees provide cooling effects and improve human health). Because most of the benefits are not directly monetized, they are often undervalued in urban planning and land related investment decision-making. This course attempts to address this lacuna by providing students with the necessary knowledge and skills to recognize the importance of ecosystem services in urban areas, to revitalize degraded urban landscapes and promote resilient and sustainable cities. Through a combination of theoretical frameworks, case studies, and hands-on exercises, students will explore methods for restoring urban sites, implementing nature-based solutions, and designing multifunctional urban spaces.

Enabling urban renewal and greening requires urban planning and land management professionals who grasp the multifaceted challenges of cities

through a holistic process that integrates environmental science, urban planning, social equity, and economic viability. This is the purpose of our transdisciplinary course: “*Trigger Change! Innovations in Sustainable Urban Land Restoration*”. SULR goes beyond simply cleaning up contaminated sites or regenerating brownfields; it encompasses a holistic strategy to create healthier, more resilient, and equitable urban environments. It is a key strategy for creating more sustainable and resilient cities in the face of increasing urbanization and the corresponding environmental challenges.

Examples of innovation and land restoration in the urban context include:

- ▶ Companies that develop and deploy technology to facilitate restoration, often by improving efficiency and lowering costs;
- ▶ Companies that sell products to the end consumer, often recycling materials from their restoration activities;
- ▶ Companies that sponsor restoration projects;
- ▶ Municipalities that embrace innovation in urban planning through urban renewal, greening, through innovative public, private partnerships (modified from Faruqi et al., 2018).

This transdisciplinary course integrates principles from ecology, urban planning, environmental engineering, development and policy studies, preparing students for diverse career paths in sustainability, environmental management and urban development. It is transdisciplinary as it goes beyond academic disciplines to integrate knowledge from experts, practitioners, and local communities, creating new frameworks. Students will gain the tools and knowledge to become change-makers in sustainable urban development, whether through research, policy-making, or community engagement.

1.2 Purpose of this manual and course outline

This manual will help you to design your own transdisciplinary course. It offers brief overviews of each module and submodule units to promote innovation in teaching SULR innovation with the aim of achieving more SULR planning and liveable cities. The manual also provides pedagogical guidance for more interactive and engaging teaching methods. These are now recognized as more effective in promoting learning that is long-lasting and sustainable as compared to the more classic lecture-only style.

This document is thus meant to supplement the development of course materials by:

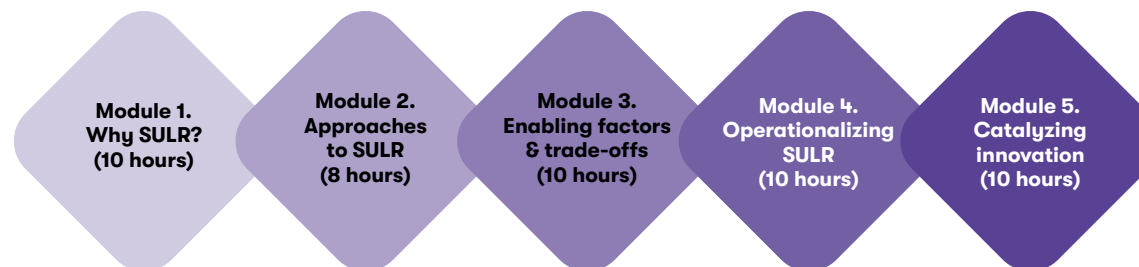
- ▶ Summarizing course objectives;
- ▶ Introducing key frameworks and bringing them together;
- ▶ Defining macro, meso and micro scales of action in different sectors/ areas.

The “*Trigger Change! Innovations in Sustainable Urban Land Restoration*” course is available to any higher education institution, or training institution that teaches urban planning / landscape architecture/ urban geography/ land administration and related courses worldwide. It is organised in five modules:

o **Module 1: Why SULR?** This module provides the course rationale and a succinct overview of cities and urban processes in the context of global challenges, specifically regarding urban land, land degradation neutrality and the need for a transition to SULR.

- o **Module 2: Approaches to SULR.** This module covers SULR entry points, approaches and principles at various scales, from strategic to local planning, including entry points for Nature-based Solutions (NbS) in cities.
- o **Module 3: Enabling SULR.** This module enables students to develop critical thinking about the factors that may hinder or enable SULR, notably: stakeholders, finance, policies, related governance approaches and trade-offs between these factors.
- o **Module 4: Operationalizing SULR.** This module covers state-of-the-art tools for assessing, planning, decision making, implementing and monitoring SULR, including land degradation assessment tools, citizen science and decision-making tools, and considerations related to data and ethics.
- o **Module 5: Catalyzing SULR through Innovation.** The final module of our course is dedicated to fostering innovation in SULR, with an overview of the business model canvas and value proposition for urban planning, whether for communities/ civil society, public, private or public-private-partnerships.
 - Creative formative and summative assignments are proposed to foster creative thinking and eco-entrepreneurship.
 - Ten case studies on applications of SULR from urban contexts worldwide (see Section 5).

Figure 1. Course modules and corresponding number of “contact” or teaching hours (see Section 4 for details on each module and for the course overview).



1.3 Target audience

The “**Trigger Change! Innovations in Sustainable Urban Land Restoration Course**” equips students with the necessary knowledge and skills to identify, analyze, and implement innovative solutions for restoring degraded urban lands.

This material is designed for lecturers who will teach the course, either fully or partially, and for students studying urban planning, landscape architecture, urban geography and allied courses in the built environment at any university worldwide. The course encourages a systems thinking approach to innovation in SULR, helping to support urban renewal and SULR planning. It also contributes to achieving several Sustainable Development Goals (SDGs), especially SDG 11, which aims “to make cities and human settlements more inclusive, safe, resilient, and sustainable”, and SDG 15 which aims “to halt and reverse land degradation”, amongst other things.

The course covers the impacts of land degradation in urban environments, sustainable land management (SLM) practices, and urban planning approaches that promote urban land restoration and sustainability in general. Students will be introduced to these concepts and solutions, supported by historical and contemporary examples from cities worldwide. They will learn how to integrate and develop strategies for integrating innovative practices into different scales and city sectors, fostering resilient and sustainable cities.

Consequently, the course learning objectives are:

- ▶ Understand the value of ecosystem services provided by healthy urban land and the purpose of SULR;
- ▶ Recognize the principal processes of urban land degradation and analyze their impacts in urban environments;
- ▶ Distinguish the challenges and benefits for restoring urban land;
- ▶ Guide students in how to research, plan and agree solutions for urban land restoration in different contexts.

Graduates are expected to find employment in the following fields:

- o **Planning related**, i.e., land use planners, urban planners, policy analysts, public policy, non-profit advocacy: they evaluate, develop and implement

land use policies that promote SULR practices while balancing economic, environmental and social needs at regional, metropolitan and urban and local scales; in particular, they plan the integration of land uses and sustainable practices into urban development, alongside drafting policies or regulations that support SULR.

- o **Design and culture related**, i.e., landscape architects: they create plans to transform open space while considering SULR; in particular, they integrate natural systems into urban and rural environments to promote biodiversity and social and natural systems health, implement sustainable stormwater management techniques, utilize vegetation and contouring techniques to stabilize soils, enhance habitats for wildlife and improve ecosystem functions, work with communities to incorporate local knowledge and values into open space change and foster stewardship and implement NbS to achieve SULR.
- o **Life science related**, i.e., environmental scientists, agronomists, biologists, soil scientists: they conduct research and assessments to understand the need and the impact of SULR; in particular, they focus on agricultural practices, soil health, crop production to enhance the sustainability of land use, investigate forested landscapes to maintain biodiversity, protect resources, work to protect and restore habitats, study species and ecosystems to support biodiversity conservation, analyze soil composition and health to recommend practices that sustain soil fertility and prevent degradation, even in urban settings.
- o **Natural resources related**, i.e., urban foresters and farmers: they implement SULR through forestry, water management and farming operations in a variety of environments, which in recent years have been expanded to include the urban or urban-rural mixed land use environments, as well as urban environmental departments, with responsibility for managing open / green / green-blue spaces.
- o **Land Administration and management related**, i.e. land officers, land administrators, land use allocation committees of governments, deeds registrars: These are professionals responsible for land titling, land taxation, managing cadasters and land registries. They are custodians of data and information on urban land ownership, manage huge land ownership

databases which enables them to identify owners of degrading or degraded lands. They are invaluable for holding such land owners to account and encouraging them to take responsibility and necessary action for urban land restoration (modified from the International Federation of Landscape Architects, unpublished document, 2025).

1.4 Course developers

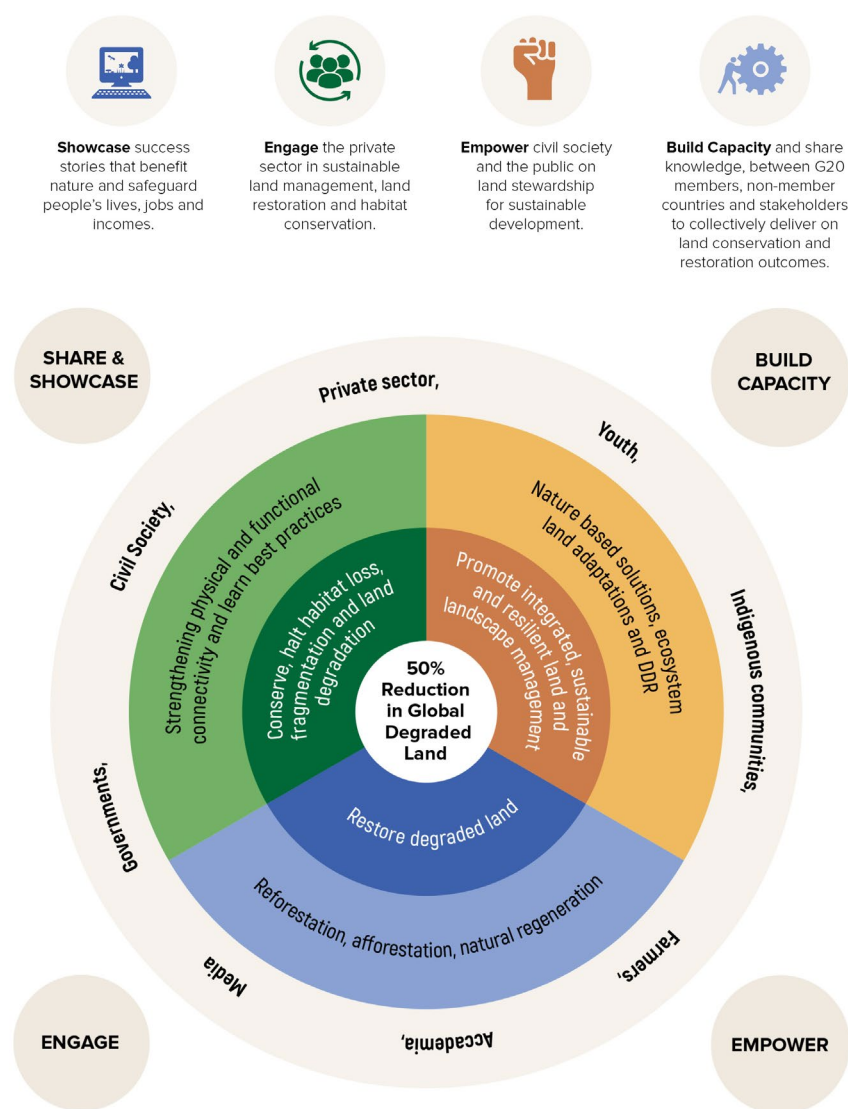
The mandate of the United Nations Convention to Combat Desertification/ Group of 20 Global Land Initiative / (UNCCD/G20 GLI) is to enable the G20 countries to achieve their ambition to reduce land degradation by 50 per cent by 2040 (UNCCD/G20 GLI, 2023). The G20 GLI, with oversight from the United Nations Convention to Combat Desertification (UNCCD), focuses on capacity-building, engaging the private sector and civil society, and showcasing success. The Convention seeks to support countries in addressing desertification/land degradation and drought (DLDD) (www.unccd.int).

The G20 GLI is comprised of four pillars:

1. Showcase success stories that benefit nature and safeguard people's lives, jobs and incomes;
2. Engage the private sector in SLM, land restoration and habitat conservation;
3. Empower civil society and the public on land stewardship for sustainable development;
4. Build capacity and share knowledge between G20 members, non-member countries and stakeholders to collectively deliver on land conservation and restoration outcomes (UNCCD/G20 GLI, 2023).

One of the main facets of pillar 4 is to build capacity by developing university curricula on land restoration to be included in agriculture, forestry, urban planning and mining programmes, globally, with a target of 1,000 universities and extension programmes worldwide so that future generations are fully familiar with the science, technology and economics of land restoration and habitat conservation (UNCCD/G20 GLI, 2023) (Fig. 2).

Figure 2. G20 Global Land Initiative/UNCCD four pillars of action



Under the leadership of the UNCCD/G20 GLI, this course has been developed by an international and interdisciplinary team to make the course materials diverse, representing different urban area types, sizes, and spread across ecosystems.

The partners involved in developing the course include:

Main developers:

- o United Nations University, Institute for Environment and Human Security (UNU-EHS)
- o TH Köln, University of Applied Sciences Cologne, Germany

Contributing institutions:

- o Ain Shams University, Egypt (ASU)
- o Kenyatta University, Kenya (KU)
- o Universitas Gadjah Mada, Indonesia (UGM)
- o University of Cuenca, Ecuador (UC)
- o Food and Agriculture Organisation – Green Cities Initiative (FAO - GCI)

With inputs by:

- o Oxford University, United Kingdom
- o Rivers State University, Nigeria
- o International Federation of Landscape Architects (IFLA)
- o SALAR International, Sweden

1.5 Course guiding principles:

Course philosophy: The course aims to provide an interactive learning environment which emphasizes active learning while providing a teaching foundation for professors who may not be familiar with all topics;

Course vision: The course is integrated into most urban planning/ landscape architecture/ urban geography/land administration programmes worldwide, supporting a significant increase in liveable cities through SULR and innovative practices;

Course learning outcomes

By the end of this course, students should be able to:

- ▶ Understand the value of ecosystem services provided by healthy urban land and the purpose of SULR;
- ▶ Recognize the principle process of urban land degradation and analyze their impacts in urban environments;
- ▶ Distinguish the challenges and benefits for restoring urban land;
- ▶ Identify innovative SULR solutions in different contexts;
- ▶ Create their own SULR business plan, or innovative proposal to a municipality which includes SULR components.

Students should be able to list and understand the benefits of SULR, while also acquiring critical thinking skills to recognize trade-offs, barriers (e.g. policies which create disincentives for innovation in sustainable agriculture) and factors that enable land restoration (e.g. subsidies and tax breaks promoting sustainable agriculture for land restoration);

Course level: The course materials are appropriate as an elective course for students enrolled in upper-level urban planning / landscape architecture/ urban geography/ land administration or related programmes (Bachelor's or Master's level);

Course length: Each submodule corresponds to roughly one hour of teaching (or contact hour with students) (equivalent to 48 hours of teaching materials); however, this can be shortened or lengthened depending on the requirements of each university;

Course content: Each PowerPoint lecture is around 40 minutes long and supplemented by suggested student exercises, suggested readings and supplementary materials, such as short instructional videos, adding up to 60 minutes per sub-module. Depending on the length of each lesson at the university, lectures can be shortened or lengthened;

Assignments: There are short quizzes, student discussion questions, short optional assignments and a proposed final assignment at the end of Module 5. Professors thus need to modify the assignments based on their university's requirements and are invited to adapt the assignments to their context;

Course materials: Materials are freely available to university professors or other institutions teaching urban planning / landscape architecture /urban geography and keen to be at the forefront of teaching innovation in SULR;

Credits: The course is intended to be equivalent to three credits in the European Credit Transfer System (3 ECTS), however professors are free to shorten or lengthen the course according to their university requirements;

Course Structure: The course has a modular structure and is organized into five modules (of which 48 submodules, 60 minutes each), which can be taught individually or as a package, together with a didactic note encouraging interactive teaching methods. As each module may be used separately, there may be some repetition to ensure that each module has a coherent narrative;

Target audience: Students in this course are enrolled in an urban planning/ landscape architecture / urban geography/ land administration or related topic and better equipped to seek employment opportunities with government agencies as urban planners or land administrators, in non-profit organizations or as restoration entrepreneurs, with a broader perspective on land as a vehicle for liveable cities.

1.6 Terminology

Adaptive management: Adaptive management is a structured, iterative process of robust decision-making in the face of uncertainty, with an aim to reduce uncertainty over time via system monitoring. In this way, decision-making simultaneously meets one or more resource management objectives and, either passively or actively, accrues information needed to improve future management (adapted from Holling, 1978; Walters, 1986).

Ecosystem restoration: “The process of halting and reversing degradation, resulting in improved ecosystem services and recovered biodiversity. It encompasses a wide continuum of activities, depending on local conditions and societal choice” (UNEP, 2021:7). Furthermore, land restoration focuses on land, while ecosystem restoration goes beyond land and encompasses entire ecosystems, which can include marine ecosystems as well.

Ecosystem services: Ecosystem services are the benefits people obtain from ecosystems. This includes things like clean water, pollination, and climate regulation (MEA, 2005). IPBES uses the term “nature’s contributions to people” which allows for a more holistic view of the interconnectedness of people and nature.

Innovation: The United Nations Innovation Network defines “innovation” as doing something new and different, whether solving an old problem in a new way, addressing a new problem with a proven solution, or bringing a new solution to a new problem (UNSSC, n.d.).

Land: The terrestrial portion of the biosphere that comprises the natural resources (soil, near surface air, vegetation and other biota, and water), the ecological processes, topography, and human settlements and infrastructure that operate within that system’ (IPCC, 2022a).Further to the IPCC 2022a definition, our course emphasizes the interrelated socio-ecological processes, interactions, and structures that occur within land.

Land degradation: “Land degradation is the reduction or loss of the biological or economic productivity and complexity of rainfed cropland, irrigated cropland, or range, pasture, forest and woodlands resulting from a combination of pressures, including land use and management practices” (UNCCD, 2022: 2).

Land Degradation Neutrality: “A state whereby the amount and quality of land resources necessary to support ecosystem functions and services to enhance food security remain stable, or increase, within specified temporal and spatial scales and ecosystems” (UNCCD, 2022:2).

Land restoration: “The process of repairing degraded or damaged land to restore ecological services, biodiversity and productivity.” (Leimu-Brown, 2025, this publication) ; see below definition of sustainable urban land restoration (SULR).

Megacity: An urban agglomeration with 10 million inhabitants or more (United Nations, Department of Economic and Social Affairs, Population Division (2019)

Nature-based Solutions (NbS): “Actions aimed at protecting, conserving, restoring, and sustainably managing natural or modified terrestrial, freshwater, coastal, and marine ecosystems, which address social, economic and environmental challenges effectively and adaptively, while simultaneously providing human well-being, ecosystem services, resilience and biodiversity benefits.” (UNEA 5.2, 2022). We consider that NbS actions contribute to achieving SULR, notably by avoiding, reversing urban land degradation and renewing urban land.

Peri-urban areas: Dynamic transition zones that have intense interaction between rural and urban economies, activities, households and lifestyles. Neither fully rural or urban (IPCC, 2022b).

Social-ecological systems (SES): Also known as coupled human–environment systems or coupled human and natural systems, is a powerful way to look at how changes in society and the environment are connected. They are interdependent and linked complex adaptive systems of people and nature, which are nested across scales. This reflects that people are part of and shape ecosystems, from local to global scales and depend on them to provide services for human well-being and societal development. Social–ecological interactions play out against a backdrop of global change and other temporal dynamics. (Fischer et al. 2015) .

Sustainable land management (SLM): The use of land resources, including soils, water, animals and plants, for the production of goods to meet changing human needs, while simultaneously ensuring the long-term productive potential of

these resources and the maintenance of their environmental functions (WOCAT, 2007).

Sustainable Urban Land Restoration (SULR): SULR combines ecological and social processes which contribute to avoiding, reducing, reversing urban land degradation or renewing urban land, therefore improving ecosystem services while meeting socio-economic needs. The major measures employed for urban land restoration include urban ecosystem restoration methods such as reforestation, urban rewilding, blue-green infrastructure, brownfield remediation, implementing NbS and designing multifunctional urban spaces (UNU-EHS, 2025, this publication).

Systems thinking: It links the natural, physical, human, cultural, and social environments. It reinforces the primacy of the relationship between elements and the flow of materials and energy rather than individual elements. It recognizes interdependencies and interconnections between policies and actions, achieving multiple benefits in outcomes that address multiple issues. Thus it replaces linear and positivist directions in urban planning (after Habitat III, 2016).

Urban land degradation (ULD): A negative trend in land conditions within urban areas, driven by direct or indirect human-induced processes, including urbanization dynamics. It manifests as a long-term reduction or loss of the quality, functionality, and value of ecosystem services, negatively affecting socio-economic needs. Urbanization is causing land degradation mainly due to changes in land use and the growing demand for resources (UNU-EHS, 2025, this publication).

Urban land system functions: They represent the complex interactions between human activities and ecological processes within urban environments. These functions result from the dynamic coupling of social and ecological systems, reflecting how urban spaces simultaneously can support human well-being and ecological integrity. Land social functions encompass how urban spaces support human activities, social interactions, and cultural values, e.g., economic activities, housing, transportation, infrastructure development, industrial zones, and social and recreational functions. Land ecological functions refer to the ecosystem processes maintained in urban green and blue spaces, including biogeochemical processes that regulate energy flows and biological

communities. When these functions bring benefits to people, they are known as ecosystem services, e.g. food and fresh water provision (UNU-EHS, 2025, this publication).

Urban management: Process that aims to synchronize and merge public and private initiatives to address the primary challenges urban inhabitants encounter, with the goal of creating a city that is fairer and more sustainable. Urban actors develop, implement, coordinate, and evaluate integrated strategies, considering the private sector's objectives and citizens' interests, within a higher-level government-defined policy framework, with the aim of achieving SULR development (van Dijk, 2006).

Urban planning: Conventionally, this term refers to a more constrained and specific type of planning with a spatial or geographical component. The major

purpose is to create a spatial structure for activities or land uses that enhances the existing pattern. This sort of planning is also known as 'physical' planning; spatial planning, a more objective and precise term, has occasionally coexisted with urban, regional, and town planning. This can include the plan-making and the spatial representation of it (Hall & Tewdwr-Jones, 2020).

Urban systems: Urban systems refer to two interconnected systems-- first, the comprehensive collections of city elements with multiple dimensions and characteristics: a) encompass physical, built, socio-economic-technical, political, and ecological subsystems; b) integrate social agent/constituency/processes with physical structure and processes; and c) exist within broader spatial and temporal scales and governance and institutional contexts; and second, the global system of cities and towns (IPCC, 2022b).



2. From urban land degradation to Sustainable Urban Land Restoration through innovative practices

2.1 Urban land degradation

Land degradation has been a persistent issue since the Neolithic era (10,000 - 7,000 years ago), when early agricultural practices began altering natural landscapes. In recent centuries, urbanization has significantly intensified land degradation, disrupting ecosystems on a global scale. Since 1850, approximately 35% of human-caused CO₂ emissions have been linked to land-use change and degradation, reinforcing the connection between these processes and climate change (IPCC, 2022a). The degradation of land manifests through declining land conditions, often difficult to measure due to the complexities of defining a baseline. This phenomenon not only contributes to climate change but is also exacerbated by it, creating a self-reinforcing cycle that threatens ecosystems, carbon sinks, and climate stability. Key drivers of land degradation include the expansion of agricultural fields, unsustainable farming and forestry practices, urban sprawl, infrastructure development, and extractive industries (IPCC, 2022a; IPBES, 2018). The effects of degradation extend beyond terrestrial environments, impacting marine and freshwater ecosystems, biodiversity, and human livelihoods, particularly in developing countries where an estimated 1.3 to 3.2 billion people are affected (UNCCD, 2022). These challenges are closely linked to poverty and food insecurity, further hindering economic stability and national development efforts (Local Governments for Sustainability/ICLEI, 2017). Given its far-reaching implications, land degradation is a key focus within the SDGs, particularly in target 15.3, which seeks to halt and reverse degradation by 2030. However, there is still a lack of systematic approaches at national and subnational levels to effectively tackle this issue.

Urbanization, as one of the most transformative processes of the modern era, has thus played a major role in both economic progress and environmental decline. Rapid urban growth and urban sprawl have become recognized as significant drivers of land degradation, contributing to issues such as industrial pollution, excessive construction, waste disposal, and the expansion of impervious surfaces that disrupt water cycles and soil health. As cities continue to expand, the pressure on land resources is increasing, exacerbating land-use conflicts and environmental degradation. Projections indicate that urban areas will triple in size by 2050 (IPCC, 2022c), making sustainable urban land restoration an urgent matter.

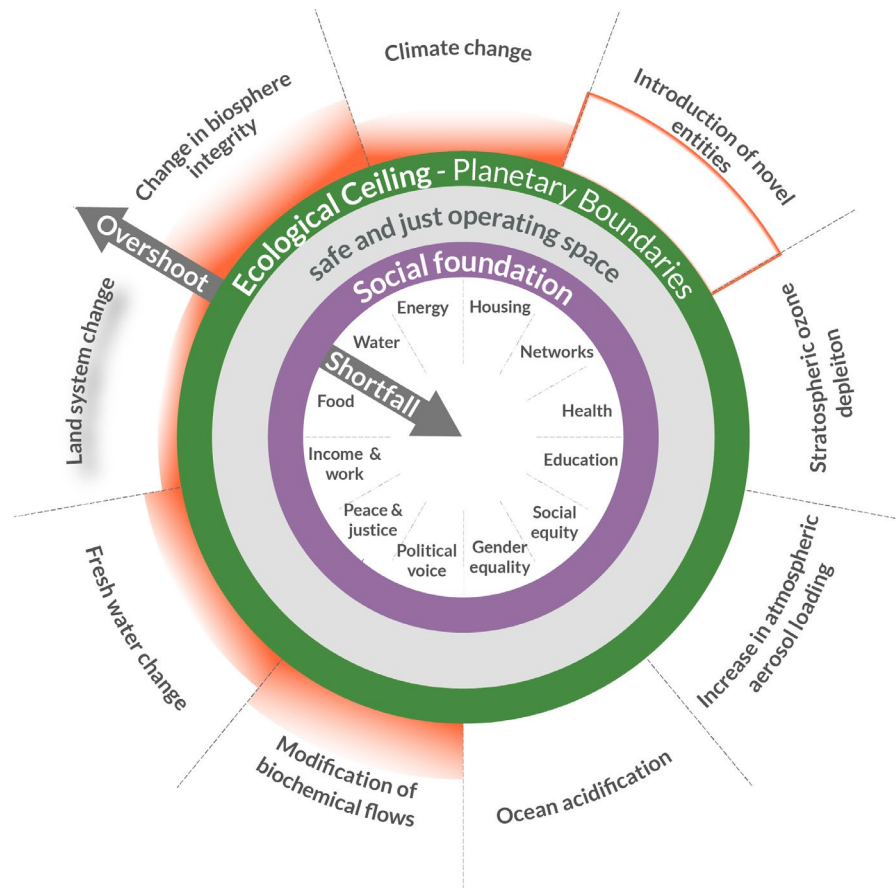
2.2 Course conceptual underpinnings

Urgent action to combat land degradation is essential to prevent escalating emissions, preserve carbon sinks and align with global temperature targets of limiting warming to 1.5°C or 2°C. Coordinated efforts are needed to mitigate its impacts, address human-driven causes and enhance ecosystem adaptability (IPCC, 2022a). UNCCD provides a valuable framework to guide these efforts, emphasizing the critical role of localized governance in achieving land degradation neutrality (UNCCD, 2022).

According to the Planetary Health Check, our planet has already crossed 6 out of 9 planetary boundaries, beyond which lie unacceptable environmental degradation and potential tipping points in Earth systems (Richardson et al. 2023, Steffen et al. 2015, Rockström et al. 2019 and Tomalka et al. 2024) (Fig. 3). These include: CO₂ concentration, biosphere integrity, land-system change, freshwater change, biogeochemical flows and novel entities. The three remaining are stratospheric ozone depletion, atmospheric aerosol loading and ocean acidification (Richardson et al. 2023, Steffen et al. 2015, and Rockström et al. 2009).

Without intervention, land degradation will continue to contribute to unacceptable environmental degradation while undermining natural resource-based livelihoods and societal resilience. Unacceptable environmental degradation includes: loss of soil permeability and fertility, loss of biodiversity, water scarcity, increased risk of infrastructure failure, water supply and quality issues, pollution, diminished local food resources, health problems due to poor quality air and heat stress, reduced resilience to disasters due to flooding and landslides; overall diminished well-being, and in specific cases, increased rates of unsafety in neglected urban areas.

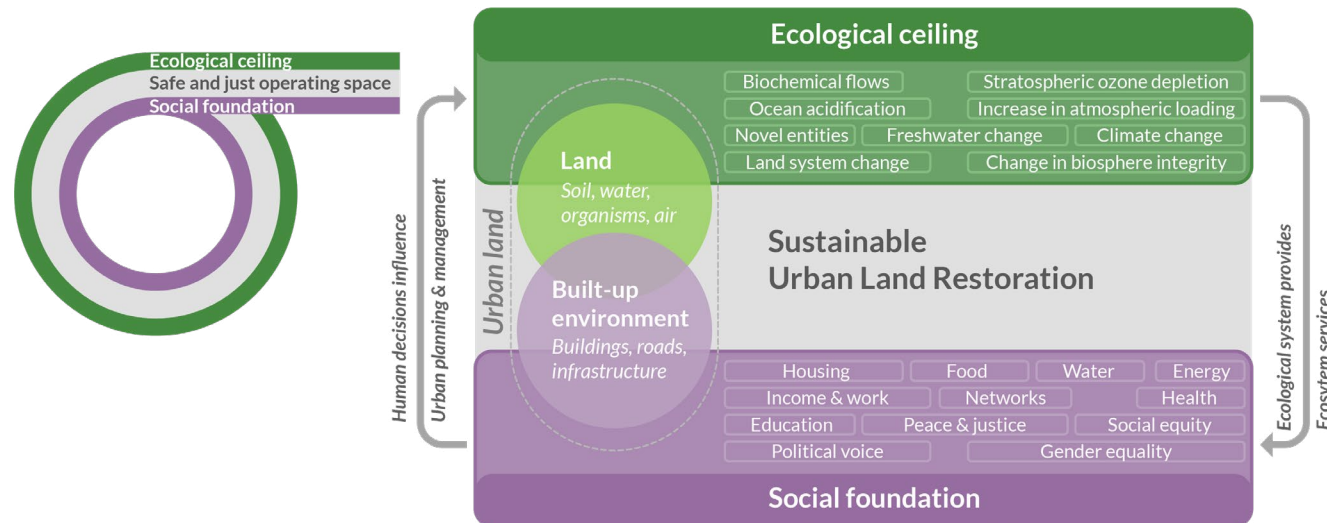
Figure 3. The doughnut of planetary boundaries and social foundation
© UNU-EHS/ A. Ramos, (2025) adapted from Raworth (2017)



Prominent economist, Kate Raworth refers to the latter in terms of “safe operating space” where basic human needs, or “life’s essentials” (i.e. food, housing, healthcare and political voice) are met (Raworth, 2017). Raworth’s “doughnut economics” conceptual framework includes 12 dimensions of the social foundation of internationally agreed minimum standards of well-being. The planetary boundaries define the ecological ceiling and the doughnut framework explains the social foundation.

We apply the planetary boundaries / doughnut framework concept to urban areas with the understanding that urban areas are influenced by human decisions and land use change, while the social system (e.g. health, social connectivity, economic underpinnings) depends on the ecological system for human well-being. Examples of ecosystem services in urban areas include trees for reducing urban heat islands, or as gathering places for families; wetlands for reducing water pollution or as a buffer for excess rainwater; urban forests or waterways as connectors for wildlife; or soil microorganisms in a patch of land for urban community kitchen gardens. The ecological and human systems are thus highly interconnected, especially in urban areas (Fig. 4).

Figure 4. SULR is the intersection in the management of ecological and social processes in urban areas, more specifically between land (e.g. soil, water, organisms, air, etc) and the built-up environment (e.g. buildings, roads, etc). The aim of SULR is to ensure as many ecosystem services as possible, while meeting as many socio-economic needs as possible, thus maximizing the conditions and qualities of the “safe and just operating space” © UNU-EHS/ A. Ramos, (2025) adapted from Raworth, (2017).



Our course explores how SULR can address both the ecological ceiling and the social foundation for urban areas. Having defined:

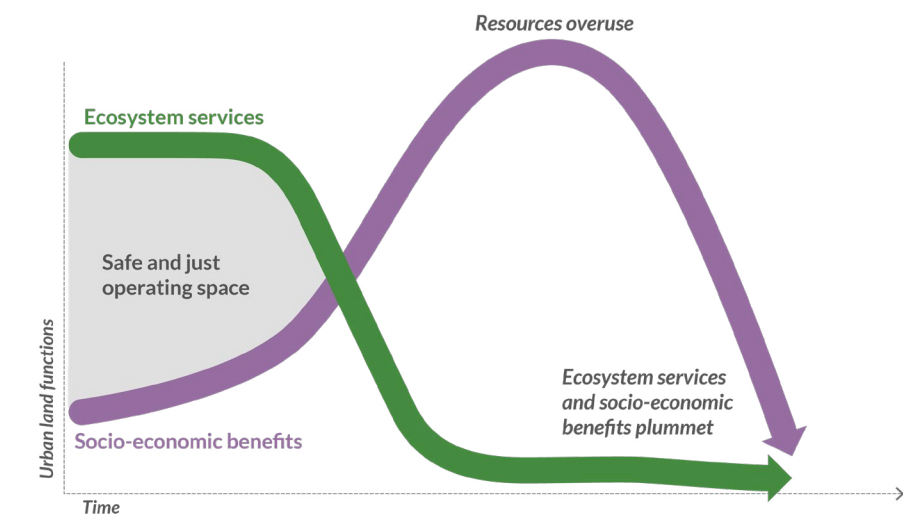
- ▶ **the problem** within the scope of this course (urban land degradation and its contribution to “planetary boundaries”);
- ▶ **the course vision:** supporting a significant increase in liveable cities through SULR and innovative practices, then:
- ▶ How does our course propose to **achieve this vision?**

So far, there is neither a universally agreed definition or approach to SULR. However, for the “Trigger Change! Innovations in Sustainable Urban Land Restoration” course, we have modified a few key frameworks from related disciplines.

Urban areas are often associated with development and human activity, however, portions of urban land—such as parks, gardens, green roofs, wetlands, and other green and blue spaces (among urban spaces typologies)—can play critical roles in maintaining ecological balance and contributing to environmental sustainability. Restoring urban land and improving the conditions for social and ecological interdependent functions will improve human well-being and ecological integrity in urban areas, providing more benefits to people.

Following the logic of the “planetary boundaries” (Rockström *et al.*, 2009) and “doughnut economics” (Raworth, 2017), the following diagram illustrates how both ecosystem services and socio-economic benefits decline in urban areas due to unsustainable urban development, when only economic and human needs are being pursued. At some point, human needs are no longer met when ecosystem services decline as air and water pollution, heat, poor waste management, etc., create adverse conditions for human health and well-being (Fig 5).

Figure 5. ULD results in short-term socio-economic gains, which eventually plummet as there are few ecosystem services to support them. Once the ULD tipping point is reached, the “safe and just operating space” diminishes considerably (© UNU-EHS/ A. Ramos, 2025).



SULR is thus about sustainable urban planning and actions that stay within the “safe and just operating space” (Raworth, 2017) to satisfy both ecological and social processes. This being said, we accept that **maintaining ecological integrity in an inner city may not be a realistic goal to achieve**. However, there are options for improving ecosystem services and human well-being even in dense inner city areas. Innovation in SULR refers to a host of public, private and public-private initiatives that catalyze action in urban planning, landscape architecture, start-ups, financial mechanisms for greater SULR planning and more liveable cities and peri-urban areas.

The underpinnings for our course conceptual framework are guided by the “Land Degradation Neutrality” (LDN) and “Mitigation and Conservation Hierarchy” (MCH) frameworks, explained in more detail below. Both frameworks distinguish between different types or hierarchies of actions that can guide the urban land restoration course development and SULR measures.

Table 1: SULR course conceptual framework based on the Land Degradation Neutrality and Mitigation Conservation Hierarchy frameworks (Feng et al. 2022; Milner-Gulland et al., 2021).

Land Degradation Neutrality Feng et al., 2022	Mitigation Conservation Hierarchy Milner-Gulland et al., 2021	SULR action
Avoid new degradation and contributing impacts	Retain and refrain - Avoiding from causing negative impacts	Avoid – Avoid urban land degradation and its negative impacts, including by proactive and long-term restoration planning
Reduce new land degradation	Minimize and Reduce Impacts - decreasing the damages that are already generated	Reduce – Reduce impacts from and expansion of already generated damage
Reverse past degradation by restoring and rehabilitating	Restore and Remediate Impacts - Restoring impacted ecosystems	Reverse – Restore and remediate past damage by restoration action
	Renew - Renewing our relationship with nature through offsets and proactive conservation measures	Renew – Building awareness and capacities towards urban land restoration

The LDN conceptual framework, developed by UNCCD, aims to maintain or enhance land-based natural capital and ecosystem services by addressing declining health and productivity of global land resources. LDN involves sustainable land management practices to avoid degradation and reverse it through restoration or rehabilitation, with priorities outlined in a hierarchy: **Reverse** past degradation by restoring and rehabilitating, **Reduce** new land degradation, **Avoid** new degradation and contributing impacts (Feng et al., 2022).

The MCH is a framework for bringing together different human actions that have net positive effects on nature (Milner-Gulland et al., 2021). The Mitigation Hierarchy compensates for losses experienced, while the Conservation Hierarchy recognizes conservation potential; together, they generate a beneficial impact on the environment. The MCH can help different types of stakeholders to make decisions that will have long-lasting effects on nature and the environment. It can be used to make policy, plan businesses, and help people make choices as consumers (decision-making at different levels), through four steps (**Retain and refrain, Minimize and Reduce Impacts - Restore and Remediate Impacts - Renew**). It also contributes to dealing with the loss of ecosystem services caused by human development actions over time (Milner-Gulland et al., 2021).

The SULR course conceptual framework is a combination of the LDN and MCH frameworks with four types of actions:

Avoid – Avoid urban land degradation and its negative impacts, including by proactive and long-term restoration planning (e.g. avoiding encroaching on wetlands in peri-urban zones).

Reduce – Reduce land degradation (e.g. put in place urban policies that maintain urban green spaces).

Reverse – Restore and remediate past damage through restoration action (e.g. restoring a riverway for flood risk reduction and recreation). [See our Barcelona case study.](#)

Renew – Building awareness and capacities for SULR (e.g. community groups are involved in visioning urban renewal and in undertaking the restoration works). [See our Argentina case study.](#)

We use the term “**Sustainable Urban land Restoration - SULR**” as an overarching concept to encompass these four types or levels of intervention. As mentioned above, **fully restoring an ecosystem in an urban context may not be realistic** and one might rightly question: restore to which state and at what point in time, especially considering the dynamic changes of ecosystems under the current climate change scenario, **hence our course conceptual framework** (Fig. 6).

Figure 6. Our course conceptual framework defines urban land as the interconnections between land (within the ecological ceiling) and the built-up environment (within the social foundation). SULR promotes actions (avoid, reduce, reverse, renew) within a “safe and just operating space” © UNU-EHS/ A. Ramos (2025) adapted from Raworth (2017) and Tomalka et al. (2024)

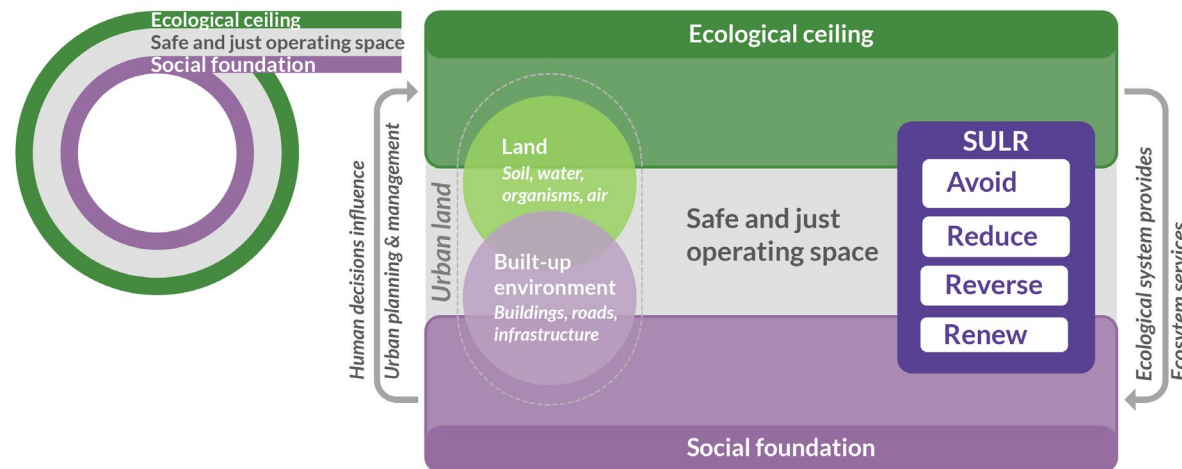
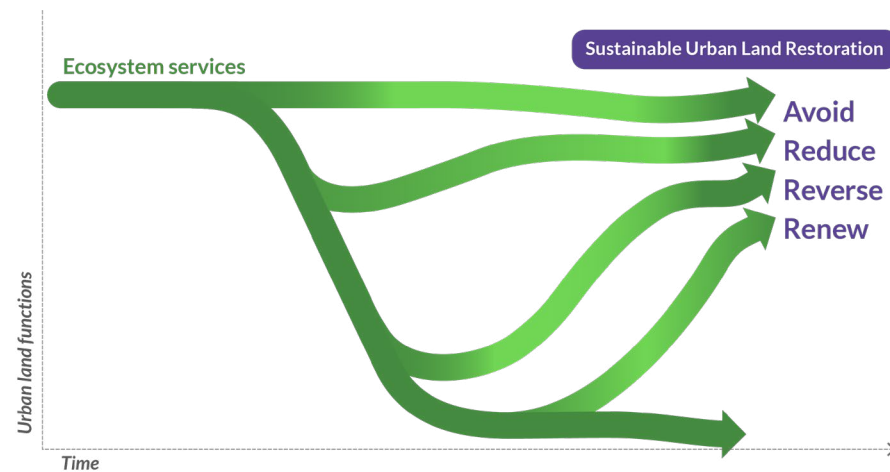


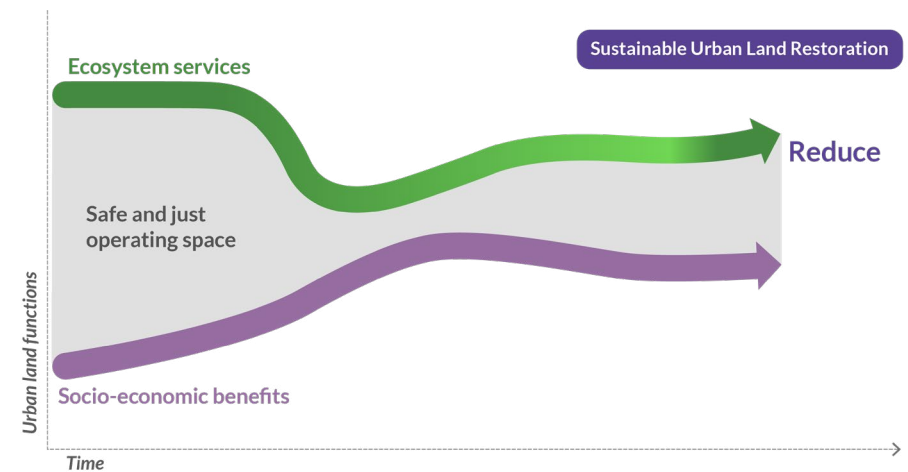
Fig 7. illustrates the four types of SULR actions : whereas AVOID reduces the loss of ecosystem services, REDUCE assumes that there will be losses but tries to reduce the downward trend in ecosystem services, REVERSE assumes even worse losses but seeks to recapture some ecosystem services, RENEW assumes a strong loss in ecosystem services but seeks to renew some services, even at the micro or nano scale (see next section on scale).

Figure 7. SULR conceptual framework with four types of actions (avoid, reduce, reverse, renew) in regards to ecosystem services in urban areas (© UNU-EHS/ A. Ramos, 2025, based on N. Leimu-Brown, unpublished).



Furthermore, Fig 8. illustrates a “reduce” land degradation scenario and one that is likely very common in urban areas in regards to ecosystem services and socio-economic benefits. Such benefits are expected to increase as land degradation is reduced, within the “safe and just operating space” (Raworth, 2017).

Figure 8. SULR scenario: “reduce” land degradation increases both ecosystem services and socio-economic benefits within a “safe and just operating space” (Raworth, 2017) (© UNU-EHS/ A. Ramos, 2025, based on N. Leimu-Brown, unpublished).



2.4 Scale

SULR is a multi-level and cross-scale process, with multiple measures and levels of complexity. SULR in particular requires taking into account scale issues and the connections between levels of organization (Cash *et al.*, 2006) (Fig. 9).

Figure 9. Cross-scale and multilevel considerations (© UNU-EHS / A. Ramos, 2025, adapted from Bennett *et al.* (2017); Cash *et al.* (2006).

	Ecological	Spatial	Jurisdictional	Institutional	Management	Urban Planning	Social organization	Knowledge
Global	Earth System	Global	International	Standards	Standards	Global Agendas	Humankind	General
Macro	Biomes	Bio-geographic and climatic regions	National Regional	Constitutions	Guidelines	Program on National scale	Large Communities	↑
Meso	Communities	Watersheds Landscapes	Municipality/ Metropolitan/ City	Laws and regulations	Strategies	Strategic development plan Land Management plan Sectoral plan	Large Communities	
Micro	Populations	Micro-watersheds	Localities	Instruments	Projects	Neighborhood plan Projects	Small Communities	
Nano	Organisms	Patches	Property	Operating rules	Activities	Construction / technical plan	Individuals/ Households	Local

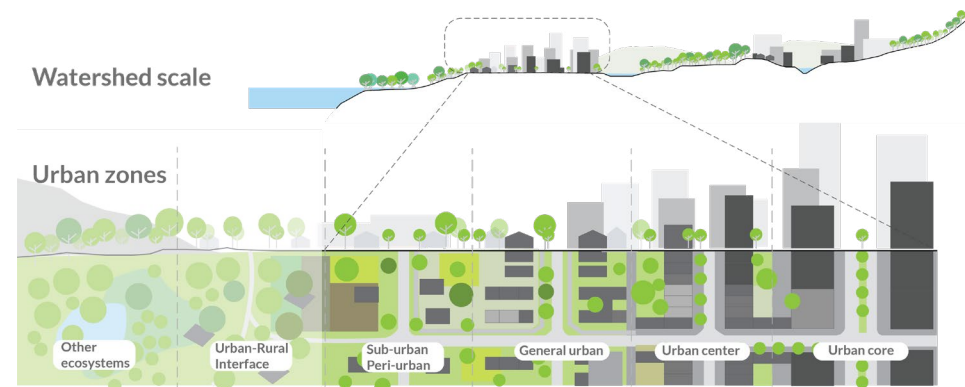
SULR can take place and will differ at various scales:

- ▶ Inner city patches: small patches of land or rooftops may be restored to foster micro ecosystems and reduce soil sealing (e.g. community garden plots, rooftop gardens for vegetables or shade);
- ▶ Inner city/ peri-urban neighborhoods: entire neighborhoods may be renewed with certain ecological functions restored and improved socio-economic benefits (e.g. neighborhood parks are created or restored for shade, soil permeability, ecological connectivity and places for social interactions);

- ▶ Metropolitan outlying areas /watershed: ecological integrity may be fully restored to outlying areas to ensure improved water quality, improve water infiltration, ecological connectivity and socio-economic benefits (e.g. provincial or state parks, low density natural areas that lie within a watershed upon which a city might depend for its drinking water).

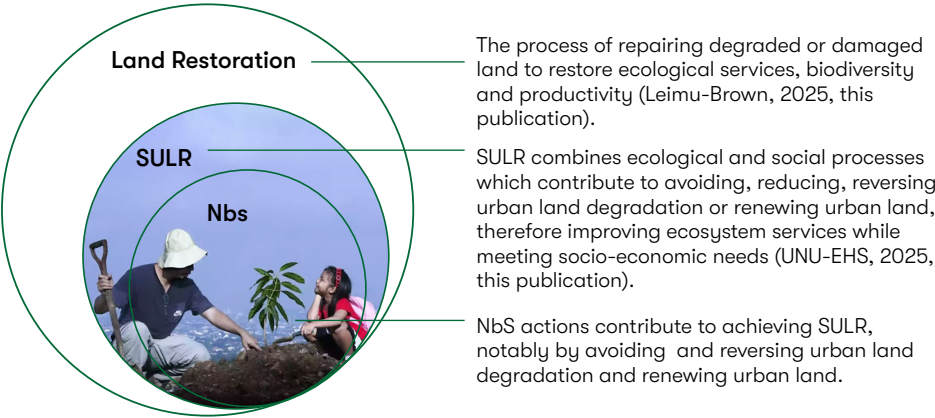
Furthermore, scale is referred to in multiple contexts, from geographic scales to types of institutions and management, among others. Thus, in this course the level of organization or scale is always identified so that it is clear to which type of scale or level it corresponds *i.e.*, macro (administrative) scale (Fig 10).

Figure 10. The range of scales and types of land restoration that might take place in an urban context from nano/micro interventions (e.g. renewal) in the urban core to more natural ecosystems (e.g. avoid or reduce). (© UNU-EHS / L. Vellozo and A. Ramos, 2025, adapted from Duany Plater-Zyberk & Company and World Bank Group)



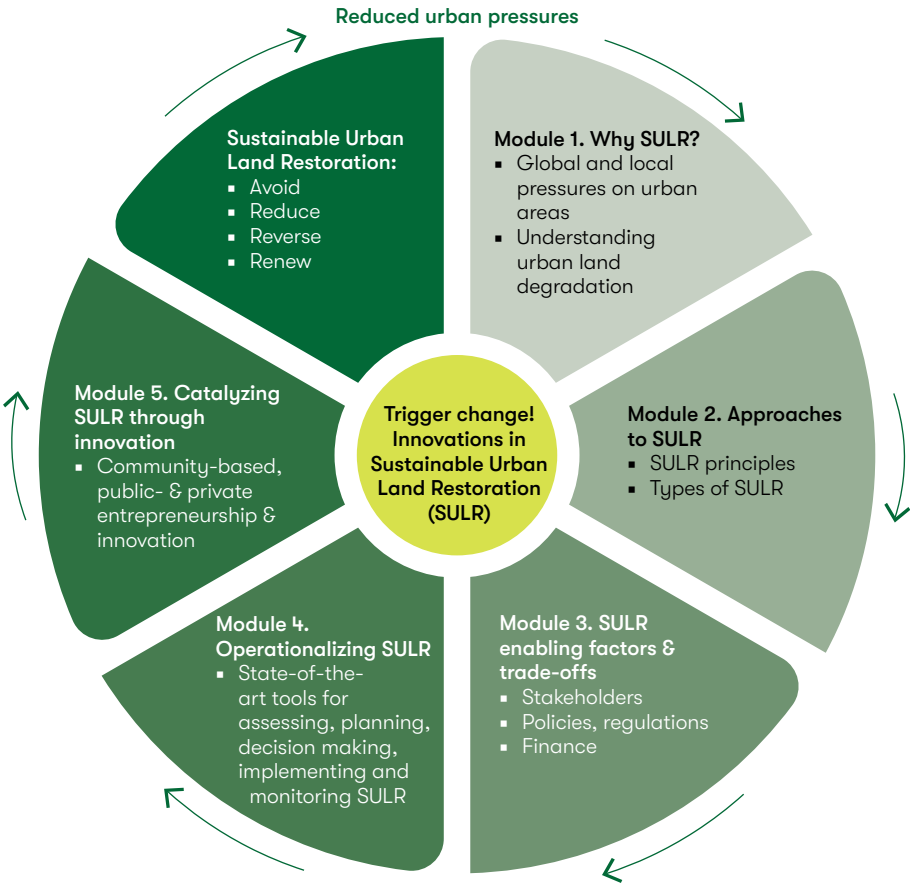
Integral design principles and NbS can be employed to implement SULR actions and improve urban spaces. The four types of SULR actions can be aligned and synergies and entry points can be found with different scales of planning and management instruments. NbS actions contribute to achieving SULR, notably by avoiding and reversing urban land degradation and renewing urban land (Fig 11).

Figure 11. NbS actions contribute to achieving SULR, notably by reversing and avoiding urban land degradation and renewing urban land (Photo © UNU-EHS / TUC / N. Victoriano).



We conclude this section with a graph illustrating the “Trigger Change! Innovations in Sustainable Urban Land Restoration” course flow (Fig 12). The figure starts with Module 1: the drivers and challenges facing urban areas; Module 2: Approaches to SULR; Module 3: SULR enabling factors & trade-offs; Module 4: Operationalizing SULR; Module 5. Catalyzing SULR through innovation; and finally the four SULR types of action (avoid, reduce, reverse, renew) which would ultimately lead to reduced urban pressures.

Figure 12. “Trigger Change! Innovations in Sustainable Urban Land Restoration” course flow





3. Innovative didactics and competencies for “Triggering Change!”

3.1 Introduction – a transdisciplinary approach to trigger change!

Teaching new combinations of topics (SULR as a subset of SLM, land degradation, eco-entrepreneurship and innovation for sustainable urban land restoration) requires a systems approach to teaching, which goes beyond interdisciplinarity, *i.e.* transdisciplinarity. The underlying message of this course is that unsustainable practices must be changed, which is only possible if scientists and practitioners (policymakers, decision-makers, resource users, farmers, etc.) cooperate closely and transcend traditional disciplinary boundaries. The transdisciplinary approach is synonymous with “knowledge co-production” or “social learning”. This requires methodological and social skills as well as a positive attitude towards transformation (Herweg *et al.*, 2021: 20-21). According to Pohl and Hirsch Hadorn (2007), **transdisciplinarity** – in which different disciplines work from practice with the participation of various stakeholders – **is appropriate for addressing: ill-defined problems**, *i.e.* problems that are difficult to define clearly, also referred to as ill-designed or ill-structured problems; and **wicked problems**, which are so complex that they are difficult or impossible to solve.

A problem- and solution-oriented learning approach offers multiple opportunities for transformative moments to arise. Experiential learning under real-life conditions enables deep-level learning to occur (Herweg *et al.*, 2021: 25):

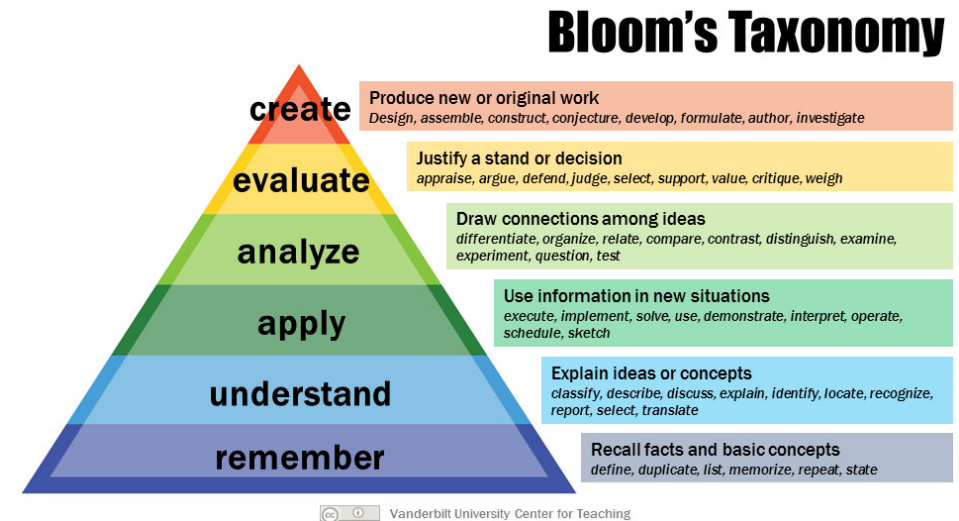
- ▶ Work on a real-world issue, with complex, current problem settings;
- ▶ Be solution-oriented in your approach to this issue;
- ▶ Bring together students from various disciplines;
- ▶ Allow students to interact with actors in the field (experiential learning), outside the classroom.

Transdisciplinary learning prepares students to cooperate and communicate with scientific disciplines other than their own, as well as with other societal actors, in order to develop solutions for ill-defined and wicked problems. Of particular interest is “**case study-based learning**” (CSBL), a blend of “problem-

based learning” and transdisciplinary aspects, such as encounters with societal actors. CSBL focuses on a real-world situation, with real actors, real problems, and real solutions – and is thus more likely to create spaces and opportunities for transformative moments to arise than by using a theoretical situation.

This is consistent with “Bloom’s taxonomy” of learning outcomes, which has been revised from the original emphasis on two learning domains: cognitive (knowledge) and affective (attitude) (Bloom 1956, in Herweg *et al.* 2021). The revised taxonomy focuses on six levels: remember, understand, apply, analyze, evaluate and create (Anderson and Krathworth, 2001, in Herweg *et al.*, 2021) (Fig. 13). This revised version uses action verbs rather than nouns and incorporates the abilities to apply, analyze, evaluate and create.

Figure 13. Revised Bloom’s taxonomy (Anderson and Krathworth, 2001, in Herweg *et al.*, 2021) CC BY 2.0



Thus, **teaching can also be understood as empowerment**. Our course, “Trigger Change! Innovations in Sustainable Urban Land Restoration” course was developed by pedagogic specialists with the aim of assisting course instructors in promoting more active learning and ensuring that students are better equipped and able to “think outside the box”, re-evaluate traditions and systems, analyze systems as they evolve and create innovative outcomes.

Teaching for change may include any of the following teaching tips:

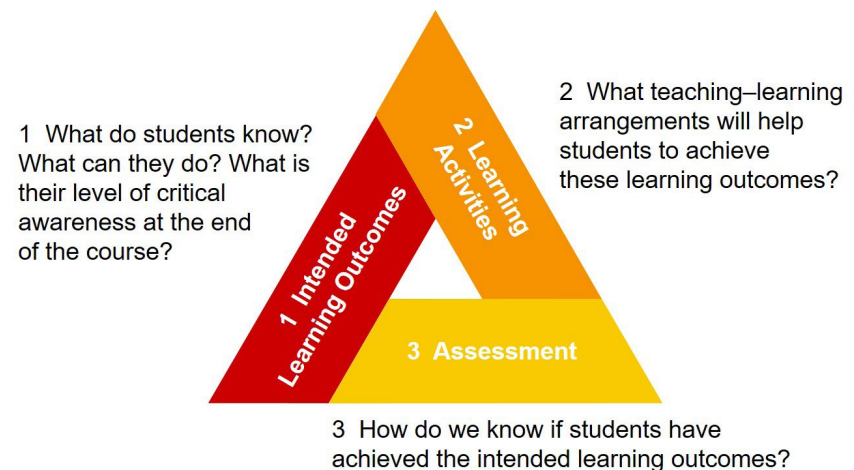
- ▶ Working within a real-world context with complex problem settings; being solution-oriented;
- ▶ Bringing together students from various disciplines;
- ▶ Allowing students to interact with actors in the field, e.g. through experiential learning outside the classroom;
- ▶ Building on participation and cooperation;
- ▶ Meeting practitioners;
- ▶ Adopting a long-term perspective;
- ▶ Having students play a more active role;
- ▶ Focusing on jointly developed solutions;
- ▶ Using emotions as a lever of change – do not ignore them!

The following sections provide additional guidance to instructors looking to deliver high-quality education that moves their students into this more active realm.

3.2 Teaching tips and tools

a. Constructive alignment is a guiding principle of high-quality educational programmes. It requires coherence between the three core elements – learning outcomes (competences), learning activities (teaching–learning arrangements), and assessment (of competences) (Herweg *et al.*, 2021) (Fig 14). Assessments are typically either formative assessments (e.g. to ensure that they have understood the course materials) or summative assessments (e.g. to evaluate student performance at the end of the course).

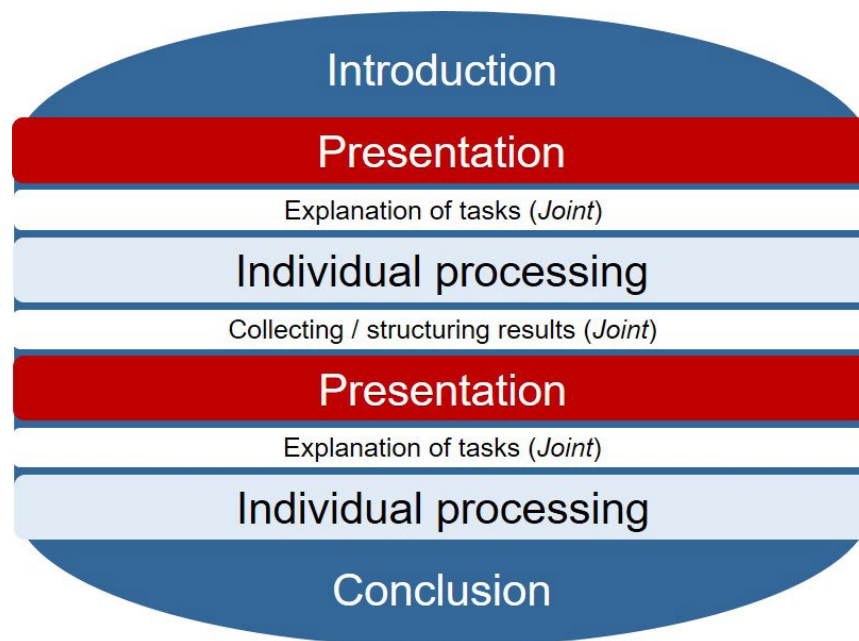
Figure 14. Constructive alignment in teaching (Herweg *et al.*, 2021).



b. Cognitive activation and the 20-minute rule

The human working memory has a naturally limited attention span, usually leading to much of the audience becoming at some point restless or tired. The point at which this occurs will vary and may at least partly depend on the quality of a lecture. Decades ago, research led to the 20-minute rule, which is still widely observed today. German didactics describe this as “rhythmizing”, meaning varying activities within a teaching sequence after an average of twenty minutes (Fig. 15).

Figure 15. The “sandwich” 20-minute interactive learning approach for more sustainable learning outcomes (Herweg et al., 2021).



c. Activating a large audience

A wide range of teaching methods trigger cognitive activation. Try to keep students as engaged as possible: they learn more when they actively work through content. This is also possible in the lecture theatre. Instead of the usual “Any questions?” at the end (which often goes unanswered), have small groups discuss a question for three minutes (“buzz groups”) or let them work on a small task and then discuss the results in the plenary. Other options include having students come up with advantages and disadvantages of an issue and comparing them afterwards, or asking learners for their opinions, assumptions, or prior experiences before presenting a particular topic. Such active breaks work well, even with a large audience.

d. Activating smaller groups

Of course, activating smaller groups in seminars is much easier than motivating a larger audience.

- ▶ Alternate a 20-minute presentation with an equally long sequence allowing students to process the information;
- ▶ Limit working groups to four people;
- ▶ Write down assignments based on the learning outcomes; project the assignments onto the wall in larger groups;
- ▶ Some ideas for assignments: Write a short synopsis on the topic;
- ▶ Set an examination task for other students;
- ▶ Draw an overview graphic.

e. Strategies for cognitive activation

Use pre-testing, which has the added effect of reactivating prior knowledge;

Before your lecture, let your students brainstorm on the question: “What do I already know about the topic?”; Have participants formulate questions about the material in advance; Follow the 20-minute rule in lecturing scenarios. Consider the sandwich principle, particularly in seminars;

Use buzz groups, i.e. small-group discussions of simple questions in-between two presentations;

Have learners answer “why?” as often as possible; let them research causes, reasons and origins in small discussion groups’;

Let them find differences and similarities between new content or old and new content (sameness analysis) (Fig. 8) (Herweg et al. 2021).

Cognitive activation is essential for learning gain. Successful learning can only be achieved through active engagement with content. Therefore: **less teaching; more learning.**

f. Cooperative Learning

Although many people may remember group work as being unsatisfactory for various reasons, statistical meta-analyses have found cooperative learning to have a considerable impact on learning. Moreover, it strengthens other factors such as self-confidence and social integration, and reduces the likelihood of dropping out of the course altogether (see also Hattie, 2015).

Think – Pair – Share

Teaching can take place in various social settings, primarily the following four: individual work – partner work – group work – plenary. Not every subject can be covered in all settings. The social setting and choice of teaching strategy are closely related. The “think – pair – share” phase model is a simple sequence starting at the individual level, then moving to small groups, and finally to the plenary. This is the basic pattern of many cooperative learning scenarios.

Think = individually analyse a topic;

Pair = draw individual conclusions and discuss with a partner;

Share = present the results of think and pair to the plenary.

Tip: Present the working tasks before announcing the altered social setting, otherwise they can easily get lost as learners search for partners when preparing for group work (Herweg et al., 2021).

g. Infrastructure for successful cooperative learning

Mundane factors such as infrastructure are also important. Furniture that cannot be moved and a lack of space strongly hinder cooperative learning. If necessary, request another classroom. However, discussions of a few minutes between three people are also possible in the lecture theatre, and technology can also be used to expand the scope. For group work to be effective, certain conditions and strategies for effective group work must be taken into account:

- Clear assignments and tasks;
- Defined roles and positive interdependence;
- Suitable infrastructure;
- Appropriate group size;
- Clear process and open learning scenarios.

h. Condensing content

Varying your learning sequences and adding student activation windows automatically reduces your own inputs. But even without this, experience shows that you always have too much material and too little time.

In short – The **4Rs**: The following four steps can be used to select the learning content:

1. Research: Find content in relevant and valid sources.

2. Reflect: What information does this particular group of learners need as starting points to complete the assignments (i.e. to master the situations)?

3. Reduce: Filter content based on what is exemplary and relevant (must – should – could).

4. Represent: Decide how to best represent your content: through a direct encounter (field excursion, discussion with practitioners, etc.), audiovisual aids (videos, photos, sound, etc.) or text (Herweg et al. 2021).

i. Guidelines for designing learning scenarios with high self-direction

The main goal of “self-directed learning” is the ability to act autonomously, an important feature for future agents of change and innovators!

- ▶ Define action-oriented goals;
- ▶ Ensure appropriate workloads;
- ▶ Balance self-learning periods and contact phases;
- ▶ Manage expectations and responsibility;
- ▶ Discuss how to manage “free riders” by establishing guidelines for “good group work”;
- ▶ Establish clear rules on how group work (and individual contributions) will be assessed;
- ▶ Provide constructive criticism with feedback that leads to greater responsibility and empowerment (Herweg et al. 2021).

j. Field visits

The field visit requires adequate advance preparation, both in terms of logistics and participant preparation. It is better to omit the field visit altogether than to use module time to undertake a touristic outing without clear links to module content. In the event that community groups or other stakeholders are consulted during the visit, it is important to ensure that they are invited to actively participate in discussions and are met with respect, considering their own traditions, norms and values. When planning the overall training module

programme, organizers need to assess the feasibility of including a field visit and identify an appropriate objective and context for the visit as the availability of appropriate field visit sites depends very much on the national and local context. It may be difficult to identify field visit locations, or field visits may not be possible or even too costly. Instead, instructors can substitute with case studies, or videos included in the course materials and engage students in a discussion on the pros and cons to each.

Conducting a field visit

1. Preparation session

A short session where essential information relating to the field visit (content, aims and logistics) is provided to participants (*i.e.* the day prior to the visit). If practical work will be undertaken during the visit, instructions are provided and clarified as appropriate;

2. Field visit

The duration of the visit varies depending on the context and organizers should identify an appropriate time in the module (*i.e.* last module day or in the middle of the module);

3. Reflection session

Upon completion of the field visit, a feedback and reflection session should be held to analyse the visit, and/or allow groups to report back on their results (if practical work was included).

References

- ▶ Anderson L.W. and D.R. Krathwohl. (2001). *A taxonomy for learning, teaching, and assessing: a revision of Bloom's taxonomy of educational objectives*. New York, NY, US: Longman.
- ▶ Bennett, N. J., Roth, R., Klain, S. C., Chan, K., Christie, P., Clark, D. A., Cullman, G., Curran, D., Durbin, T. J., Epstein, G., Greenberg, A., Nelson, M. P., Sandlos, J., Stedman, R., Teel, T. L., Thomas, R., Veríssimo, D., and C. Wyborn. (2017). Conservation social science: Understanding and integrating human dimensions to improve conservation. *Biological Conservation*, 205, 93–108. <https://doi.org/10.1016/j.biocon.2016.10.006>
- ▶ Bloom, B.S. (1956). *A Taxonomy of Educational Objectives Handbook I The Cognitive Domain*. Longman, Green Co., New York.
- ▶ Cash, D. W., Adger, W. N., Berkes, F., Garden, P., Lebel, L., Olsson, P., Pritchard, L., and O. Young (2006). Scale and Cross-Scale Dynamics: Governance and Information in a Multilevel World. *Ecology and Society*, 11(2). <http://www.jstor.org/stable/26265993>
- ▶ Faruqi, S., Wu, A., Brolis, E., Ortega, A.A. and A. Batista. (2018). *The Business of Planting Trees, A Growing Investment Opportunity*, Washington D.C.: The World Resources Institute and the Nature Conservancy: <https://www.wri.org/research/business-planting-trees-growing-investment-opportunity>
- ▶ Feng, S., Zhao, W., Zhan, T., Yan, Y., and P. Pereira.(2022). Land degradation neutrality: A review of progress and perspectives. *Ecological Indicators*, 144, 109530. <https://doi.org/https://doi.org/10.1016/j.ecolind.2022.109530>
- ▶ Habitat III. (2016). *Glossary of the Habitat III Preparatory Process and the Conference*. <https://habitat3.org/wp-content/uploads/Habitat-III-Glossary.pdf>
- ▶ Fischer, J., Gardner, T. A., Bennett, E. M., Balvanera, P., Biggs, R., Carpenter, S., Daw, T., Folke, C., Hill, R., Hughes, T. P., Luthe, T., Maass, M., Meacham, M., Norström, A. V., Peterson, G., Queiroz, C., Seppelt, R., Spierenburg, M., and J. Tenhunen. (2015). Advancing sustainability through mainstreaming a social–ecological systems perspective. *Current Opinion in Environmental Sustainability*, 14, 144–149. <https://doi.org/10.1016/j.cosust.2015.06.002>
- ▶ Habitat III. (2016). *Glossary of the Habitat III Preparatory Process and the Conference*. <https://habitat3.org/wp-content/uploads/Habitat-III-Glossary.pdf>
- ▶ Hall, P., and M. Tewdwr-Jones. (2020). *Urban and Regional Planning* (Sixth Edition). Routledge, Taylor & Francis Group.
- ▶ Hattie, J. (2015). The applicability of Visible Learning to higher education. *Scholarship of Teaching and Learning in Psychology* 1(1): 79–91.
- ▶ Herweg, K., Tribelhorn, T. Lewis, A.L., Providoli, I., Trechsel, L.J., and C. Steinböck. (2021). *Transdisciplinary Learning for Sustainable Development, Sharing Experience In Course And Curriculum Design*, Bern, Switzerland: Center for Development and Environment, University of Bern with Bern Open Publishing (BOP): https://boris.unibe.ch/157343/8/150dpi_online_E_tdLearnSD.pdf
- ▶ Holling, C.S. (1978). *Adaptive environmental assessment and management*. John Wiley & Sons.
- ▶ Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). (2018). *The IPBES assessment report on land degradation and restoration*. Montanarella, L., Scholes, R., and A. Brainich. (eds.). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, Bonn, Germany. 744 pages. <https://doi.org/10.5281/zenodo.3237392>
- ▶ IPCC. (2022a). *Land degradation*. In: *Climate Change and Land: IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems*. Cambridge University Press; 2022:345–436. <https://doi.org/10.1017/9781009157988.006>
- ▶ IPCC. (2022b). Glossary. In: Intergovernmental Panel on Climate Change (IPCC) (Ed.), *Climate Change 2022 – Impacts, Adaptation and Vulnerability* (pp. 2897–2930). Cambridge University Press. <https://doi.org/10.1017/9781009325844.029>

- ▶ IPCC. (2022c). Urban systems and other settlements. In *Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (pp.861-952). Cambridge University Press. <https://doi.org/10.1017/9781009157926.010>
- ▶ Local Governments for Sustainability (ICLEI) (2017). Land degradation and cities: The essential role of local and regional government. ICLEI Briefing Sheet - 2017, No. 3 - Land Degradation and Cities. <https://iclei.org/e-library/land-degradation-and-cities-the-essential-role-of-local-and-regional-governments/>
- ▶ Millennium Ecosystem Assessment (MEA). (2005). *Ecosystems and Human Well-being: Synthesis*. Island Press: <https://www.millenniumassessment.org/documents/document.356.aspx.pdf>.
- ▶ Milner-Gulland, E. J., Addison, P., Arlidge, W. N. S., Baker, J., Booth, H., Brooks, T., Bull, J. W., Burgass, M. J., Ekstrom, J., zu Ermgassen, S. O. S. E., Fleming, L. V., Grub, H. M. J., von Hase, A., Hoffmann, M., Hutton, J., Juffe-Bignoli, D., ten Kate, K., Kiesecker, J., Kümpel, N. F., ... and J. E. M. Watson. (2021). Four steps for the Earth: mainstreaming the post-2020 global biodiversity framework. *One Earth*, 4(1), 75–87. <https://doi.org/https://doi.org/10.1016/j.oneear.2020.12.011>
- ▶ Pohl C., and G. Hirsch Hadorn. (2007). *Principles for Designing Transdisciplinary Research*. Munich, Germany: oekom. https://www.oekom.de/_files_media/titel/leseproben/9783865810465.pdf
- ▶ Raworth, K. (2017). *Doughnut Economics : Seven Ways to Think Like a 21st-century Economist*. London: Random House.
- ▶ Richardson, K., Steffen, W., Lucht, W., Bendtsen, J., Cornell, S.E., Donges, J.F., Drüke, M., Fetzer, I., Bala, G., von Bloh, W., et al. (2023). Earth beyond six of nine planetary boundaries. *Sci. Adv.* 2023, 9, eadh2458 <https://www.science.org/doi/10.1126/sciadv.adh2458>
- ▶ Rockström, J., Steffen, W., Noone, K., Persson, Å., Chapin, F.S., III, Lambin, E.F., Lenton, T.M., Scheffer, M., Folke, C., Schellnhuber, H.J., et al. (2009) A Safe Operating Space for Humanity. *Nature* 2009, 461, 472–475 <https://www.nature.com/articles/461472a>
- ▶ Steffen, W., Richardson, K., Rockström, J., Cornell, S.E., Fetzer, I., Bennett, E.M., Biggs, R., Carpenter, S.R., De Vries, W., De Wit, C.A., et al. (2015) Planetary Boundaries: Guiding Human Development on a Changing Planet. *Science* 2015, 347, 1259855 <https://www.science.org/doi/10.1126/science.1259855>
- ▶ Tomalka, J., Hunecke, C., Murken, L., Heckmann, T., Cronauer, C., Becker, R., Collignon, Q., Collins-Sowah, P., Crawford, M., Gloy, N., Hampf, A., Lotze-Campen, H., Malevolti, G., Maskell, G., Müller, C., Popp, A., Vodounhessi, M., Gornott, C., Rockström, J. (2024). *Stepping back from the precipice: Transforming land management to stay within planetary boundaries*. Potsdam, Germany: Potsdam Institute for Climate Impact Research. DOI: <https://doi.org/10.48485/pik.2024.018>.
- ▶ United Nations Convention to Combat Desertification/ Group of 20-Global Land Initiative (UNCCD/G20 GLI). (2023). *Implementation strategy and work plan*. UNCCD: Bonn. <https://g20land.org/implementation-strategy/>
- ▶ UNCCD. (2022). Sustainable Development Goals indicator metadata. <https://unstats.un.org/sdgs/metadata/files/Metadata-15-03-01.pdf>
- ▶ United Nations, Department of Economic and Social Affairs, Population Division, (2019). *World Urbanization Prospects: The 2018 Revision*. ST/ESA/SER.A/420. United Nations, New York .DOI: <https://doi.org/10.18356/b9e995fe-en>
- ▶ United Nations Environment Assembly (UNEA) UN Environment Assembly (UNEA 5.2) Resolutions <https://www.unep.org/resources/resolutions-treaties-and-decisions/UN-Environment-Assembly-5-2>
- ▶ United Nations Environment Programme (UNEP). (2021). *Becoming #GenerationRestoration: Ecosystem restoration for people, nature and climate*. <https://www.unep.org/resources/ecosystem-restoration-people-nature-climate>
- ▶ United Nations System Staff College (UNSSC). (no date). *Innovation toolkit - getting started*. https://www.unssc.org/sites/default/files/toolkit_getting_started_guide.pdf

- ▶ van Dijk, M. P. (2006). *Managing Cities in Developing Countries*. <https://www.e-elgar.com/shop/gbp/managing-cities-in-developing-countries-9781845428808.html>
- ▶ Walters, C.J. (1986). *Adaptive management of renewable resources*. Macmillan.
- ▶ World Overview of Conservation Approaches and Technologies (WOCAT). (2007). *Where the land is greener – case studies and analysis of soil and water conservation initiatives worldwide*. Eds: Hanspeter Liniger and William Critchley. https://www.wocat.net/documents/87/where_the_land_is_greener_WEB.pdf
- ▶ World Overview of Conservation Approaches and Technologies (WOCAT). (2019). *SLM Practices: Technologies and Approaches*: <https://www.wocat.net/en/global-slm-database/slm-practices-technologies-and-approaches/>

Important links

- ▶ Scientific conceptual framework for Land Degradation Neutrality. A report of the Science-Policy Interface <https://www.unccd.int/resources/reports/scientific-conceptual-framework-land-degradation-neutrality-report-science-policy>
- ▶ Science-Policy Brief: integrated land use planning and integrated landscape management to implementing Land Degradation Neutrality <https://www.unccd.int/resources/brief/science-policy-brief-integrated-land-use-planning-and-integrated-landscape>
- ▶ The contribution of integrated land use planning and integrated landscape management to implementing Land Degradation Neutrality: Entry points and support tools | UNCCD
- ▶ Technical Guide on the Integration of the Voluntary Guidelines on the Responsible Governance of Tenure of Land <https://www.unccd.int/resources/publications/technical-guide-integration-voluntary-guidelines-responsible-governance>



4. Course module descriptions



Figure 16. Course overview

1. Why SULR?	2. Approaches to SULR	3. Enabling SULR	4. Operationalizing SULR	5. Catalyzing SULR through innovation
1.1 Challenges and big picture	2.1 City systems	3.1 Sequence and framing	4.1 Tools for planning, implementing, and monitoring SULR	5.1 Innovation in SULR
1.2 Ecological ceiling	2.2 Planning rationale	3.2 Enabling environment and co-benefits	4.2 Data, Data Literacy and Data Governance for SULR	5.2 Developing your SULR Innovation Model
1.3 Social foundation	2.3 Frameworks, approaches, narratives-entry points for SULR	3.3 Actors and Stakeholders in SULR	4.3 Data requirements for SULR	5.3 SULR Value Proposition
1.4 Safe and just urban areas	2.4 Operationalizing LDN through land use management and planning	3.4 Multi-Level Governance in SULg	4.4 Overview of tools and methods for assessing Urban Land Degradation	5.4 Operationalizing SULR value proposition
1.5 Urban Land Degradation	2.5 Thematic approaches SULR-blue-green-brown spaces	3.5 Co-creation, visioning and goal setting	4.5 Mapping Urban Land Degradation	5.5 SULR feasibility
1.6 Sustainable cities – principles & frameworks	2.6 Nature-based Solutions in cities	3.6 Policy instruments	4.6 Urban planning and design tools	5.6 Communication: Scaling SULR Innovations
1.7 Urban Typologies	2.7 Design principles	3.7 Finance instruments	4.7 SULR decision design tools	5.7 Good practices I - Locally-led and Grassroot Initiatives
1.8 Urban ecosystem services	2.8 Theory of change (Exercise)	3.8 Challenges and handling conflict	4.8 Citizen science and local empowerment	5.8 Good practices II: Private Sector
1.9 Sustainable Urban Land Restoration (SULR)		3.9 Urban stocktaking and action prioritization	4.9 Monitoring and Evaluation (M&E) in SULR	5.9 Good practices III: Public Sector
1.10 Identifying Urban Land Restoration Demands (Exercise)		3.10 Assess the status quo - identify actors and resources needed (Exercise)	4.10 Assess, implement and monitor a SULR intervention (Exercise)	5.10 Pitching your Sustainable Urban Land Restoration Innovation (Final assignment)

Module 1. Why Sustainable Urban Land Restoration?

Description

Urban areas continue to develop and grow rapidly; by 2050, 68% of the world population will be urban. Urban areas exert high pressure on land resources through their intensive land use, resulting in increasing challenges from accelerating land degradation. Hence, understanding the critical role of sustainable urban land restoration for sustaining critical land functions is urgent. Unsustainable urban expansion, resource overuse, and land use change not only drive degradation but also contribute to crossing planetary boundaries and failing to meet essential social needs.

This module uses Raworth's (2017) "doughnut of social and planetary boundaries" as a guiding conceptual framework to examine how sustainable urban land restoration can help urban areas address main environmental and social challenges to operate within ecological limits, and foster safe, just and resilient cities. The module presents environmental and socio-economic components of cities and their interlinkages. It also explores different framings for defining urban areas, urban ecosystem services, and causes of urban land degradation in order to discover opportunities for urban land restoration.

Core competencies

Academic knowledge:

- ▶ The doughnut of social and planetary boundaries framework, consisting of an ecological ceiling, socio-economic foundation, and safe and just operating space concepts, applied to urban contexts;
- ▶ Sustainable cities principles and frameworks, the pillars of sustainability and the qualifiers used to promote sustainable urban planning;
- ▶ Urban land as a social-ecological system, providing ecosystem services and fundamental benefits for the well-being of people;
- ▶ Types of SULR actions (avoid, reduce, reverse land degradation and renew the urban-nature-city relationship), based on the land degradation neutrality and mitigation conservation frameworks;

Skills:

- ▶ Relate the doughnut of social and planetary boundaries to main urban challenges and apply it as analytical approach;
- ▶ Understand urban land as a system with ecological and social components and processes that are interlinked in multiple ways;
- ▶ Connect urban land degradation causes, related consequences and drivers, and contextualize them in local urban spaces;
- ▶ Identify the most important environmental and social challenges in urban areas and how SULR can help to solve them;
- ▶ Determine the need for SULR according to degradation characteristics and context.

Attitudes:

- ▶ Perspective: Identifying SULR relevance at global, national, urban and local scales;
- ▶ Sense-making: Current and future urban challenges and the need to act.

Authors

Module coordination: UNU-EHS

Developers:

(UNU-EHS)

Alejandra Ramos Gálvez (MSc)
Leticia Domingos Vellozo (MSc)
Dr. Simone Sandholz

Syed Muhammad Jaffar Abbas (MSc)

Andrea Ortiz Vargas (MSc)

Lucas de Carvalho Turmena (MSc)

Dr. Yvonne Walz

Nathan Fridtjof Schultze-Florey

(ASU)

Dr. Aya Elsisy

Prof. Dr. Marwa Khalifa

(Oxford University)

Prof. Nick Leimu-Brown

(THK)

Prof. Dr. Johannes Hamhaber

(UNCCD-G20)

Dr. Karen Sudmeier-Rieux

1.1 Challenges and big picture

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Nick Leimu-Brown (Oxford University)

Karen Sudmeier-Rieux (UNCCD-G20)

Andrea Ramirez Agudelo (UNU-EHS)

Learning objectives

- ▶ Understand the current planetary boundaries overshoot, social needs shortfall and related urban challenges;
- ▶ Recognize the needed balance between planetary boundaries and social foundations for equitable and sustainable urban development;
- ▶ Distinguish SULR role in addressing land degradation and urban challenges.

Summary

Part 1. Why should we talk about sustainable urban land restoration?

Our current way of living and building cities is not sustainable; we are facing planetary boundaries overshoot and a social needs shortfall. Unsustainable development and management practices, land use change and land degradation have contributed to crossing the land system boundary. The doughnut framework of social and planetary boundaries offers a compass for achieving urban sustainability by balancing ecological limits and social foundations, resulting in a safe and just operating space.

Part 2. Main environmental and social urban challenges

Urban areas worldwide face severe environmental and socio-economic challenges, such as urban sprawl, urban population growth, inadequate housing, air pollution, water scarcity and pollution, waste contamination, and biodiversity loss, which jeopardize resilience and sustainability. SULR could help solve these urban issues.

Part 3. SULR relevance

SULR can contribute to safe and just operating cities and neighborhoods within the planetary and social foundations. SULR efforts go from local urban projects to more strategic planning without losing sight of the global impacts. SULR contributes to sustainable urban development, addressing environmental degradation and urban challenges.

Part 4. Case Study—Barcelona

The Barcelona case study showcases SULR as an evolving process, marked by challenges and innovative solutions over time. The case study illustrates how a city can adopt sustainable urban land restoration strategies to enhance ecological integrity, improve urban living conditions, and adapt to future challenges.

What is this course about?

This course provides a structured exploration of SULR through five interconnected yet flexible modules.

Lessons learned

- ▶ Unsustainable urban development can be a driver of land degradation;
- ▶ Considering the doughnut of social and planetary boundaries as a compass for urban development and urban interventions helps balance environmental concerns and social benefits;
- ▶ Getting to a safe and just operating space in cities will require land restoration efforts;
- ▶ SULR addresses critical urban environmental and social challenges, offering a strategic pathway to enhance ecosystem services, urban resilience and sustainability.

Student questions and exercises

- ▶ What are the main urban challenges where you live?
- ▶ Are they somehow linked to land degradation?
- ▶ Overlap land degradation with additional information, such as the number of people living in slums, the income of the population, or the ecological footprint. What do you observe?

See Case Study in Section 5.1: Barcelona Metropolitan Area, Besòs River: Sustainable Urban Land Restoration

Also see our documentary film, “Green Revival: From the River to the Heart of the City, Barcelona, Spain”: <https://youtu.be/Uml0NY8P9ac?si=d0MLbyzVUoYNkkrR>

References/resources

- DEAL (2022). *Doughnut Economics Action Lab. Principles and Guidelines. Principles & Guidelines | DEAL*
- IPBES (2018). *The IPBES assessment report on land degradation and restoration*. Montanarella, L., Scholes, R., and Brainich, A. (eds.). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, Bonn, Germany. 744 pages. <https://doi.org/10.5281/zenodo.3237392>
- IPCC (2022). *Land degradation*. In: Intergovernmental Panel on Climate Change (IPCC) (Ed.), *Climate Change and Land* (pp. 345–436). Cambridge University Press. <https://doi.org/10.1017/9781009157988.006>
- Raworth, Kate. (2013-2025). *Exploring Doughnut economics*. <https://www.kateraworth.com/doughnut/>
- Richardson, K. Steffen, W., Lucht, W., Bendtsen, J., Cornell, S.E., Donges, J.F., Drüke, M., Fetzer, I., Bala, G., von Bloh, W., Feulner, G., Fiedler, S., Gerten, D., Gleeson, T., Hofmann, M., Huiskamp, W., Kummu, M., Mohan, C., Nogués-Bravo, D., et al., (2023). Earth beyond six of nine planetary boundaries. *Sci. Adv.* 9, eadh2458 DOI:10.1126/sciadv.adh2458
- Tomalka, J., Hunecke, C., Murken, L., Heckmann, T., Cronauer, C., Becker, R., et al. (2024). *Stepping back from the precipice: Transforming land management to stay within planetary boundaries*. [Stepping back from the precipice: Transforming land management to stay within planetary boundaries :: PIKpublic](https://www.pikpublications.org/publication/stepping-back-from-the-precipice-transforming-land-management-to-stay-within-planetary-boundaries)

1.2 Ecological ceiling

Developed by

Alejandra Ramos (UNU-EHS)

Nick Leimu-Brown (Oxford University)

Simone Sandholz (UNU-EHS)

Learning objectives

- ▶ Understand the concept of planetary boundaries—ecological ceiling, particularly land system change;
- ▶ Analyze the role of ecosystem restoration in addressing land degradation and promoting healthy ecosystems;
- ▶ Identify the most important environmental challenges in cities and how SULR can help to solve them.

Summary

Part 1. Introduction - Ecological ceiling in urban areas

The concept of planetary boundaries provides a framework for understanding the ecological limits within which humanity can safely operate. These boundaries, such as climate change and land system change, serve as an “ecological ceiling” that cities must respect to ensure sustainable development. Urban areas have a high demand for resources and big ecological footprints that strain the “ecological ceiling” through, *i.e.*, habitat fragmentation, heat island effects, and waterway pollution.

Part 2. Ecosystem Restoration

This section provides a timeline about the history of the environmental movement, explaining the UN Decade on Ecosystem Restoration and its principles. Ecosystem restoration is defined, and sustainable urban land restoration can be

understood as part of this large worldwide effort. In urban ecosystems, efforts to restore should contribute to more nature-positive urban areas.

Part 3. SULR relevance

Restoration in urban areas is of high priority because they are the locus of some of the most severe environmental degradation issues that SULR can help to mitigate. Main urban environmental problems are air pollution, water scarcity, water pollution, flooding, heat stress, landslides, waste contamination, and biodiversity loss.

Lessons learned

- ▶ The concept of the ecological ceiling highlights the urgent need for cities to operate within planetary boundaries;
- ▶ Cities need to shift from being ecological burdens to being regenerative spaces that contribute positively to the environment;
- ▶ While restoration efforts are often associated with rural or natural landscapes, urban areas, as hubs of environmental degradation, require immediate and focused restoration strategies;
- ▶ SULR directly addresses the environmental crises faced by cities, from air and water pollution to extreme weather impacts.

Student questions and exercises

- ▶ Are you aware of the biodiversity status of the place where you live?
- ▶ What are the most urgent environmental challenges in your city?

See Case Study in Section 5.2. Beirut: Reclaiming urban green spaces and restoring hope, Lebanon

References/resources

- Aram, F., E. Higuera García, E. Solgi and S. Mansournia. (2019). Urban green space cooling effect in cities. *Heliyon* 5(4): e01339.
- Canteiro, M., Arellano-Aguilar, O., Bravo, J.E.B. and L. Zambrano (2023). Urban green spaces and their relationship with groundwater quality: the case of a shallow aquifer in the south of Mexico City. *Sustain. Water Resour. Manag.* 9, 156. <https://doi.org/10.1007/s40899-023-00935-x>
- Carla S.S. Ferreira, Rory P.D., Walsh, A. and J.D. Ferreira. (2018). Degradation in urban areas, *Current Opinion in Environmental Science & Health*, Volume 5, 19-25, <https://doi.org/10.1016/j.coesh.2018.04.001>.
- Fairbairn, A.J., Meyer, S.T., Mühlbauer, M. et al. Urban biodiversity is affected by human-designed features of public squares. *Nat Cities* 1, 706–715 (2024). <https://doi.org/10.1038/s44284-024-00126-5>
- FAO, IUCN/CEM, SER. (2021). *Principles for ecosystem restoration to guide the United Nations Decade 2021-2030*. <https://openknowledge.fao.org/handle/20.500.14283/cb6591en>
- Hou, D., Al-Tabbaa, A., O'Connor, D. et al. (2023). Sustainable remediation and redevelopment of brownfield sites. *Nat Rev Earth Environ* 4, 271–286. <https://doi.org/10.1038/s43017-023-00404-1>.
- Laurance, W. F. and J. Engert (2022). Sprawling cities are rapidly encroaching on Earth's biodiversity. *Proceedings of the National Academy of Sciences* 119(16): e2202244119.
- Shi, X., Mao, D., Song, K., Xiang, H., Li, S., and Z. Wang. (2024). *Effects of landscape changes on water quality: A global meta-analysis*. *Water Research* 260: 121946. <https://doi.org/10.1016/j.watres.2024.121946>
- Tomalka, J., Hunecke, C., Murken, L., Heckmann, T., Cronauer, C., Becker, R., Collignon, Q., Collins-Sowah, P., Crawford, M., Gloy, N., Hampf, A., Lotze-Campen, H., Malevolti, G., Maskell, G., Müller, C., Popp, A., Vodounhessi, M., Gornott, C., Rockström, J. (2024). *Stepping back from the precipice: Transforming land management to stay within planetary boundaries*. Potsdam, Germany: Potsdam Institute for Climate Impact Research. DOI: <https://doi.org/10.48485/pik.2024.018>.

1.3 Social foundation

Developed by

Simone Sandholz (UNU-EHS)
Aya Elsisy (ASU)
Alejandra Ramos (UNU-EHS)

Learning objectives

- Understand the need to create safe and just operating spaces that balance socio-economic and ecological needs;
- Understand the role of cities in national and global economies, link economic priorities to urban impacts and identify implications on urban development, land uses and critical land resources;
- Understand the role of politics and policy for urban development, land uses and land resources;
- Reflect on social consequences from political and economic decisions;
- Being able to link economic and political decisions/priorities to urban impacts and land uses.

Summary

Part 1. Socio-economic needs as determining factor

After introducing the basics of urban economic and political aspects in doughnut approach, Part 1 dives into the role of cities and how they are economic hubs within countries and regions. It describes the role of development paradigms for urban development, how far these were/are shaped by economic thinking, and ends on describing how far a different paradigm can change planning, based on a case study.

Part 2. Economic dimensions of sustainable urban development

Part 2 focuses on the economic dimensions of sustainable urban development. Starting from fleshing out current economic development across world regions and between rural and urban areas, implications on land use, ownership and access to different user groups are described. It highlights how seeing land as a predominantly economic asset can contribute to degradation, and lack of accessibility. It ends on theorising differences in land use and access by shifting economies. In doing so the module draws on different theoretical approaches including political economy and global cities.

Part 3. Political dimensions of sustainable urban development

Part 3 links economic to political dimensions by presenting and questioning power hierarchies, including due to available capital, and implications of economic power on political decision-making. Actor types involved in urban governance are listed, pointing at their potentially conflicting interests and role they could take in restoration (governance). Finally, negative social impacts that could go along with land use decisions are discussed.

Part 4. Reflections on territoriality

Part 4 summarizes the previous parts and links them, drawing on the concept of territoriality. It ends on reflections around pathways to mainstream land restoration into urban policy-making.

Lessons learned

- ▶ Global economic systems have impacts on national and urban economies and land use/resources in cities;
- ▶ Rethinking economies - following doughnut economics - can be beneficial for land resources/restoration;
- ▶ Land is a valuable economic resource in itself, and decisions over land use are including economic aspects and political priorities;
- ▶ Shifting economic systems and political priorities will have implications on land use and protection/restoration of land resources.

Student questions and exercises

The submodule picks up on the doughnut framework introduced previously and focuses on the socio-economic needs that guide urban development.

It starts from a reflection by asking:

- ▶ Which social foundations (categories in inner circle) do you see connected to politics and economy?
- ▶ Which planetary boundaries (categories in inner circle) do you see connected to politics and economy?
- ▶ Are there some that are particularly linked? How?

The module includes another student exercise where students are asked to identify actors in urban land restoration, based on a figure identifying actor groups. Guiding questions are:

- ▶ Which actors within these groups would you see involved in urban land restoration and related governance processes?
- ▶ What could be the particular interests of these actors?

Further questions that could spark discussions are:

- ▶ How far do dominant economic models influence the way we plan our cities?
- ▶ How far do these economic models determine the way we see and value land resources

References/resources

Crisp, R., Waite, D., Green, A., Hughes, C., Lupton, R., MacKinnon, D., and A. Pike. (2023). 'Beyond GDP' in cities: Assessing alternative approaches to urban economic development. *Urban Studies*, 61(7), 1209-1229. <https://doi.org/10.1177/00420980231187884>

Ferretto, A., Matthews, R., Brooker, R., and P. Smith. (2022). Planetary Boundaries and the Doughnut frameworks: A review of their local operability. *Anthropocene* 39:100347 <https://doi.org/10.1016/j.ancene.2022.100347>

Goodwin, K. (2021). *Designing the Doughnut: A Story of Five Cities* <https://doughnuteconomics.org/stories/designing-the-doughnut-a-story-of-five-cities>

Khmara, Y. and J. Kronenberg. (2022). Urban degrowth economics: making cities better places for living, working, and playing. *Local Environment*, 28(3), 304–321. <https://doi.org/10.1080/13549839.2022.2136638>

Turner, R. A. and J. Wills. (2022). Downscaling doughnut economics for sustainability governance, *Current Opinion in Environmental Sustainability* 56:101180. <https://doi.org/10.1016/j.cosust.2022.101180>

Additional references:

Elden, S. (2010). Land, terrain, territory. *Progress in Human Geography*, 34(6), 799–817. <https://doi.org/10.1177/0309132510362603>

Tao, Q., Tao, Y., Huang, C., Ou, W., Bryan, B.A. and C.L. Archibald. (2024). The role of ecosystem services within safe and just operating space at the regional scale. *Landsc Ecol* 39, 134. <https://doi.org/10.1007/s10980-024-01934-9>

1.4 Safe and just urban areas

Developed by

Syed Muhammad Jaffar Abbas (UNU-EHS)

Simone Sandholz (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Learning objectives

- ▶ Understand urban vulnerabilities affecting marginalized communities;
- ▶ Analyze the concepts of environmental justice and just transitions and its relevance in urban settings;
- ▶ Distinguish how SULR can contribute to safe and just urban areas.

Summary

Part 1. Understanding Vulnerability in Cities

Part 1 begins by defining urban vulnerability and examining the socio-economic and environmental factors that contribute to it. It then introduces key concepts such as urban poverty and land tenure, providing a foundation for understanding how urban spaces can be more equitably built and shared.

Part 2. Environmental Justice and Just Transitions

Environmental and climate justice address the unequal distribution of environmental burdens and benefits among different communities, often exacerbated in urban areas. Marginalized populations disproportionately face pollution, climate risks, and limited access to green or healthy open public spaces. The principles of just transitions aim to achieve sustainability while ensuring equity and inclusivity, particularly for vulnerable groups.

Part 3. SULR contributing to safe and just urban areas

Emphasis is placed on integrating a justice approach into restoration actions and urban planning and development to create cities that are not only environmentally sustainable but also socially equitable and fair. The case study in Argentina is an example of how an initiative that started addressing social challenges was also an opportunity to address environmental problems.

Lessons learned

- ▶ Socio-economic and environmental factors intertwine to create and perpetuate urban vulnerability;
- ▶ Poverty, insecure land tenure, and lack of access to essential infrastructure disproportionately affect marginalized communities;
- ▶ Environmental degradation does not impact all urban residents equally. Vulnerable communities often bear the heaviest burdens, from exposure to pollution to the absence of green spaces;
- ▶ Integrating environmental justice into urban policies and planning can ensure that sustainability efforts do not exacerbate existing inequalities but rather work for fair distribution of environmental benefits;

- ▶ There is a need for participatory planning, where affected communities have a voice in shaping the urban environments in which they live.
- ▶ The integration of a justice-based approach in SULR efforts can transform degraded areas into safe, accessible, and inclusive urban spaces.

Student questions and exercises

- ▶ Can you identify vulnerable areas and populations in your city?
- ▶ Is spatial segregation/fragmentation recognizable in your city? Where? What are the causes?
- ▶ Is land degradation currently affecting the urban area where you live? How?
- ▶ If you would like to address this degradation, where would you start? Would you begin by addressing social or environmental concerns?

See Case Study in Section 6.3 Buenos Aires - sustainable urban land restoration opportunities in informal settlements, Argentina

References/resources

Birkmann, J., Cardona, O.D., Carreño, M.L. et al. (2013.) Framing vulnerability, risk and societal responses: the MOVE framework. *Nat Hazards* 67, 193–211. <https://doi.org/10.1007/s11069-013-0558-5>

Dodman, D., Hayward, B., Pelling, M., Castan Broto, V., Chow, W., Chu, E., ... and G. Ziervogel. (2022). *Cities, Settlements and Key Infrastructure*. In H.-O. Pörtner et al. (Eds.), *Climate Change 2022: Impacts, Adaptation and Vulnerability* (pp. 907–1040). Cambridge University Press. <https://doi.org/10.1017/9781009325844.008>

Haderer, M. (2020). Revisiting the Right to the City, Rethinking Urban Environmentalism: From Lifeworld Environmentalism to Planetary Environmentalism. *Social Sciences*, 9(2), 15. <https://doi.org/10.3390/socsci9020015>

Markowitz, G. and R. D. Bullard (eds) (2007). The Quest for Environmental Justice: Human Rights and the Politics of Pollution. *Hum Ecol* 35, 257–258 <https://doi.org/10.1007/s10745-006-9093-2>

Sandholz, S., Mirwald, M., Abbas, S. M. J., and N. Cleetus. (2024). *Barriers to action: Gaps in financing for a just transition, social protection, and loss and damage in cities*. United Nations University Institute for Environment and Human Security (UNU-EHS). https://collections.unu.edu/eserv/UNU:9873/C40_Cities_and_UNU-EHS_2024_Barriers_to_action.pdf

1.5 Urban Land Degradation

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Nick Leimu-Brown (Oxford University)

Johannes Hamhaber (THK)

Marwa Khalifa (ASU)

Karen Sudmeier-Rieux (UNCCD-G20)

Learning objectives

- ▶ Understand urban land as a complex social-ecological system;
- ▶ Identify urban land degradation (ULD) causes and consequences;
- ▶ To validate the need of land restoration efforts in urban settlements.

Summary

Part 1. Introduction

This section focuses on defining land and explaining related concepts. Understanding land as a system and how it has transformed over time in parallel with human activities is crucial. Land degradation has accompanied humans since the widespread adoption of agriculture around 10,000 to 7,500 years ago. This is also why it is difficult to determine a previous state before land

degradation occurred when thinking of restorative efforts, especially in urban land systems.

Part 2. Urban Land

Recognizing urban land as a social-ecological system comprising various interconnected components is crucial. Urban environments present specific challenges. We highlight the dynamics and connections cities have with other areas to meet their demands.

Part 3. Urban Land degradation

In cities, land degradation can occur due to pollution, poor waste management, loss of land's economic and productive value, loss of vegetation cover, deforestation, fires, and landslides, and due to processes such as soil sealing and compaction, erosion and subsidence, and water table disruption. It can be understood as an interconnected spiral of losses, ranging from the value of urban land to the loss of ecosystem services. The urban land degradation-restoration figure is presented—to be related with the next submodule.

Lessons learned

- ▶ Land degradation has a detrimental effect on the quality of life for people;
- ▶ Land degradation is expected to worsen due to factors such as urban expansion, current consumption patterns, unsustainable exploitation of resources, and climate change;
- ▶ The strong interconnection between urbanization, land degradation, and climate change highlights the need for integrated solutions;
- ▶ Addressing land degradation and restoring degraded land is a pressing necessity to safeguard biodiversity, ecosystem services and well-being.

Student questions and exercises

- ▶ How would you define land?

- ▶ How has land been transformed world-wide over time in parallel to human activities?
- ▶ What causes land degradation in urban areas?
- ▶ Is ULD the same in different climate regions and urban settlements?

References/resources

- Elden, S. (2010). Land, terrain, territory. *Progress in Human Geography*, 34(6), 799–817. <https://doi.org/10.1177/0309132510362603>
- Ellis, E. C. (2015). Ecology in an anthropogenic biosphere. *Ecological Monographs*, 85(3), 287–331. <https://doi.org/10.1890/14-2274.1>
- IPCC. (2022). *Land degradation: Chapter 4. In Intergovernmental Panel on Climate Change IPCC (Ed.), Climate Change and Land: IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems* (pp. 345–436). Cambridge University Press. <https://doi.org/10.1017/9781009157988.006>
- Qadeer, M. A. (1981). The Nature of Urban Land. *The American Journal of Economics and Sociology*, 40(2), 165–182. <http://www.jstor.org/stable/3486582>
- Seto, K. C.-Y., and A. Reenberg. (Eds.). (2014). *Strüngmann forum reports. Rethinking global land use in an urban era*. The MIT Press.
- UNCCD. (2024). SDG indicator metadata: Indicator 15.3.1: Proportion of land that is degraded over total land area. <https://unstats.un.org/sdgs/metadata/>
- United Nations Statistics Division (UNSD), a division of the Department of Economic and Social Affairs (DESA). (2024) The Sustainable Development Goals Report. SDG11 Sustainable cities and communities— [SDG Indicators](#)

1.6 Sustainable cities – principles & frameworks

Developed by

Lucas Turmena (UNU-EHS)

Leticia Domingos Vellozo (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Learning objectives

- ▶ Recognize the origins of sustainable development;
- ▶ Identify the premises of sustainable urban planning;
- ▶ Recognize international frameworks related to Sustainable Urban Development (SUD).

Summary

Part 1. Sustainable urban development

SUD derives from the definition of sustainability and refers to the practices and approaches to plan, build and manage cities, considering the balance of economic, environmental and social needs. Sustainable urban development is oriented to the needs of present and future generations, without compromising the health of ecosystems and promoting social equity.

Part 2. Sustainable Cities

What does it mean to plan a sustainable city? According to UN-Habitat, five qualifiers have been identified to promote Sustainable Urban Planning: Compact, inclusive, resilient, integrated and connected.

Part 3. International Frameworks

Global goals and international agendas framing and guiding sustainable urban development. Their implementation timeline, focus and key points.

Lessons learned

- ▶ Sustainability is often referred to as having multiple dimensions, organized in three pillars: social equity, economic growth and environmental protection;
- ▶ Sustainable Urban Development refers to practices and approaches to planning, building and managing cities, considering the balance of economic, environmental and social needs;
- ▶ International frameworks provide a shared roadmap for sustainable urban development, helping cities align local policies with global goals.

Student questions and exercises

- ▶ In what ways do past and present power structures influence global environmental policies?
- ▶ How can participation increase transparency in planning processes?
- ▶ Can you identify specific goals, targets or principles from each framework that align with SULR efforts?

References/resources

Birgisdóttir, H., Bjørn, A., Branny, A., Clausen, C., Fanning, A., Fetzer, I., Francart, N., Grcheva, L., Heide, M., Lassen, E., Leonardsen, J., Moberg, F., Nørgaard, R., Pham, D., Rasmussen, K.K., Raworth, K., Ryberg, M. and E. Sahan. (2023). *Doughnut for Urban Development: A Manual*. Edited by Dani-Hill-Hansen and Kasper Guldager Jensen. Copenhagen. The Danish Architectural Press. <https://doughnuteconomics.org/tools/doughnut-for-urban-development-manual-and-tools>

Escobar, A. (1994). *Encountering development: The making and unmaking of the third world*. Princeton University Press.

Leach, M., Reyers, B., Bai, X., Brondizio, E. S., Cook, C., Díaz, S., Espindola, G., Scobie, M., Stafford-Smith, M., and S.M. Subramanian. (2018). Equity and sustainability in the Anthropocene: a social–ecological systems perspective on

their intertwined futures. *Global Sustainability*, 1(e13). <https://doi.org/10.1017/sus.2018.12>

Purvis, B., Mao, Y. and D. Robinson. (2019). Three pillars of sustainability: in search of conceptual origins. *Sustain Sci* 14, 681–695 <https://doi.org/10.1007/s11625-018-0627-5>

UN-Habitat. (2018). *International Guidelines on Urban and Territorial Planning Handbook*. <https://unhabitat.org/international-guidelines-on-urban-and-territorial-planning-ig-utp-handbook>

1.7 Urban Typologies

Developed by

Leticia Domingos (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Lucas Turmena (UNU-EHS)

Learning objectives

- Understand city as a complex concept with many definitions, depending on the region and historical time;
- Identify the urban and rural continuum;
- Recognize components of urban growth and its historical and current trends.

Summary

Part 1. Cities

This part covers the foundational concepts of urban settlements, defining what constitutes a city, urban agglomerations, and the various types of cities based on scale, function, and level. It highlights the critical distinctions between urban and rural environments. Urban landscapes are mentioned, consisting of built environments, infrastructure, and natural elements. The urban-rural continuum

reflects the fluid transition between densely built urban areas and sparsely populated rural zones, which are vital for understanding the complexities of contemporary cities.

Part 2. Urbanization Processes

Multifaceted processes of urbanization, including trends that have shaped cities over time. The focus is on population growth through natural increase and migration, as well as the accompanying land use changes. Additionally, this section covers the economic development reflected in sector shifts and spatial expansion, providing a global perspective on the interplay between these factors and urban land use transformation.

Part 3. Types of Cities

This part covers diverse characteristics of cities, emphasizing how size, population, density, and spatial extension interplay with economic functions, climate regions, and ecosystems to shape urban landscapes. Urban environments can be very different across world regions, facing different opportunities and challenges. Cities have different development paths.

Lessons learned

- Cities are diverse, shaped by factors such as size, ecosystems, economic activities, population dynamics, and much more. Each city faces its own set of challenges and opportunities that require tailored solutions specific to local conditions;
- Solutions should be context-specific but can benefit from shared knowledge and practices. Cities with similar characteristics—such as those dealing with water scarcity or rapid urbanization—can exchange best practices to address common issues;
- By learning from one another, keeping in mind different contexts and trajectories, cities can create more sustainable, resilient urban environments that promote equity and address unique challenges in innovative ways.

Student questions and exercises

- ▶ How would you define a city?
- ▶ What can we learn from animal-built habitats to create cities that work with nature instead of against it?
- ▶ What cities are similar to yours? In which way? How do they also differ?

References/resources

Metcalf, T. R. (2013). *Colonial Cities*. Oxford University Press.

OECD (2020). *OECD Regions and Cities at a Glance 2020*, OECD Publishing, Paris, <https://doi.org/10.1787/959d5ba0-en>

Potter, R., Binns, T., Elliott, J., Nel, E., and D. Smith. (2018). *Geographies of Development: An Introduction to Development Studies* (4th ed.). Routledge. <https://doi.org/10.4324/9781315759319>

UN-Habitat (2020). *What is a city?* https://unhabitat.org/sites/default/files/2020/06/city_definition_what_is_a_city.pdf

UN-Habitat. (2024). *World Cities Report 2024*. https://unhabitat.org/sites/default/files/2024/11/wcr2024_-_full_report.pdf

1.8 Urban ecosystem services

Developed by

Andrea Ortiz Vargas (UNU-EHS)

Yvonne Walz (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Nick Leimu-Brown (Oxford University)

Learning objectives

- ▶ Identify the characteristics of urban ecosystems;
- ▶ Relate ecosystem services to urban systems and urban land;
- ▶ Recognize the importance of urban ecosystems for the well-being of the people in urban areas.

Summary

Part 1. Introduction

This section outlines the main functions provided by urban lands and highlights their importance for urban areas.

Part 2. Urban Ecosystems

The idea of urban ecosystems is explained; some examples are given to show how they depend on human management and interactions (social-ecological systems SES), as well as how they interact with technological aspects. Technological concerns regarding the built-up environment are often considered part of the social system.

Part 3. Urban Ecosystem Services

Part 3 introduces the different categories of ecosystem services and explains how cities benefit from ecosystem services provided by areas far away (urban teleconnections). It also establishes a connection between ecosystem services and people's well-being.

Lessons learned

- ▶ Urban ecosystems and urban land provide functions and ecosystem services fundamental for the well-being of people;
- ▶ Cities also have important ecosystems; thus, we should conserve them;
- ▶ Cities can thrive while ensuring the conservation of ecosystems and land.

Student questions and exercises

- ▶ Which examples of urban ecosystems come to your mind?
- ▶ Which sort of interactions among the social and ecological systems do you think can happen when we have urban crops?
- ▶ What benefits could we have if we consider a socio-technological subsystem as fundamental in cities?
- ▶ Can you give examples of the benefits that, e.g., wetlands provide to urban populations?
- ▶ What about ecosystems you identify in your city?

Exercise: Here is a green park in a city

- ▶ Which kind of ES can you identify?
- ▶ How do you think these ES influence the life of the people in cities? - show image
- ▶ Change image - And now....
 - o Which kind of ES can you identify?
 - o Which challenges people in the city might face with a decrease of ES supply?
- ▶ Change image And now....
 - o Which kind of ES can you identify?
 - o Which challenges people in the city might face with a decrease of ES supply?

References/resources

Alberti, M. (2005). *The effects of urban patterns on ecosystem function. International regional science review*, 28(2), 168-192. <https://doi.org/10.1177/0160017605275160>

Liu, J., Yang, W., and S. Li. (2016). Framing ecosystem services in the telecoupled Anthropocene. *Frontiers in Ecology and the Environment*, 14(1), 27-36. <https://doi.org/10.1002/16-0188.1>

McPhearson, T., Cook, E. M., Barbés-Blázquez, M., Cheng, C., Grimm, N. B., Andersson, E., ... and T.G. Troxler. (2022). A social-ecological-technological systems framework for urban ecosystem services. *One Earth*, 5(5), 505-518.

MEA: Lee, M., and S. Diop. (2009). *Millennium ecosystem assessment. An Assessment of Assessments: Findings of the Group of Experts Pursuant to UNGA Resolution 60/30*, 1, 361. <https://www.millenniumassessment.org/en/index.html>

TEEB: Perrings, C., Baumgärtner, S., Brock, W. A., Chopra, K., Conte, M., Costello, C., ... and A. Xepapadeas. (2009). *The economics of biodiversity and ecosystem services. Biodiversity, ecosystem functioning, and human wellbeing: an ecological and economic perspective*. Oxford University Press, Oxford, 230-247.

Ziter, C. (2016). The biodiversity–ecosystem service relationship in urban areas: a quantitative review. *Oikos*, 125(6), 761-768. <https://doi.org/10.1111/oik.02883>

1.9 Sustainable Urban Land Restoration

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Leticia Domingos Vellozo (UNU-EHS)

Nick Leimu-Brown (Oxford University)

Johannes Hamhaber (THK)

Marwa Khalifa (ASU)

Karen Sudmeier-Rieux (UNCCD-G20 GLI)

Learning objectives

- ▶ Understand the principles and objectives of the Land Degradation Neutrality Framework (LDN) and the Mitigation and Conservation Hierarchy (MCH);
- ▶ Link both frameworks principles for sustainable urban land restoration (SULR) conceptualization.

Summary

Part 1. Land Degradation Neutrality (LDN)

This section explains LDN as a goal set by the UNCCD to ensure no net loss of healthy land resources by 2030. It details the principles of avoiding, reducing, and reversing land degradation and introduces the neutrality principle as a framework for balancing ecosystem losses and gains.

Part 2. Mitigation-Conservation Hierarchy (MCH)

MCH can be used as a decision-making framework for stakeholders, emphasizing the need for transformative changes to address biodiversity loss. It outlines four key steps for mitigating impacts on ecosystems and restoring ecological integrity.

Part 3. Sustainable Urban Land Restoration (SULR)

SULR is understood as a process that aims to restore urban land systems by enhancing the functionality, quality, and value of its components in a gradient manner. The International Union for Conservation of Nature (IUCN) Restoration Barometer, which examines terrestrial ecosystems and human interactions, lists specific actions to restore urban ecosystems. Urban zones can determine the placement of these actions.

Lessons learned

- ▶ LDN and MCH are frameworks that guide efforts to tackle land degradation and biodiversity loss. Their principles can be adopted for sustainable urban land restoration action;

- ▶ SULR aims to revive social-ecological integrity and maximize the benefits nature provides to people;
- ▶ Restored urban land systems retain their intrinsic cultural, economic, and functional values while continuing to deliver essential benefits through ecological processes within ecosystems;
- ▶ Restoring and rehabilitating degraded urban areas can revive ecosystems and enhance the quality of life for residents;
- ▶ SULR practices contribute to sustainable urban development and meeting social needs within planetary boundaries.

Student questions and exercises

- ▶ How do SULR actions contribute to sustainable urban development?
- ▶ Can you identify efforts going on in your city or neighborhood that could be considered as SULR actions? Explain why

References/resources

Cowie, A. L., Orr, B. J., Castillo Sanchez, V. M., Chasek, P., Crossman, N. D., Erlewein, A. et al. (2018). Land in balance: The scientific conceptual framework for Land Degradation Neutrality. In *Environmental Science & Policy* 79, pp. 25–35. <https://doi.org/10.1016/j.envsci.2017.10.011>

FAO and UNCCD. (2022). *Technical Guide on the Integration of the Voluntary Guidelines on the Responsible Governance of Tenure of Land*. <https://www.unccd.int/resources/publications/technical-guide-integration-voluntary-guidelines-responsible-governance>

Feng, S., Zhao, W., Zhan, T., Yan, Y., and P. Pereira. (2022). Land degradation neutrality: A review of progress and perspectives. *Ecological Indicators*, 144, 109530. <https://doi.org/https://doi.org/10.1016/j.ecolind.2022.109530>

IUCN (2022). Restoration Barometer. A guide for governments. <https://restorationbarometer.org/knowledge-hub/restoration-barometer-a-guide-for-governments/>

Milner-Gulland, E. J., Addison, P., Arlidge, W. N. S., Baker, J., Booth, H., Brooks, T., Bull, J. W., Burgass, M. J., Ekstrom, J., zu Ermgassen, S. O. S. E., Fleming, L. V., Grub, H. M. J., von Hase, A., Hoffmann, M., Hutton, J., Juffe-Bignoli, D., ten Kate, K., Kiesecker, J., Kumpel, N. F., ... and J. E. M. Watson. (2021). Four steps for the Earth: mainstreaming the post-2020 global biodiversity framework. *One Earth*, 4(1), 75–87. <https://doi.org/https://doi.org/10.1016/j.oneear.2020.12.011>

UNCCD. (2023). *Science-Policy Brief: integrated land use planning and integrated landscape management to implementing Land Degradation Neutrality*. <https://www.unccd.int/resources/brief/science-policy-brief-integrated-land-use-planning-and-integrated-landscape>

1.10 Identifying Sustainable Urban Land Restoration Demands (Exercise)

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Learning objectives

- ▶ Identify and analyze urban land degradation in your neighborhood or city;
- ▶ Identify challenges, vulnerable areas and ULD causes and consequences.

Summary

Select a neighborhood that is in need of urban land restoration - analyze and look into the underlying reasons requiring restoration.

Part 1. Define your urban land system

Define your urban land system (could be the neighbourhood or quarter of the city you live in or your whole city).

Part 2. Identify urban land degradation where you live

Identify ULD according to different causes. Mention related consequences. How are those causes affecting your neighborhood? Where are the most visible effects? What urban spaces can reflect ULD? Examples of urban spaces that can reflect ULD include streets, sidewalks, other infrastructure, public spaces, and buildings. Who is the most affected? Relate the causes and consequences to a driver that may have triggered them. What could have triggered the ULD you had identified? Could it have been a land use change, land take, neighborhood expansion, or something more specific to your neighborhood?

Part 3. Create a collage

Create a collage or drawing with pictures of your city or neighborhood, linking ULD drivers, causes, and consequences.

Lessons learned

- ▶ Urban land systems are dynamic systems influenced by land use, infrastructure, social interactions, and environmental factors, impacting inhabitants well-being;
- ▶ Urban land degradation effects are recognizable in urban spaces; they can provide proof of the current situation according to different contexts;
- ▶ Recognizing interconnections of ULD causes, consequences, and drivers is crucial for identifying areas in need of urban land restoration.

Teaching notes

- ▶ The lecturer introduces the exercise, explaining the importance of understanding urban land degradation and restoration;
- ▶ Students select a neighborhood or city, preferably one where they live or are assigned one with which the lecturer is familiar and can provide detailed information on.

Module 2. Approaches to Sustainable Urban Land Restoration

Description

The module introduces cities as systems of systems, where land use, infrastructure, governance and social dynamics are deeply interconnected. Understanding urban environments through systems thinking enables a holistic approach to SULR by identifying key entry points for interventions and integration with existing urban development actions. The module also explores global frameworks – such as the Rio Conventions, the SDGs, the New Urban Agenda, the Sendai Framework for Disaster Risk Reduction (SFDRR), and the Paris Agreement – to highlight opportunities for mainstreaming land restoration efforts into broader sustainability goals on international and (sub-)national levels.

Students will learn how increasingly popular NbS can serve as an approach to urban regeneration. A focus on land degradation neutrality, sustainable land management, and land-use planning equips students with an overview to avoid, reduce, and reverse urban land degradation. The module also examines diverse measures, namely NbS, engineering, and remediation techniques, to transform degraded urban spaces into multifunctional areas, by enhancing blue-green spaces, repurposing urban brownfields, and improving environmental and social resilience.

Core competencies

Academic knowledge:

- ▶ Urban areas as systems of systems, applying a system thinking approach;
- ▶ Potential SULR entry points on different scales and links to prominent urban narratives;
- ▶ Existing international frameworks, their key goals, involved actors, and entry points for SULR;
- ▶ Nature-based Solutions as an approach sharing many goals with land restoration and that is applicable in urban areas;
- ▶ Conceptualization of SULR interventions at two levels: 1) strategic and 2) local level.

Skills:

- ▶ Critically analyze restoration approaches, strategies and measures that integrate social and ecological dimensions, and their contribution to the sustainable transformation of urban environments;
- ▶ Identify cross-sectoral and interdisciplinary solutions that align with SULR principles and practical urban planning strategies;
- ▶ Develop an understanding of urban NbS, diverse NbS measures and how to select the best-suited approach for a given context;
- ▶ Identify and apply design principles for SULR interventions;
- ▶ Understand benefits of long-term strategizing and develop a theory of change in the field of SULR.

Attitudes:

- ▶ Complexity awareness: Recognizing the interconnectedness of urban systems, impacts of degradation, land use planning and management and possibilities for restorative urban spaces;
- ▶ Creativity: Developing context-specific and adaptive SULR interventions.

Authors

Module coordination: UNU-EHS

Developers:

(UNU-EHS)

Alejandra Ramos Gálvez (MSc)
Dr. Simone Sandholz
Leticia Domingos Vellozo (MSc)

Nivya Pazhayakatt Cleetus (M.Plan)

Dr. Yvonne Walz

Lisa Hartmann (MSc)

Nathan Fridtjof Schultze-Florey

(ASU)

Dr. Aya Elsisy

Prof. Dr. Marwa Khalifa

(THK)

Prof. Dr. Johannes Hamhaber

(CU)

Arch. Fabiana Torres Espinoza

Cristina Peralta Peñaloza (MSc)

Dr. Mónica González Llanos

Dr. Ximena Salazar Guamán

(UNCCD-G20 GLI)

Dr. Karen Sudmeier-Rieux

(IFLA)

Dr. Alessandro Martinelli

(SALAR International)

Helena Ohlsson (MSc)

2.1 City systems

Developed by

Johannes Hamhaber (TH Köln, Germany)

Alejandra Ramos (UNU-EHS)

Fabiana Torres Espinoza (UC—Territorium Research Group, Ecuador)

Learning objectives

- ▶ Conceptualize urban areas as systems of systems;
- ▶ Understand how urban systems are interconnected and relate to national and global challenges;
- ▶ Reflect around the importance of incorporating a systems thinking approach for SULR.

Summary

Part 1. Systems Thinking

A systems thinking approach is crucial for achieving SULR in urban areas.. This approach helps understand elements, events, and processes in relation to broader systems. Instead of causing less damage, we should design and build with nature, considering how the built-up environment and urban dynamics affect other ecosystems, ourselves and the planet. Systems thinking embraces this awareness and promotes a more sustainable approach to urban development.

Part 2. The city as a machine

Understanding the city as a machine and considering the elements that allow it to function and be efficient is crucial. Cities can differentiate between the following main sectors: water, energy, waste, food, health, land, and some of their associated attributes. This is also referred to as urban metabolism.

Part 3. The city as an ecosystem

The city is an open system, with subsystems and depends on resources from other regions. It has geographical, administrative and ecological dynamics beyond its scale. The city is embedded in a region and an ecosystem.

Part 4. The city as a community

This section introduces how cities are organized, who the actors are, and how these complex relationships affect how we view and manage city resources, sectors and infrastructure according to different jurisdictions and governance levels. Urban governance is the exercise of authority to manage common affairs. It is comprised of mechanisms, processes and instruments and involves actors and institutions from the private, public (state), as well as civil sectors.

Lessons learned

- ▶ Systems thinking can be considered as a tool for managing resources in cities and among these resources is urban land and its use;
- ▶ City systems approaches determine the qualities that are of primary concern and can guide management processes. These approaches can be sector-based, scale-based or governance-based;
- ▶ Understanding cities as systems of systems allows a more comprehensive view of urban environments, recognizing the interconnectedness among its multiple components, its structures and impacts;
- ▶ Systems thinking is necessary for implementing restorative and regenerative urban land practices.

Student questions and exercises

- ▶ How would you define an urban system? Where would you put the boundaries?
- ▶ Do you know which ecosystems are nearby where you live?

- What advantages do you see for SULR actions in understanding cities as systems of systems?

References

Bai, X., Surveyer, A., Elmqvist, T., Gatzweiler, F. W., Güneralp, B., Parnell, S., Prieur-Richard, A. H., Shrivastava, P., Siri, J. G., Stafford-Smith, M., Toussaint, J. P., and R. Webb. (2016). *Defining and advancing a systems approach for sustainable cities*. In *Current Opinion in Environmental Sustainability* (Vol. 23, pp. 69–78). Elsevier B.V. <https://doi.org/10.1016/j.cosust.2016.11.010>

Chapin, F. S., Kofinas, G. P., and C. Folke. (2009). *Principles of ecosystem stewardship: Resilience-based natural resource management in a changing world*. In *Principles of Ecosystem Stewardship: Resilience-Based Natural Resource Management in a Changing World*. Springer New York. <https://doi.org/10.1007/978-0-387-73033-2>

GIZ (2015). *Unpacking Metropolitan Governance for Sustainable Development Discussion Paper*. <https://unhabitat.org/unpacking-metropolitan-governance-for-sustainable-development>. German Federal Ministry for Economic Cooperation and Development (BMZ), Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) and United Nations Human Settlements Programme (UN-Habitat)

Holling, C. (2001). *Understanding the Complexity of Economic, Ecological, and Social Systems*. *Ecosystems* 4, 390–405. <https://doi.org/10.1007/s10021-001-0101-5>

Jackson, M. (2003). *Systems thinking : creative holism for managers*. <https://archive.org/details/systemsthinkingc0000jack/page/n5/mode/2up>

Reed, B. (2007). *Shifting from ‘sustainability’ to regeneration*. *Building, Research and Information*, 35(6), 674–680. <https://doi.org/10.1080/09613210701475753>

2.2 Planning rationale

Developed by

Mónica González Llanos (UC, Territorium Research Group, Ecuador)

Ximena Salazar Guamán (UC, Territorium Research Group, Ecuador)

Cristina Peralta Peñaloza (UC, Territorium Research Group, Ecuador)

Fabiana Torres Espinoza (UC, Territorium Research Group, Ecuador)

Learning objectives

- Analyze the impacts of urbanization processes at various scales and their contribution to ULD, recognizing the necessity for strategic planning interventions;
- Evaluate the limitations of conventional urban planning approaches and develop comprehensive regulatory frameworks that address demographic and climate change challenges in urban land restoration;
- Identify sustainable urban planning methodologies and territorial management instruments to create resilient urban ecosystems that effectively respond to environmental challenges.

Summary

Part 1. Urban land degradation and the need for planning

This section establishes a framework for understanding urbanization processes across local to global scales, categorizing them into typologies. Through reflective exercises, it examines the cumulative terrestrial impacts of these processes and their sustainability implications. This foundational approach underscores the necessity for strategic planning interventions to address urban land degradation before exploring more specific methodologies in subsequent sections.

Part 2. Scope and challenges of urban planning facing urban land degradation

This section explores the critical challenges of urban planning in addressing land degradation through a comprehensive lens. It begins by examining conventional planning approaches and their limitations, then delves into the pivotal role of urban planning in responding to evolving demographic and climate dynamics. It emphasizes the necessity of developing robust regulatory frameworks that anticipate future scenarios, constructing forward-looking strategic visions, and establishing clear regulations for urban land restoration. A key focus is aligning planning tools and methodologies directly with the complex challenges of urban land degradation, recognizing that traditional planning methods are insufficient to address the multifaceted environmental and social transformations occurring in urban landscapes.

Part 3. Sustainable Urban Planning

This section focuses on sustainable urban planning as a transformative approach to territorial development. It explores various approaches to territorial planning that move beyond traditional methodologies, emphasizing holistic and integrated strategies for urban sustainability. The section outlines the systematic process of developing sustainable urban planning, which involves comprehensive analysis, strategic thinking, and innovative problem-solving. Central to this part is the examination of land management instruments that promote sustainability, providing practical tools and frameworks for urban planners to effectively address environmental challenges, optimize resource use, and create resilient urban ecosystems that can adapt to changing social and ecological conditions.

Lessons learned

- Urbanization processes occur across multiple scales with distinct typologies, creating cumulative impacts on land that require integrated planning approaches to mitigate degradation;

- Effective urban land restoration demands forward-looking regulatory frameworks that anticipate demographic and climate changes, moving beyond traditional planning methods to address environmental and social transformations;
- Sustainable urban planning requires holistic territorial management instruments that balance comprehensive analysis with innovative problem-solving to create adaptable urban ecosystems.

Student questions and exercises

Pedagogical Approach

- Interactive exercises (e.g., mapping urban changes)
- Conceptual models (5-Cities Model)
- Practical implementation strategies
- Participatory planning techniques

Recommended Teaching Strategies

- Encourage student reflection on local urban experiences
- Use comparative analysis between current and sustainable city models
- Promote critical thinking about urban transformation
- Integrate community participation concepts

Exercise Design: Planning for Sustainable Urban Land Restoration

- Personalizes urban planning challenges
- Encourages creative problem-solving
- Develops systems thinking
- Connects theoretical concepts to practical solutions

References/resources

Amado, M. P. (2005). The Operative Process in Sustainable Urban Planning. In O. Kungolos (Ed.), *Wit Transactions on Ecology and the Environment: Sustainable Development and Planning II* (Vol. I, pp. 181-191). WIT Press, London.

Catanese, A. J., and J.C. Snyder. (1988) *Urban Planning* (2nd ed.) McGraw-Hill, Inc.

Haines, A. (2001). Using Visioning in a Comprehensive Planning Process. https://www3.uwsp.edu/cnr-ap/clue/documents/publicprocesses/using_visioning_in_comprehensive_planning_process.pdf

Herrick, J. E., Duniway, M. C., Pyke, D. A., Bestelmeyer, B. T., Wills, S. A., Brown, J. R., Karl, J. W., and K. M. Havstad. (2012). A holistic strategy for adaptive land management. *Journal of Soil and Water Conservation*, 67(4), 105A-113A. <https://doi.org/10.2489/jswc.67.4.105a>

Kakulu, I. I. and C. Atakora. (2024). Land Use and Spatial Planning Tools to address land degradation and Promote Land Restoration [Conference presentation]. FIG Working Week, Accra, Ghana, May 19-24, 2024

Seto, K. C. & Dhakal, S. (Coord). (2014). Human settlements, infrastructure, and spatial planning. In O. Edenhofer, R. Pichs-Madruga, Y. Sokona, J. C. Minx, E. Farahani, S. Kadner, K. Seyboth, A. Adler, I. Baum, S. Brunner, P. Eickemeier, B. Kriemann, J. Savolainen, S. Schlömer, C. Von Stechow and T. Zwickel (Eds.) *Climate Change 2014 Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, United Kingdom and New York.

2.3 Frameworks, approaches, narratives - entry points for Sustainable Urban Land Restoration

Developed by

Simone Sandholz (UNU-EHS)

Leticia Domingos Vellozo (UNU-EHS)

Lisa Hartmann (UNU-EHS)

Yvonne Walz (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Nathan Fridtjof Schultze-Florey (UNU-EHS)

Learning objectives

- ▶ Be able to identify potential entry points for SULR at different scales;
- ▶ Understand key global frameworks, and how their goals could be aligned with sustainable urban development and land restoration;
- ▶ Be familiar with nature-based solution as an approach sharing many goals with land restoration, benefiting the implementation of many global frameworks;
- ▶ Get an overview on prominent urban narratives and their links to SULR.

Summary

Part 1. Key global frameworks and their links to SULR

After reflecting on key global and urban challenges, Part 1 defines entry points and how the concept can help to identify opportunities to mainstream SULR in existing actions. It then provides an overview of key global frameworks, their goals and actors, namely the Rio Conventions, the SDGs, the New Urban Agenda, the SFDRR, and the Paris Agreement. Consequently, key entry points for SULR in the different frameworks are described, followed by highlighting the benefits of coherent implementation by making use of overarching approaches such as NbS.

Part 2. Nature-based solutions as integrative approach

Part 2 starts from introducing NbS as a recognized concept with multiple benefits, implemented in many sectors. Starting from fleshing out protective functions, benefits of NbS in different ecosystems are described. The chapter ends by describing different strategic entry points for NbS in national, sectoral and local level policy-making, as well as highlighting NbS as strategic entry point for restoration, including in urban areas.

Part 3. Entry points for SULR in prominent urban narratives

Part 3 introduces key urban narratives, namely ecological urbanism, smart growth, new urbanism, transit-oriented development, compact city and sponge city. For each concept an implementation case is presented, fleshing out the particular benefits. The chapter ends on reflections on the particular potentials to introduce SULR in either approach, followed by a summary of the sub-module summarizing SULR entry points on different scales and in different actions.

Lessons learned

- ▶ There are multiple entry points to SULR in international frameworks, even if their core foci might not be on urban development or land restoration, but on climate action, risk reduction, biodiversity or others, which however share some common goals;

- ▶ NbS as cross-sectoral approach can facilitate collaboration between different stakeholders and support land restoration, under the Rio Conventions as well as other national and international frameworks and agreements;
- ▶ Prominent urban narratives follow dedicated sustainability goals, which can be good – yet different – entry points for SULR.

Student questions and exercises

The submodule focuses on already existing actions in the form of international frameworks, approaches, and urban narratives and their potential to support or include SULR. Hence, throughout the presentation the lecturer could trigger discussions by asking:

- ▶ How do you rate the overlap of this framework or policy with SULR?
- ▶ Do you have ideas how to strategically use these synergies to promote SULR?
- ▶ How relevant do you think these frameworks are in your country or city?

The module includes a student exercise in Part 3 where students are asked:

- ▶ What do those planning approaches have in common? What is the most striking difference between them?
- ▶ Can you identify other examples of projects that correlate to the principles of the planning approaches we have discussed?

References/resources

Keith, M., Birch, E., Buchoud, N.J.A., Cardama, M., Cobbett, W., Cohen, M., Elmqvist, T., Espey, J., HAjer, M., Hartmann, G., Matsumoto, T., Parnell, S., Revi, A., Roberts, D.C., Saiz, E., Schwanen, T., Seto K.C., Tuts, R., and M. van der Pütten. (2023). A new urban narrative for sustainable development. *Nat Sustain* 6, 115–117 <https://doi.org/10.1038/s41893-022-00979-5>

Salama, A. M., Patil, M. P., Elsemellawy, A. N., Abudib, H. H., Almansor, N. A., MacLean, L., and K. van Riel. (2024). People–Place Narratives as Knowledge Typologies for Social Sustainability: Cases from Urban Contexts in the Global South. *Buildings*, 14(4), 1001. <https://doi.org/10.3390/buildings14041001>

Sandholz, S., Wannewitz, M., Moure, M., and M. Garschagen. (2020). *Costs and benefits of (in)coherence: Disaster Risk Reduction in the Post-2015-Agendas*. United Nations University- Institute for Environment and Human Security. <https://collections.unu.edu/view/UNU:7638>

Song, Y., Kirkwood, N., Maksimović, Č., Zheng, X., O'Connor, D., Jin, Y., and D. Hou. (2019). Nature based solutions for contaminated land remediation and brownfield redevelopment in cities: A review. *Science of the Total Environment* 663:568-579, <https://doi.org/10.1016/j.scitotenv.2019.01.347>

Walz, Y., Nick, F., Higuera Roa, O., Nehren, U. and Z. Sebesvari. (2021). *Coherence and Alignment among Sustainable Land Management, Ecosystem-based Adaptation, Ecosystem-based Disaster Risk Reduction and Nature-based Solutions*. United Nations University - Institute for Environment and Human Security. <http://dx.doi.org/10.53324/MWGP9896>

2.4 Operationalizing Land Degradation Neutrality through Sustainable Land Management and Integrated Land Use Planning

Developed by

Leticia Domingos Vellozo (UNU-EHS)

Lisa Hartmann (UNU-EHS)

Learning objectives

- Understand the principles of SLM and its role in achieving
- LDN, particularly in urban contexts;
- Identify key tools and methodologies of Integrated Land Use Planning (ILUP) that support sustainable urban land management and conflict resolution;

- Analyze how ILUP, SLM, and LDN interact to balance urban development, ecological sustainability, and land restoration through case studies and global frameworks.

Summary

Part 1. Operationalizing LDN

LDN aims for no net loss of productive land, ensuring that unavoidable degradation is compensated with restoration. While originally applied at national and rural levels, LDN is critical for cities facing rapid expansion, soil sealing, and ecosystem degradation. The Neutrality Principle requires balancing losses and gains through avoidance, reduction, and restoration measures to maintain urban land productivity. Over 100 countries have set LDN targets under the UNCCD framework, integrating land sustainability into national policies.

Part 2. Sustainable Land Management (SLM)

SLM involves land-use practices that balance human needs with long-term ecosystem sustainability. Traditionally applied to rural areas, SLM principles—such as land conservation, green infrastructure, and climate adaptation—are now essential for sustainable cities. SLM Strategies include afforestation, soil erosion control, sustainable forest management, urban green infrastructure, and integrated water systems. SLM has a role in LDN, which is to provide practical solutions to achieve neutrality, ensuring land remains productive while cities expand. Landscape architecture, ecological urbanism, and community-based land management can help operationalize SLM in cities.

Part 3. Integrated Land Use Planning (ILUP)

ILUP is a decision-making tool that ensures land use is allocated efficiently, balancing development with environmental and social sustainability. It identifies the best land-use combinations, resolves conflicts, enhances ecological connectivity, and operationalizes both LDN and SLM, including GIS-based land use mapping, environmental impact assessments, participatory land-use planning, and tenure security frameworks.

Lessons learned

- ▶ LDN provides a measurable framework for sustainable urban land use, but achieving it requires proactive planning regarding trade-offs;
- ▶ SLM strategies can be effectively adapted from rural to urban contexts, supporting resilience and land restoration;
- ▶ ILUP is essential for integrating LDN and SLM into urban decision-making, ensuring that land resources are used efficiently and equitably.

Student questions and exercises

- ▶ Which land commitments does your country have?
- ▶ SLM has traditionally been applied in rural and agricultural contexts. What challenges arise when adapting these practices to urban environments?
- ▶ Which SLM strategies (e.g., afforestation, soil conservation, green infrastructure) are most effective in cities, and why?
- ▶ What are the most critical tools in ILUP for securing land tenure and resolving conflicts among stakeholders?

See Case Study, Section 5.5 Cuenca: Shinshin creek renovation, Ecuador

References/resources

Hartmann, L. Hansohm, J., Vellozo, I., Walinder, E., Orr, B.J., Andreeva, O. and Y. Walz. (2024). *The contribution of land and water management approaches to Sustainable Land Management and achieving Land Degradation Neutrality*. Bonn: United Nations University - Institute for Environment and Human Security. <https://collections.unu.edu/view/UNU:9640>

M.J. Sanz, J. de Vente, J.-L. Chotte, M. Bernoux, G. Kust, I. Ruiz, M. Almagro, J.-A. Alloza, R. Vallejo, V. Castillo, A. Hebel, and M. Akhtar-Schuste. (2017).

Sustainable Land Management contribution to successful land-based climate change adaptation and mitigation: A Report of the Science-Policy Interface. Bonn: United Nations Convention to Combat Desertification. <https://www.unccd.int/resources/reports/sustainable-land-management-contribution-successful-land-based-climate-change>

Orr, B.J., A.L. Cowie, V.M. Castillo Sanchez, P. Chasek, N.D. Crossman, A. Erlewein, G. Louwagie, M. Maron, G.I. Metternicht, S. Minelli, A.E. Tengberg, S. Walter, and S. Welton. (2017). *Scientific Conceptual Framework for Land Degradation Neutrality. A Report of the Science-Policy Interface*. United Nations Convention to Combat Desertification (UNCCD), Bonn, Germany. https://www.unccd.int/sites/default/files/documents/2017-08/LDN_CF_report_web-english.pdf

P.H. Verburg, G. Metternicht, E. Aynekulu, X. Deng, S. Herrmann, K. Schulze, F. Akingemi, N. Barger, V. Boerger, F. Dosdogru, H. Gichenje, M. Kapović-Solomon, Z. Karim, R. Lal, A. Luise, B.S. Masuku, E. Nairesiae, N. Oettlé, A. Pilon, O. Raja, N.H. Ravindranath, R. Ristić and G. von Maltitz. (2022). *The Contribution of Integrated Land Use Planning and Integrated Landscape Management to Implementing Land Degradation Neutrality: Entry Points and Support Tools. A Report of the Science-Policy Interface*. UNCCD, Bonn, Germany. <https://www.unccd.int/resources/reports/contribution-integrated-land-use-planning-and-integrated-landscape-management>

UNCCD. (2022). *Global Land Outlook 2nd edition*. UNCCD. <https://www.unccd.int/resources/global-land-outlook/glo2>

Walz, Y., Nick, F., Higuera, Roa, O., Nehren, U. and Z. Sebesvari. (2021). *Coherence and Alignment among Sustainable Land Management, Ecosystem-based Adaptation, Ecosystem-based Disaster Risk Reduction and Nature-based Solutions*. Bonn: United Nations University - Institute for Environment and Human Security. <http://collections.unu.edu/view/UNU:8495>

2.5 Thematic approaches to Sustainable Urban Land Restoration—blue-green-brown urban spaces

Developed by

Nivya Pazhayakatt Cleetus (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Learning objectives

- ▶ Identify the applications of urban greening, water-sensitive design and brownfield restoration for enhancing ecosystem services and urban spaces;
- ▶ Understand the main benefits of related projects for urban areas;
- ▶ Analyze the interconnected roles of restoring and improving blue, green, and brown urban spaces in promoting urban livability by integrating different techniques.

Summary

Part 1. Introduction

Different measures can address degradation, ranging from policy integration, zone and climate coding, NbS, engineering, remediation, and specific techniques. Key priorities include creating and enhancing blue and green spaces and remediating and repurposing urban brownfields. These blue-green-brown urban space classifications can help picture how mixed uses connect environmental concerns, activities, and needs in cities. Enhanced blue-green-brown spaces boost environmental health, economic opportunities, and aesthetic qualities, while also contributing to greater mental health and a higher quality of life in urban areas.

Part 2. Urban greening

Urban greening refers to the process of introducing and enhancing green spaces, which are characterized by the presence of vegetation. These spaces improve environmental quality, biodiversity, and residents' well-being and among their main benefits are reducing urban heat island effects by providing shade and cooling and enhancing biodiversity by creating habitats for wildlife. Green spaces include green infrastructure (as a network), urban forests, green belts, green corridors, urban farms, and more. Urban green typologies may vary.

Part 3. Water sensitive design

Urban water sensitive design incorporates sustainable water management strategies to improve water quality, reduce flooding, promote water reuse, and enhance ecological resilience. Furthermore, the focus is on creating or enhancing blue urban spaces, which encompass blue infrastructure (as a network), drainage systems, urban wetlands, urban rivers, canals, and water bodies. The case study—Ecological and Social Recovery of Nairobi River—Michuki Park, Kenya—is presented to illustrate the relevance of restoring blue-green spaces in urban areas.

Part 4. Brownfield restoration

Brownfield restoration refers to the process of rehabilitating and repurposing contaminated or underutilized industrial and commercial sites that have been affected by hazardous substances, pollutants, or previous industrial activities. These sites are often abandoned or neglected due to environmental concerns and legal liabilities. Techniques such as soil myco-restoration and fungal bioremediation technology can achieve this. Urban brown spaces include landfills, contaminated plots, abandoned plots, and derelict land.

Lessons learned

- ▶ Improving blue-green-brown urban spaces helps us enhance ecosystem functions and integrate nature into city layouts. They provide multiple benefits to urban life (ecosystem services;

- ▶ Spatially just urban areas require systemic integration of greening, water-sensitive design, and brownfields restoration in planning. The integration will also contribute to urban resilience;
- ▶ Examples worldwide provide valuable insights by incorporating nature into the built-up environment and restoring degraded urban land systems;
- ▶ Initiatives to enhance blue-green-brown urban spaces can align with urban planning strategies that are pursuing sustainable urban development.

Student questions and exercises

- ▶ How are the blue, green, and brown spaces where you live? Are they close to where you live? Could you map them in your neighborhood? What would you improve?
- ▶ Template for exercise: Fill in the boxes according to what you would improve

See Case Study in Section 5.8 Nairobi: Ecological and Social Recovery of Nairobi River- Michuki Park, Kenya

References/resources

Carmona, M. (2010). Contemporary Public Space, Part Two: Classification. *Journal of Urban Design*, 15(2), 157–173. <https://doi.org/10.1080/13574801003638111>

FAO (2025). *Urban and periurban agriculture*. <https://openknowledge.fao.org/server/api/core/bitstreams/0aa558d4-57c7-498d-87f7-b9e37577882f/content/src/html/food-safety-considerations-for-agriculture-within-urban-spaces.html>

FAO (2023). *Urban Forests. A global perspective*. <https://openknowledge.fao.org/items/57a6df3d-0ca7-442c-a19a-fc098d108fae>

Scott, M., Lennon, M., Haase, D., Kazmierczak, A., Clabby, G., and T. Beatley. (2016). *Nature-based solutions for the contemporary city/Re-naturing the city/Reflections on urban landscapes, ecosystems services and nature-based solutions in cities/Multifunctional green infrastructure and climate change*

adaptation: brownfield greening as an adaptation strategy for vulnerable communities?/Delivering green infrastructure through planning: insights from practice in Fingal, Ireland/Planning for biophilic cities: from theory to practice. *Planning Theory & Practice*, 17(2), 267–300. <https://doi.org/10.1080/14649357.2016.1158907>

UNCCD (2022). *Global land outlook. Land restoration for recovery and resilience. Second edition*. <https://www.unccd.int/resources/global-land-outlook/global-land-outlook-2nd-edition>

2.6 Nature-based Solutions in cities

Developed by

Simone Sandholz (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Karen Sudmeier-Rieux (UNCCD-G20 GLI)

Nathan Fridtjof Schultze-Florey (UNU-EHS)

With contributions from IFLA

Learning objectives

- ▶ Know what urban nature-based solutions (NbS) are and what makes urban application of NbS special;
- ▶ Understand diverse NbS measures for urban application and how to select the best-suited NbS for a given context;
- ▶ Understand urban NbS implementation processes and the benefits and challenges of NbS;
- ▶ Raise awareness on the social aspects of NbS and how to overcome challenges for implementing NbS in vulnerable urban settings;
- ▶ Be aware of the links between NbS, land restoration and urban planning.

Summary

Part 1. Overview of urban NbS

Part 1 of the module starts with introducing NbS and how they can be used in different ecosystems, including cities. This is followed by an overview on different NbS measures, from very small-scale to larger scale application, and their potential uses in risk reduction, climate change adaptation, urban open spaces and recreation, and many others. The part concludes by reflecting on how to choose the best-suited NbS for a given task and location.

Part 2. Implementation of urban NbS

Part 2 introduces NbS design principles, following a 10-step approach. Next, a series of slides reflects on the current status of NbS implementation in urban areas, presented in form of six key lessons around availability, linkages to grey infrastructure, economic feasibility, growing popularity, and multiple benefits. Lastly, one lesson learned highlights the potential to draw on a vast history of urban and housing designs from many world regions that draw on NbS even before the concept became popular in modern planning.

Part 3. Social aspects of urban NbS

Part 3 focuses on social aspects, starting from highlighting challenges that come when implementing urban NbS. It highlights the challenges and potentials of NbS implementation in vulnerable contexts such as urban marginal and informal settlements. Finally, it reflects on the need of developing locally accepted and culturally-sensitive approaches to make them long-term sustainable and to unlock transformative potentials of urban NbS.

Part 4. Linking urban planning, Nature-based Solutions and land restoration

The concluding part broadens the scope by comparing and linking urban NbS to urban restoration, as both approaches have many overlaps and benefits for urban inhabitants. The module ends with a summary showing the linkages between the different approaches, how they are guided by overall urban development goals, and what their contribution to overarching goals such as the land degradation neutrality principles are.

Lessons learned

- ▶ There are multiple NbS available for application in urban areas, as single measure, as combination of different NbS, or as hybrid approaches linked to grey infrastructures such as dams or dikes;
- ▶ Well-designed NbS can be economically reasonable and provide multiple co-benefits, for example for climate adaptation, risk reduction, and human well-being;
- ▶ There are overlaps between NbS and land restoration measures, including in urban areas, however they are not yet used systemically in planning;
- ▶ Well-planned urban NbS and land restoration measures contribute to sustainable and resilient cities, and to larger development goals beyond the urban area.

Student questions and exercises

The submodule provides a broad overview on urban NbS, how they are applied and what their multiple benefits are. Students will get an overview on the width of urban NbS by lecturers, with the aim to equip them with the skills to consider NbS in urban planning and to choose appropriate urban NbS measures. Lecturers are supported by case studies and examples backing the lessons learned presented. While parts I and II are designed more as a presentation providing basic learnings around urban NbS, parts III and IV offer the potential for critical reflections.

Part 3 ends on the need to design culturally-sensitive approaches, coming with the opportunity for a group discussion or group work to collect cultural no-go's and/or desirable features of NbS in the city/country the lecture takes place, based on a 5-step approach being presented.

Furthermore, the final slide conceptualizing linkages between urban development, land restoration and NbS allows for a final reflection around:

- ▶ In your perspective, what are key benefits of urban NbS for building sustainable and resilient cities?

- How can the commonalities of urban NbS and land restoration be used strategically?

References/resources

IUCN. (2024). *Nature-based Solutions for cities*, online repository, link: <https://iucn.org/our-work/topic/nature-based-solutions-cities>

Kabisch, N., Frantzeskaki, N. and R. Hansen. Principles for urban nature-based solutions. (2022). *Ambio* 51, 1388–1401 <https://doi.org/10.1007/s13280-021-01685-w>

Kibii, C., Guerra, F., Bananayo, Bonera, P.B. and S. Sandholz. (2025). Nature-Based Solutions for Climate Change Adaptation and Resilience in Urban Informal Settlements: insights from Kibera, Kenya and Villa 20, Argentina. *Nature-Based Solutions* 7:100216. <https://doi.org/10.1016/j.nbsj.2025.100216>

Rink, D., and C. Schmidt. (2021). Afforestation of Urban Brownfields as a Nature-Based Solution. Experiences from a Project in Leipzig (Germany), *Land* 10(9):893. <https://doi.org/10.3390/land10090893>

Song, Y., Kirkwood, N., Maksimović, Č, Zheng, X., O'Connor, D., Jin, Y., and D. Hou. (2019). Nature based solutions for contaminated land remediation and brownfield redevelopment in cities: A review. *Science of The Total Environment* 663:568–579. <https://doi.org/10.1016/j.scitotenv.2019.01.347>

2.7 Design principles

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

With contributions from SALAR International

Learning objectives

- Understand the main principles for developing a SULR plan (strategic level) or project (local level);
- Acknowledge the need for nature-inclusive, regenerative and distributive design.

Summary

Part 1. Introduction

Various urban instruments at different scales can incorporate SULR actions. The presented principles are generic and try to encompass considerations for planning at a strategic level and projects at a local scale.

Part 2. SULR Planning principles

The principles stress the importance of meeting urban needs within the ecological ceiling, making synergies, taking proactive and adaptable actions, and encouraging people from different fields to work together to make sure that urban settings can adapt as needed. Jurisdictional and administrative scales need to be taken into account, as well as the distribution and location of urban functions and ecosystem services provision.

Part 3. SULR Local project principles

Nature-inclusive design is essential for SULR. Projects should avoid harmful practices like unnecessary resource extraction, reduce impacts such as urban sprawl and impermeable surfaces, and restore underutilized spaces by enhancing ecosystem services. Renewal involves prioritizing conservation, fostering healthy human-nature relationships, and building with nature to establish sustainable urban environments. Context-appropriate solutions need to be considered. Projects demonstrating the relevance of multifunctional spaces for socio-ecological systems integrity are shown.

Part 4. Exercise

A template to identify SULR actions is provided. The template can be used to analyze an existing urban space or as a guide to start drafting the main SULR actions and principles for a new intervention.

Lessons learned

- ▶ It is necessary to understand specific urban land systems, use local materials, native plants, and community knowledge for context-appropriate SULR projects and actions
- ▶ Applying the SULR principles requires a holistic understanding of the urban environment and a commitment to sustainable design
- ▶ By aligning project-level actions with broader restoration plans, urban land restoration efforts can balance ecological and social issues
- ▶ Interventions may address immediate restoration goals while contributing to long-term city resilience

Student questions and exercises

- ▶ What is the value of nature in urban life?

Exercise

- ▶ According to the ULD you have identified in Module 1, think about how you would imagine your neighborhood. What would you change (refer to the ULD causes)? How would you change it?
- ▶ Follow the SULR actions and principles to transform the urban spaces that are affected by those causes
- ▶ Choose to focus on one of the following:
 - o Designing a plan or strategy to address the causes from a planning perspective (create strategies and break down to specific actions, or;

- o Designing an urban intervention. Improve one of the affected spaces you had identified. Layout a multifunctional, co-designed, and nature-based proposal – drawing, sketch, collage.

- ▶ You can adapt your collage from Mod 1 exercise according to how you would imagine your neighbourhood, or you can create a completely new one

References/resources

Adams, D., and C. Watkins. (2014). *The value of planning*. Royal Town Planning Institute: London, UK.

Ecoshape (n.d.) *Building with nature*. <https://www.ecoshape.org/en/>

FAO, IUCN/CEM, SER (2021). *Principles for ecosystem restoration to guide the United Nations Decade 2021-2030*. <https://openknowledge.fao.org/handle/20.500.14283/cb6591en>

SymbioCity. *Cities define the future*. <https://symbiocity.org>

UNFCCC (2024). *WORKS with Nature: Low Carbon Adaptation Techniques for a Changing World*. <https://unfccc.int/documents/644016>

van Eekelen, E. and M. Bous. (2020). *Building with Nature. Creating, implementing and upscaling Nature-based solutions*. Nai010publishers.

2.8 Theory of change

Developed by

Simone Sandholz (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Learning objectives

- ▶ Understanding what a theory of change is, how it is being constructed and where it is used;

- ▶ Identify cascading impacts and benefits of urban restoration initiatives and projects;
- ▶ Learn how to develop a theory of change in the field of urban land restoration.

Summary

Part 1. Background

Main concepts of the theory of change process, emphasizing the importance of setting long-term goals that guide urban land restoration efforts. Explores the necessary conditions required to achieve these goals, understanding how different factors must align to ensure successful outcomes.

Part 2. Theory of change application

Benefits and applications in an urban context are presented. Example: Romania's national urban policy theory of change.

Part 3. Exercise

Developing a theory of change around urban land restoration. Reflection and discussion.

Lessons learned

- ▶ A well-structured theory of change provides a roadmap for projects/ initiatives. It identifies long-term goals and how to reach them, ensuring that efforts are targeted and different measures integrated;
- ▶ The theory of change can be a suitable way for including urban land restoration into broader sustainability objectives;
- ▶ Theories of change can be a helpful planning tool that outlines needed steps to achieve goals, connecting actions to outcomes, following a system thinking perspective;
- ▶ A theory of change is essential for monitoring and evaluation, as it comes with concrete and measurable components.

Student questions and exercises

Exercise Develop a Theory of Change around sustainable urban land restoration

Step 1.

Below you find a problem description (left) and the change/impact planned, rooted in a vision of change (right side) – form small groups and note answers to below questions

Step 2.

Reflect together: What did you identify? Are there overlaps, are there or connections between your suggested activities, outputs, impacts?

References/resources

Conservation International. (2013). Constructing theories of change models for Ecosystem-based Adaptation projects: a guidance document. Conservation International. Arlington, VA. [constructing-theories-of-change-for-ecosystem-based-adaptation.pdf](https://www.conservationinternational.org/publications/constructing-theories-of-change-for-ecosystem-based-adaptation.pdf)

Deutsch, L., Belcher, B., Claus, R. and S. Hoffmann. (2021). Leading inter- and transdisciplinary research: Lessons from applying theories of change to a strategic research program. *Environmental Science & Policy* 120:29-41. <https://doi.org/10.1016/j.envsci.2021.02.009>

Weiss, C. (1995). *Nothing as Practical as Good Theory: Exploring Theory-Based Evaluation for Comprehensive Community Initiatives for Children and Families in 'New Approaches to Evaluating Community Initiatives'*. Aspen Institute.

World Bank. (2022). *Romania - Theory of Change : Romanian National Urban Policy (English)*. Romania Urban Policy Washington, D.C. : World Bank Group. [P1711760c54a68040b6e60e51669d97dd9.pdf](https://www.worldbank.org/publications/romanian-national-urban-policy-theory-of-change)

Module 3. Enabling Sustainable Urban Land Restoration

Description

This module focuses on critical thinking about enablers and barriers of SULR and links urban planning to sustainable development and decision-making processes. Students will explore four key phases of a SULR intervention: Assessment, planning strategy, operationalization, and implementation. Inclusive policies, stakeholder collaboration, supporting financial mechanisms, and land governance are described as enablers of successful restoration. Students will examine the multiple co-benefits of SULR, namely social, economic, and ecological benefits.

The module also emphasizes the role of multi-level governance, including the important roles of national governments and citizen participation. Through visioning exercises and back casting (working backward to identify the steps needed to achieve a common vision), students will learn to set clear, actionable goals for urban land restoration, aligning with global visions such as land degradation neutrality. Students will use stocktaking as a tool to create a comprehensive baseline that can help to understand strengths, weaknesses, opportunities, and challenges of SULR efforts within an urban area.

Core competencies

Academic knowledge:

- ▶ Rationale and sequence for planning SULR interventions;
- ▶ Actor typology of SULR and the role of multi-level governance for developing long-term sustainable approaches;
- ▶ Forms and levels of public, private and civil participation formats;
- ▶ Policy instruments to enable SULR, scales, common objectives with other sectors and the need for policy coherence;
- ▶ Scales and forms of financial instruments and relation to entry points for SULR, the relevance of synergies.

Skills:

- ▶ Balance challenges and enablers to SULR adoption and evaluate trade-offs between environmental and social factors;
- ▶ Identify involved actors and resources needed to develop and implement SULR strategies and analyze policy and finance instruments;
- ▶ Apply visioning techniques to set and agree on urban restoration goals;
- ▶ Adapt a theory of change, starting from a degradation state, mapping the context, setting directions and co-designing an intervention;
- ▶ Prioritize and sequence SULR actions.

Attitudes:

- ▶ Future-oriented: Long-term orientation and visioning;
- ▶ Connectedness: Engaging with multiple disciplines and actors and willingness to embrace diversity for better SULR interventions.

Authors

Module coordination: UNU-EHS

Developers:

(UNU-EHS)

Alejandra Ramos Gálvez (MSc)

Dr. Simone Sandholz

Leticia Domingos Vellozo (MSc)

Fabian Rackelmann (MSc)

Sinja Buri (MA)

Sönke Kreft (MSc)

(ASU)

Dr. Aya Elsisy

Prof. Dr. Marwa Khalifa

(THK)

Prof. Dr. Johannes Hamhaber

3.1 Sequence & framing

Developed by

Marwa Khalifa (ASU, Egypt)

Aya Elsisy (ASU, Egypt)

Learning objectives

- ▶ Understand the value of planning and the decision-making rationale related to SULR objectives;
- ▶ Analyse the different stages of SULR Planning and the importance of each stage;
- ▶ Evaluate the enablers/constraints of each stage and the roles of different stakeholders and structures in shaping the planning strategy for ULR.

Summary

Introduction: Framing SULR & The Value of Planning

This section discusses the importance of urban planning, how it is related to understanding the concept of development, who's meant to decide the aspects of planning, its strategies, and processes based on the overall vision and pursued values of SULR.

Part 1. Planning Sequence Rational of SULR

This part explains in detail the 4 different phases of sequencing the activities of SULR: assessment (including situational analysis and defining requirements for SULR), planning strategy (including scenario-building and developing scope), operationalization (including developing instruments form action mobilization and monitory control), and finally implementation (including change management and conflict resolution). Each phase is discussed from the perspective of its aim, inputs, tools, and expected outcomes. This part encourages students to engage in critically investigating targets of SULR and how to spot their impact in the physical built environment.

Part 2. Balancing Formal & Informal Structures with Decision-Support Enablers

This part discusses the dynamics between structured planning and unstructured planning, and the influence of different systems in shaping the planning cycles and effectiveness. It also reflects on the importance of data governance in terms of its accessibility and inclusivity, as well as the role of system thinking models in understanding the impact of decisions on different variables.

Lessons learned

- ▶ Sequencing SULR projects activities and framing their planning processes around objectives of SULR.
- ▶ The required data/information, tools, and activities to develop an SULR project.
- ▶ The influence of different aspects on SULR planning processes like informal structures, data governance approaches, and system thinking mapping

Student questions and exercises

Introduction: Framing SULR & The Value of Planning

Short Discussion: How to convert SULR goals into SMART objectives?

The instructor should encourage students to discuss how to convert SULR goals into SMART objectives (Specific, Measurable, Achievable, Relevant, and Time-Bound). Each student/group of students could take one of the identified goals and state at least two SMART objectives. The instructor could then invite other students to evaluate how SMART are these objectives, and what could go wrong if they are not properly identified.

Part 1: Planning Sequence Rational of SULR

Short Discussion: What aspects of Restoration are we targeting?

This discussion is a reflection on the setting target and scope mapping for the SULR projects. The instructor should encourage students to reflect on what aspects of urban land degradation could be targeted by a sustainable

restoration project. The instructor could reflect on the restorative continuum from Module 1 and make sure that students would be specific in their targets for planning.

Student Activity: How to frame SULR in a physical setting?

The goal of this activity is to allow students to be able to pinpoint what is meant by SULR and how it would affect the built environment and social well-being of communities. Depending on the delivery medium of the lecture and number of students, the instructor should choose the proper method for conducting this activity, whether in group or individual work, use boards, sticky notes, and markers, or use online tools like white board.

Steps:

- Ask students to list, draw, or make a collage from online images to “Map physical/social attributes for a built environment that would qualify from their view as an outcome of SULR”.
- Conduct group presentations to discuss similarities and differences in their work.
- Reflect on why SULR could be perceived differently and how to close the gap in understanding the concept and its operationalization in the physical setting.

Part 2: Balancing Formal & Informal Structures with Decision-Support Enablers

Short Discussion: How to navigate unstructured planning?

A reflective discussion with students about how to navigate the informal structures existing while planning for urban land restoration. This includes how to regularize, document, and capitalize on useful structures (like community bonds, proactive and self-organization dynamics, survival resilience structures, ...etc.) and how to restrain, discourage, and disempower harmful structures (like systematic racism and sexism, favouritism, social exclusion, illegal or criminal structures, ...etc.).

Student Activity: How to identify direct & indirect variables for planning activities of SULR?

The goal of this activity is to allow students to use system thinking in mapping different variables affecting the activities of SULR.

Steps:

- Choose one activity from any of the four phases discussed at the beginning of the lecture.
- Define three direct variables and three indirect variables that affect this activity.
- Define 3 direct variables and 3 indirect variables that are affected by this activity

References/resources

Adams, D. and C. Watkins. (2014). *The Value of Planning*, UK.

Bedinger, M., Beevers, L., Walker, G. H., Visser Quinn, A. and K. McClymont. (2020). Urban systems: Mapping interdependencies and outcomes to support systems thinking, *Earth's Future*, 8(3). <https://doi.org/10.1029/2019EF001389>

Khatri, V. and C.V. Brown. (2010). Designing data governance, *Communications of the ACM*, 53(1), pp. 148-152. <https://doi.org/10.1145/1629175.1629210>

UN-Habitat. (2024). *Our City Plans: An Incremental and Participatory Toolbox for Urban Planning*. Fourth Edition. Nairobi: Earthscan: for UN-Habitat. <https://unhabitat.org/our-city-plans-an-incremental-and-participatory-toolbox-for-urban-planning>

UN-Habitat. (2023). *Enabling Meaningful Public Participation in Spatial Planning Processes*. Nairobi: Earthscan: for UN-Habitat <https://unhabitat.org/enabling-meaningful-public-participation-in-spatial-planning-processes>

3.2 Enabling environment and co-benefits

Developed by

Leticia Domingos (UNU-EHS)

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Learning objectives

- ▶ Understand the key components of an enabling environment for SULR, including policy frameworks, governance, and financing mechanisms;
- ▶ Identify the multi-dimensional benefits of SULR, considering social, economic, and ecological perspectives;
- ▶ Analyze how systemic approaches in urban planning can create cascading benefits that enhance resilience, sustainability, and equity.

Summary

Part 1. Enabling Environment for SULR

Creating an enabling environment for SULR requires inclusive policies, strong institutions, financial resources, and effective land governance. Policies must integrate LDN goals with national planning frameworks while securing land tenure, as land access remains a challenge in many regions. Institutional arrangements should include cross-sectoral coordination to ensure coherence in land management decisions. Additionally, sustainable financing mechanisms—such as national budgets dedicated to LDN and funding from global entities—are essential for long-term success. A critical component of enabling environments is the science-policy interface, which strengthens decision-making through research, data collection, and monitoring systems that inform land-use policies. Finally, land governance plays a central role in restoration efforts,

ensuring transparent decision-making, equitable access to land, and balancing formal and informal governance

Part 2. Cascading effects of SULR in cities

SULR generates a range of interconnected social, economic, and ecological benefits. Socially, it enhances well-being by providing access to green spaces, fostering social cohesion, and offering educational opportunities. Economic co-benefits include job creation in the restoration sector, cost savings in public health due to improved environmental conditions and financial returns from green infrastructure investments. Ecologically, SULR contributes to climate resilience by reducing urban heat, improving air and water quality, enhancing biodiversity, and supporting ecosystem services such as carbon sequestration. The systemic nature of these benefits highlights how restoring degraded urban land can simultaneously address multiple urban challenges, reinforcing the importance of holistic planning approaches that maximize positive ripple effects across city systems.

Lessons learned

- ▶ Creating an enabling environment is crucial; multi-level governance, science-policy integration, and institutional collaboration foster a supportive environment for SULR;
- ▶ Cities are not just collections of isolated problems but interconnected networks, where targeted actions can unlock cascading benefits;
- ▶ Co-benefits of SULR highlight the interconnectedness of urban systems, demonstrating how restoration projects can address multiple urban challenges;
- ▶ A systemic approach to urban restoration helps ensure that interventions generate positive ripple effects. By strategically identifying leverage points, we can maximize co-benefits and minimize unwanted outcomes.
- ▶ Student questions and exercises
- ▶ Continue the cascading effects from where the previous diagrams stopped. Try to add another 5 elements to each.

- Pick one of the co-benefits of SULR that were presented and think of its cascading effects in the city.

See Case Study in Section 5.6 Jakarta: Evaluating Mangrove Sustainability in the context of Sustainable Urban Land Restoration, Indonesia

References/resources

Barry, M., & Augustinus, C. (2016). *Framework for Evaluating Continuum of Land Rights Scenarios*. UN-Habitat. https://unhabitat.org/sites/default/files/download-manager-files/Framework%20for%20Evaluating%20Continuum%20of%20Land%20Rights%20Scenarios_English_2016.pdf

Ding, H., Faruqi, S., Wu, A., Altamirano, J.-C., Ortega, A. A., Verdone, M., Cristales, R. Z., and R.C.A. Vergara. (2017). *Roots of Prosperity: The Economics and Finance of Restoring Land*. https://files.wri.org/d8/s3fs-public/roots-of-prosperity_0.pdf

Elmqvist, T., Setälä, H., Handel, S. N., van der Ploeg, S., Aronson, J., Blignaut, J. N., Gómez-Baggethun, E., Nowak, D. J., Kronenberg, J., and R. de Groot. (2015). Benefits of restoring ecosystem services in urban areas. *Current Opinion in Environmental Sustainability*, 14, 101–108. <https://doi.org/10.1016/j.cosust.2015.05.001>

IPBES (2019). *Summary for policymakers of the global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services*. IPBES secretariat, Bonn, Germany. 56 pages. <https://www.ipbes.net/global-assessment>

Verburg, P.H., Metternicht, G., Allen, C., Debonne, N., Akhtar-Schuster, M., Inácio da Cunha, M., Karim, Z., Pilon, A., Raja, O., Sánchez Santivañez, M., and A. Şenyaz. (2019). *Creating an Enabling Environment for Land Degradation Neutrality and its Potential Contribution to Enhancing Well-being, Livelihoods and the Environment. A Report of the Science-Policy Interface*. (UNCCD) Bonn, Germany. https://www.unccd.int/sites/default/files/2019-08/UNCCD_SPI_2019_Report_1.2.pdf

3.3 Actors and Stakeholders in SULR

Developed by

Johannes Hamhaber (TH Köln, Germany)

Learning objectives

Upon completion of the module, the participants are able to

- To distinguish actors in SULR in general from stakeholders in particular;
- To define and distinguish actors in SULR;
- To assess roles and responsibilities in SULR to stakeholders in a given case in order to assess actors' roles in SULR, their case-related contextual positions as stakeholders.

Summary

The lecture starts with a recouping of three slides from Module 1 as a point of departure, reminding the students of the different types of scales, and the concept of 'Just Transition'. A typology of actors in SULR is defined, based on a governance triangle proposed by Heldeweg and Seintier (originally for the energy system transition). The typology distinguishes three modes of governance and intermediate hybrids. It also allows to 'map' institutional actors into such a triangle, thus helping to differentiate the different actor types systematically, accordingly into three classical actors types (state, private sector, and civil society) and also hybrids thereof. This concept is then applied to SULR, allowing the students to diversify the actors typically involved in SULR with their positions, their forms of interaction, and other criteria.

Part 1. Actor typologies in SULR

SULR generally is implemented by institutional actors, such as public sector actors (e.g. urban planning department or environmental protection agency), private sector players (e.g. real estate companies) or actors from civil society (such as community-based initiatives or NGOs). For a better understanding of their roles, these actors and hybrids thereof are classified and debated with regards to their typical role in SULR.

In specific cases, these roles turn into specific interests and powers of actors. In short, their ability to affect or their exposure being affected by such a SULR initiative or project renders them as stakeholders in the case, that negotiate with each other, exert power to achieve their interests.

Part 2. Different actors' roles and positions in SULR and a typology of institutional actors in SULR according to modes of governance.

Once the institutional actors and their roles/mandates are defined, this can be applied to specific cases, turning the actors into stakeholders with vested interests and case-related power.

- ▶ Public Actors
- ▶ Private Sector Actors
- ▶ Civil Society Actors
- ▶ Hybrid Actors (Public-Private Partnerships, Social Entrepreneurship, etc.)
- ▶ Institutional Actors' Roles and Responsibilities in SULR

Part 3. Stakeholder mapping typology in SULR

Finally, several mapping tools for stakeholders are introduced, so that the students can exercise the stakeholder mapping, either in cases provided in the modules, or by the lecturer with a local case application.

- ▶ Stakeholder definitions by interest and power
- ▶ The role of legitimation in stakeholder definitions

Lessons learned

Upon completion of this submodule, students are able to:

- ▶ Grasp the actor typology according to modes of governance;
- ▶ Understand the prototypical roles of different actors in SULR;
- ▶ Assess roles and responsibilities in SULR in a given case;
- ▶ Analyze the interest and powers of stakeholders in a given SULR case.

Student questions and exercises

Exercise 1: Map the actors that are potentially relevant to the SULR into the actor typology, based on the modes of governance table.

To do this, please assess the interest type, the relation type, the position in major relations forms, and the main legitimacy, and not least the organisational form. If you can map the organizations below into several columns of the table without one column (and thus institutional order dominating), deem them as hybrid actor

Exercise 2: Map the actors that are potentially relevant to the SULR into the actor typology, based on the modes of governance table.

To do this, please assess the interest type, the relation type, the position in major relations forms, and the main legitimacy, and not least the organisational form. If you can map the organizations below into several columns of the table without one column (and thus institutional order dominating), deem them as hybrid actors.

Exercise 3: From the case study provided, make a list of the stakeholders defining their roles, interests, power, etc. Map the stakeholders into one of the three provided models of stakeholder mapping. Discuss in small groups about the location of stakeholders in the map, their ability to influence the SULR project, and their position with regards to a just transition to sustainability (as discussed in module 1.8)

References/resources

Chevalier, J. M., and D.J. Buckles. (2008). *SAS² - A Guide to Collaborative Inquiry and Social Engagement*. SAGE India, 316p.

Heldeweg, M. A., and S. Saintier. (2017). Renewable energy communities as 'socio-legal institutions': A normative frame for energy decentralization? *Renewable and Sustainable Energy Reviews* 119, 14p. <https://doi.org/10.1016/j.rser.2019.109518>

Reed, M. S., Graves, A., Dandy, N., Posthumus, H., Hubacek, K., Morris, J., Prell, C., Quinn, C. H, and L.C. Stringer (2009). Who's in and why? A typology of stakeholder analysis methods for natural resource management. *Journal of Environmental Management* 90:1933-1949. <https://doi.org/10.1016/j.jenvman.2009.01.001>

Widerberg, O., Pattberg, P., and K.E. Kristensen. (2016). *Mapping the institutional architecture on global climate change governance*. Vrije Universiteit Amsterdam, IVM, Technical Report, 42p.

3.4 Multi-Level Governance in SULR

Developed by

Johannes Hamhaber (TH Köln, Germany)

Learning objectives

Upon completion of this submodule, students are able to:

- ▶ Distinguish multi-level governance (MLG) type I and type II;
- ▶ Understand the tiers of public administration in SULR (type I);
- ▶ Differentiate the forms of private sector participation (type II);
- ▶ Know the forms of public-civil co-management in SULR (type II);
- ▶ Grasp the levels of citizens' participation in SULR projects in order to distinguish institutional designs in SULR

Summary

Part 1. Why Multi-Level Governance in SULR?

The module refers to the core concept of MLG, according to Hooghe and Marks (2003). This is to illustrate how multiple actors across different scales collaborate for a successful implementation of SULR, including both, vertical

and horizontal relations. When referring only to public sector actors in rigid institutional architectures (i.e. hierarchies of public administration), MLG type I applied as an applicable frame. When also private sector and civil society actors are involved, the architecture often becomes softer and more flexible, and horizontal links become more relevant, showing characteristics of MLG type II.

The module builds largely the three conceptual frames, namely MLG as organizing principle, and private sector participation concepts and community co-management (Carlsson and Berkes 2007) for the selected MLG type II examples. Each concept is exercised based on the SULR activities list and/or SULR case studies. These may be replaced by a case study from targeting the audience in class, selected by the lecturer. This can be applied for both sub-modules 3.3 and 3.4.

Part 2. The constitutional levels of governments and the Multi-Level Governance (MLG) typology

SULR is an activity that affects a diverse set of stakeholders. As a starting point, the major actors may be seen as belonging to the public sector, and even the central national government. Alternatively, governments have surrendered some activities and related resources to lower tiers of government, following the principle of subsidiarity. Thus, multi-level governance emerges: the governance and management of SULR will be allocated at different tiers of public administration.

Part 3. Multi-Stakeholder Governance in SULR MLG Type II: Private Sector Participation

Furthermore, SULR activities may also be delegated to the private sector (under the umbrella term “private sector participation”). This may take several forms, from simple contracts (endowing the private companies with a specific limited task), up to full privatization (giving the private sector all the assets (also the land) and all related opportunities and risks, specifically the economic ones).

Part 4. Multi-Stakeholder Governance in SULR MLG Type II: Community Co-Management in SULR

Similarly, the civil society (with NGOs on each scale, or also local communities) can become collaboration partners to the public sector. Depending on the relation between the two actor types, different formats can be established,

along a gradient of so-called co-management. Finally, the role of citizens and the public at large, not being organized in an institutional way, is reflected by a debate of citizens' participation in SULR.

Lessons learned

- ▶ MLG explains how different actors, including governments, the private sector, and civil society, collaborate across different levels to implement SULR successfully.
- ▶ By analyzing different tiers of public administration and their roles in SULR, it becomes clear that MLG Type I, with its limited tiers and exclusive responsibilities, can be effectively applied to SULR tasks.
- ▶ MLG Type II demonstrates how public-private cooperation in SULR can enhance flexibility and collaboration in managing urban restoration projects.
- ▶ Community co-management in SULR shows the value of sharing responsibilities between the state and civil society, fostering more inclusive and effective governance.

Student questions and exercises

Exercise 1:

From the activity in module 3.2, take your barometer guide actors' list, map the public sector stakeholders into the jurisdictional scales (be aware that they should fit in) and check, on which scales you may not have an actor listed. Why?

Exercise 2.

From the activity in module 3.2, take your barometer guide actors' list, map the non-public sector stakeholders into the jurisdictional scales (be aware that they should fit in) and check, on which scales you may not have an actor listed. Why? Map the private sector and civil society stakeholders into the jurisdictional scales (be aware that they may NOT fit in) and check, which scales you may have to add above, below, or in between. Why?

Exercise 3.

From the activity in module 3.3, take your barometer guide actors' list, select three to five of the activities, and please assess which of the tasks can be transferred to the private sector; which of the Public Sector Participation forms would be the best option for collaboration; for which tasks should the public sector retain more control - Why? Brief explanation of the flow of the topics, exercises, and slides. Main purpose and relation of concepts, figures and contents.

References/resources

Carlsson, L. and F. Berkes (2005). Co-management: Concepts and methodological implications. *Journal of Environmental Management*, 75(1), 65–76. <https://doi.org/10.1016/j.jenvman.2004.11.008>

Delmon, J. (2010). *Understanding options for public-private partnerships in infrastructure*. The World Bank. https://ppp.worldbank.org/sites/default/files/2022-05/WPS5173_0.pdf

Hooghe, L. and G. Marks (2003). Unraveling the central state, but how? Types of multi-level governance. *American Political Science Review*, 97(2), 233–243. https://garymarks.web.unc.edu/wp-content/uploads/sites/13018/2016/09/hooghe.marks_unravelingcentralstate.apsr_.2003.pdf

3.5 Co-creation, visioning and goal setting

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Learning objectives

- ▶ Identify co-design, co-creation and participation essential for sustainable urban planning and development;
- ▶ Integrate visioning while setting goals for an urban intervention toward SULR.

Summary

Part 1. Co-design and co-creation

Part 1 presents creative tools and participatory methods for co-creation and co-design to trigger change in SULR. It highlights the importance of inclusive processes across the spectrum of public participation, from informing to empowering actors and stakeholders.

Part 2. Visioning

Visioning exercises and processes help establish a shared future goal, while other tools such as 3D mapping (Theory U) offer a dynamic approach to visualize transformative change. These methods foster collaboration among diverse actors, enabling socially just and innovative SULR projects.

Part 3. Goal setting

Part 3 focuses on the need to steer and align urban narratives to address land degradation. By incentivizing urban narratives aligned with SULR and the land degradation neutrality principle, decision-makers can contribute to shifting mindsets and building collective support. The goal-setting process involves a

backcasting process, determining specific goals to achieve the common vision toward restoring urban land.

Lessons learned

- ▶ Participatory processes such as co-creation and co-design foster collaboration and ensure that multiple perspectives are considered. They also enable long-term change;
- ▶ Tools oriented towards possible futures help align participants around common goals and visualize future transformations;
- ▶ Visioning allows sharing goals and imagine how SULR projects could be, inspiring and motivating efforts;
- ▶ Transforming urban narratives by incentivizing SULR-related goals can shift public perceptions and encourage more sustainable urban development practices.

Student questions and exercises

- ▶ Template for Co-design process—New open space in your neighborhood that incorporates SULR actions. Who would you involve? Ideas? Topics? Are there any working groups or teams representing city sectors, institutions, or interests? What are the shared perspectives about the new open space?
- ▶ Visioning tool - What does the future look like? What accomplishments have been made? What has changed from the current state? - Vision statement
- ▶ 3-Modelling (Theory U) tool - As a group create a model with objects that represent the current state of a determined degraded urban space. Represent elements like actors, institutions, specific features of the space like vegetation, soil, buildings and their relations with objects. What would you change? Change the model according to the desired restored state. Move and add objects. What are the most important leverage points for moving from the first model to the second? Does the model represent the desired and shared future state? What do you need to get there?
- ▶ Are you aware of the “land vision” of your country?

See Case Study, Section 5.5 Cuenca: Shinshin creek renovation, Ecuador

References/resources

- Biggs, R., Preiser, R., de Vos, A., Schlüter, M., Maciejewski, K. and H. Clements (2021). *The Routledge Handbook of Research Methods for Social-Ecological Systems*. Routledge. <https://doi.org/10.4324/9781003021339>
- Cinderby, S., de Bruin, A., Cambridge, H., Muhoza, C. and A. Ngabirano (2021). *Transforming urban planning processes and outcomes through creative methods*. *Ambio*, 50(5), 1018–1034. <https://doi.org/10.1007/s13280-020-01436-3>
- Design Council. (2024). *History of the Double Diamond*. <https://www.designcouncil.org.uk/our-resources/the-double-diamond/history-of-the-double-diamond/>
- IAP2. (2018). *International Association for Public Participation*. IAP2 Spectrum of Public Participation. <https://www.iap2.org/page/pillars>
- Leino, H. and Puumala, E. (2021). What can co-creation do for the citizens? Applying co-creation for the promotion of participation in cities. *Environment and Planning C: Politics and Space*, 39(4), 781–799. <https://doi.org/10.1177/2399654420957337>
- Presencing Institute . *Theory U Leading from the emerging future*. <https://www.presencing.org/theoryu>

3.6 Policy instruments

Developed by

Simone Sandholz (UNU-EHS)
Fabian Rackelmann (UNU-EHS)
Sönke Kreft (MCII & UNU-EHS)

Learning objectives

- ▶ Distinguish policy instruments and their effectiveness in promoting sustainable urban development and restoration;
- ▶ Analyze the role of policy instruments and policy coherence in driving sustainable urban development and restoration efforts;
- ▶ Explore pathways to promote synergies between land restoration and other policy objectives using the example of social protection objectives social protection objectives.

Summary

Part 1. Urban policy instruments and links to SULR

The lecture begins with an introduction to policy instruments, and what kind of urban policy instruments exist on different scales. After giving an overview on the range of instruments, their goals, users and benefits are highlighted, followed by a critical reflection around potential policy gaps, sustainability and justice concerns. Part 1 concludes by underlining the need for developing, adjusting or broadening urban restoration policies.

Part 2. Accelerating SULR through sectoral policies

Part 2 focuses on leveraging other sectors such as climate action, ecosystem conservation, social protection, financial inclusion, and disaster risk reduction as catalysts for accelerating land restoration. After explaining common objectives of these different sectors, key policy instruments are explained. A focus is on identifying their potential to be applied to SULR, including by providing practical examples.

Part 3. Pathways to design coherent policies

The last part of this submodule explains policy coherence and its multiple benefits, exemplified by good practices. After reflecting on the common objectives of policy coherence with other approaches and sectors, different ways to leverage synergies are outlined, and critically reflected.

Lessons learned

- ▶ Urban development draws on multiple policies on various scales, which guide sustainable urban development;
- ▶ Nevertheless, there could still be policy gaps, especially for topics considered less important or emerging issues such as urban land restoration;
- ▶ Effective land restoration requires embedding within other sectoral policy instruments such as social protection, financial inclusion, and disaster risk finance. This integration ensures that land restoration is not an isolated effort but part of a comprehensive strategy, including in urban areas;
- ▶ Successful urban land restoration requires the coordination of multiple sectors, coherent approaches are more likely to come to lasting solutions with multiple benefits.

Student questions and exercises

The session begins with inputs on urban policy, followed by policy instruments from other sectors, including from land restoration. While presenting, the lecturer could trigger a discussion by asking:

- ▶ For what do we need policy instruments in urban development?
- ▶ Can you think of policy instruments to address urban land degradation?
- ▶ What kind of policy instrument are they (e.g., regulatory, information-based...)?
- ▶ What might be their benefits and challenges?

The lecture also includes a short exercise, during which students are supposed to find synergies between urban land restoration and a 2nd sector, with the aim to identify a policy with goals that benefit both sectors.

References/resources

Capano, G., and M. Howlett. (2022). Instrumentation in policy design: Policy tools – *From devices to activators*. In G. Peters & G. Lafontaine (Eds.), *Research handbook of policy design* (Vol. 2022, pp. 72–87). Edward Elgar. <https://doi.org/10.4337/9781839106606.00012>

Federal Office for Building and Regional Planning. (2000). *Urban Development and Urban Policy in Germany. An Overview*. BBSR, Bonn. https://www.bbsr.bund.de/BBSR/EN/publications/CompletedSeries/Berichte/2000_2009/Bd06UrbanDevelopment.html

OECD/UN-HABITAT/UNOPS. (2021). *Global State of National Urban Policy 2021: Achieving Sustainable Development Goals and Delivering Climate Action*, OECD Publishing: Paris. <https://doi.org/10.1787/96eee083-en>

Rackelmann, F., van der Heijden, S., Parekhelashvili, M., Panda, A., Pineda Fernandez, D. M., Bananayo, P., Sandholz, S., Kreft, S., Orr, B. J. and Y. Walz (2024). *Harmony in action for land restoration: Linking social protection, financial inclusion and disaster risk finance*. UNCCD and UNU-EHS. Bonn, Germany, <http://collections.unu.edu/view/UNU:10073>

Slobodian, L., Vidal, A. and C. Saint-Laurent. (2020). *Policies that support forest landscape restoration: What they look like and how they work*. Gland, Switzerland: IUCN <https://portals.iucn.org/library/sites/library/files/documents/2020-045-En.pdf>

3.7 Finance instruments

Developed by

Simone Sandholz (UNU-EHS)

Fabian Rackelmann (UNU-EHS)

Sinja Buri (MCII & UNU-EHS)

Sönke Kreft (MCII & UNU-EHS)

Learning objectives

- ▶ Understand financial impacts from land degradation and financial benefits from restoration;
- ▶ Understand what national to local finance mechanisms and instruments might be available for SULR;
- ▶ Understand international finance mechanisms from different sectors that can be used in support of restoration and how such budgets could be made applicable to urban areas;
- ▶ Get an overview on key finance actors involved on different scales.

Summary

Part 1. Financing urban land restoration needs and opportunities

Part 1 starts from providing an overview on financial impacts from lack of urban land restoration, and how financing land restoration would pay off. It then touches on the challenges related to financing urban land restoration, given diverging interest of different actors. The part concludes on key entry points for urban land restoration under the different international frameworks related to restoration, climate, disaster risk, biodiversity, and others.

Part 2. Financial instruments and potential users (from household to landscape/national)

Part 2 elaborates on different available financial instruments on different scales, from international to urban, and identifies involved actors and their roles. A particular focus is on access for urban actors. Based on examples the particular benefits as well as potential challenges and trade-offs of different instruments and how they could be mitigated is being discussed. Besides providing an overview on already existing financial instruments this chapter also provides a short overview on international frameworks and emerging topics and related discussions around funding mechanisms in the making.

Part 3. Accelerating SULR by systemic approaches

The concluding part advocates for synergetic approaches that bundle sectoral funding streams to multiply benefits. It explains the benefits of leveraging cross-sectoral synergies by identifying common objectives to develop coherent and cost-efficient funding streams. This is further explained by providing concrete examples advocating for linking sectoral approaches such as social protection, financial inclusion and disaster risk financing to land restoration and related finance streams.

Lessons learned

- ▶ The costs from not financing land restoration might be high, but are not necessarily accounted for;
- ▶ There are multiple international policies and frameworks that might not be geared towards SULR directly, but that would allow for including SULR goals;
- ▶ Accordingly, multiple international, national and local financial instruments are available to different beneficiaries. They might differ between countries, but there are some that can be used for SULR, by both public and private actors;
- ▶ Each one has advantages and disadvantages, including in terms of accessibility, user-friendliness, duration;
- ▶ Instruments can be combined, to create synergies and cost-effective approaches.

Student questions and exercises

The session could start with a reflection of the policy instruments introduced in the previous session, by asking:

- ▶ Which of the previously introduced land restoration policy instruments require more financial resources, which ones might even generate financial resources?

Later, the session includes a reflection exercise along the following questions:

- ▶ What kind of skills would be needed to have access to the different types and scales of finance mechanisms?
- ▶ What do you think could help to facilitate access for urban actors and SULR?
- ▶ Do you see any negative trade-offs from particular finance instruments? How could that risk be mitigated?

References/resources

Garvey, J., Grigoriadis, V., Flannery, G., Knapp, E., Gold, E., Hutchinson, G., Frewer, L.J., Brereton, P., and K.A. Byrne. (2025). Designing financial instruments for land-based ecological restoration: A review and future research Agenda, *Cleaner Production Letters* 8:q00089 <https://doi.org/10.1016/j.clpl.2024.100089>

International Bank for Reconstruction and Development / The World Bank (2022). *Scaling up Ecosystem Restoration Finance. A Stocktake Report*. International Bank for Reconstruction and Development / The World Bank, Washington. <https://documents1.worldbank.org/curated/en/099955011092213526/pdf/P17770602aad4701309adb08b084c12888c.pdf>

Mirwald, M., S. Sandholz, S.M.J. Abbas and N. Cleetus. (2024). *Building urban resilience using systemic approaches: Financing a just transition, social protection, and loss and damage in cities*, C40, 58p. https://www.c40knowledgehub.org/s/article/Barriers-to-action-Gaps-in-financing-for-a-just-transition-social-protection-and-loss-and-damage-in-cities?language=en_US

UNEP. (2024). *State of Finance for Nature: Restoration Finance Report. Growing the Green: Why and how restoration finance needs to quadruple by 2030*. Nairobi. <https://doi.org/10.59117/20.500.11822/46730>

UrbanShift. (2024). *Unlocking Urban Finance: The Vital Role of Project Preparation for City Projects* <https://www.shiftcities.org/post/unlocking-urban-finance-vital-role-project-preparation-city-projects>

3.8 Challenges and handling conflict in Sustainable Urban Land Restoration

Developed by

Leticia Domingos Vellozo (UNU-EHS)

Simone Sandholz (UNU-EHS)

Learning objectives

- ▶ Understand the constraints and trade-offs in urban land restoration and their impact on sustainability;
- ▶ Analyze the competing priorities in urban land use and how they shape decision-making;
- ▶ Explore conflict resolution tools and participatory approaches that facilitate inclusive land-use planning.

Summary

Part 1. Challenges in SULR

SULR faces multiple constraints that make implementation difficult. Land constraints arise from limited availability and high competition between residential, commercial, and industrial development, making it challenging to allocate space for restoration initiatives. Financial constraints further limit restoration efforts, as private investors are often hesitant due to uncertain returns, and remediation of brownfields requires significant investment. Additionally, technological constraints, such as the lack of environmental monitoring tools and GIS-based planning systems, hinder the ability of cities—especially in low-income regions—to design and track restoration progress effectively. Policy fragmentation exacerbates these issues, with urban planning responsibilities often spread across multiple agencies, leading to inefficiencies, weak enforcement of restoration policies, and a focus on short-term economic growth over long-term ecological planning.

Part 2. Trade-offs in urban land use

SULR involves navigating difficult trade-offs between competing social, economic, and environmental priorities. While ecological restoration can improve air quality, biodiversity, and resilience to climate change, it often contributes to gentrification, displacing lower-income residents as property values rise. Similarly, green spaces, parks, and conservation efforts compete with the demand for affordable housing and commercial development, creating conflicts over land allocation. Cities also face tensions between short-term economic development goals—such as attracting investors and expanding infrastructure—and long-term ecological objectives, which require sustained investment but offer gradual benefits. Moreover, public-private partnerships in restoration projects can lead to privatized benefits, restricting public access to regenerated spaces and reinforcing inequities. Finally, restoration efforts may unintentionally erase historical or cultural significance in formerly industrial or marginalized areas, highlighting the need to balance environmental recovery with social and cultural preservation.

Part 3. Handling conflict

Conflicts over urban land use arise from competing demands and are influenced by political, economic, social, and environmental factors. As land is a finite resource, any decision in favor of one use excludes multiple alternatives, creating tensions among stakeholders with diverse interests. Disagreements can stem from direct competition over land (e.g., conservation vs. development) or broader socio-political dynamics, such as economic disparities or shifting environmental risks. Managing these conflicts requires structured approaches, distinguishing between conflict management—which seeks to minimize negative impacts—and conflict resolution, which aims for long-term, mutually acceptable solutions. There are many planning tools that can help with these steps. These include Environmental Impact Assessments (EIAs) to compare different land uses, participatory urban planning to involve a wide range of stakeholders, and mediation or conflict transformation to deal with systemic disputes. Integrating these tools into urban governance is essential for ensuring fair and sustainable land-use decisions.

Lessons learned

- ▶ SULR faces significant challenges like competing land use demands, exclusion of vulnerable groups and financial constraints;
- ▶ Navigating Trade-offs is Essential for Balanced Outcomes, urban restoration efforts often involve trade-offs between ecological integrity and economic or social benefits;
- ▶ Effective land-use planning in cities requires conflict resolution mechanisms to balance competing demands.

Student questions and exercises

- ▶ Are there trade-offs that you find more challenging to address? Why?
- ▶ What are the main constraints that you can identify in the place that you live?

- ▶ What kind of (potential) conflicts linked to green gentrification do you see in the Al-Fustat development?
- ▶ How could they be overcome?
- ▶ Which actors do you see involved in conflict resolution?

See Case Study in Section 5.6 Jakarta: Evaluating Mangrove Sustainability in the context of Sustainable Urban Land Restoration, Indonesia

References/resources

Brown, G, Raymond, C.N. (2014). Methods for identifying land use conflict potential using participatory mapping. *Landscape and Urban Planning* 122:196-208. <https://doi.org/10.1016/j.landurbplan.2013.11.007>

IUCN. (2023). *IUCN SSC guidelines on human-wildlife conflict and coexistence* <https://portals.iucn.org/library/sites/library/files/documents/2023-009-En.pdf>

Maronedez, A.K., Mutanga, O., and M.A. Cho (2024). *Promoting inclusion in urban land use planning using participatory geographic information system (PGIS) techniques: A systematic review*. <https://doi.org/10.1016/j.jenvman.2024.123099>

Quinton, J., Nesbitt, L., Sax, D., and L. Harris (2024). Greening the gentrification process: Insights and engagements from practitioners. *Environment and Planning E: Nature and Space*, 7(4), 1893-1917. <https://doi.org/10.1177/25148486241236281>

Seto, K. C., and A. Reenberg. (2014). *Rethinking Global Land Use in an Urban Era*. Cambridge, MA: MIT Press. <http://dx.doi.org/10.7551/mitpress/9780262026901.001.0001>

3.9 Urban stocktaking and action prioritisation

Developed by

Johannes Hamhaber (TH Köln)

Alejandra Ramos-Galvez (UNU-EHS)

Simone Sandelholz (UNU-EHS)

Learning objectives

- ▶ Comprehend the role of stocktaking in reporting and decision making for SULR;
- ▶ Distinguish the urban stakeholders in data creation and ownership;
- ▶ Embed urban stocktaking into the “City as a System” concept.

Summary

Part 1. Agendas for stocktaking in SULR

Data collected through urban stocktaking is used to create a comprehensive baseline that helps planners and decision-makers understand strengths, weaknesses, opportunities, and challenges within an urban area. Understanding the current state of urban land degradation is also crucial in the case of stocktaking for SULR. The baseline is crucial for developing strategic plans, setting priorities, and making informed decisions. Stocktaking can show either a status (at one time or over time to track change in a KPI logic) or the way systems behave (systems dynamics).

Part 2. Stakeholders' roles and types of stocktaking

Stakeholders may collaborate during stocktaking, which can occur on various scales. Stakeholders' role in stocktaking, for example with municipalities, may consider data existence, availability, and reporting. These can be sidelined by private sector (environmental) reporting and citizens (science) and communities. Some examples of global finance opportunities and demand for reporting are presented.

Part 3. Stocktaking for SULR—a systems approach

Stocktaking can be carried out in a (multi-)linear data-driven decision setup, having a notion of solutions and prioritizing under a problem-solving premise. This logic leads to a multi-criteria aggregation and related trade-offs. It can also be useful for having a complex model approach, prioritizing under a system transition premise, and defining system leverages for desired shifts. Both approaches are useful depending on the needs and purpose.

Lessons learned

- ▶ Stocktaking is essential for management and governance of urban land degradation;
- ▶ It is valid and necessary to use data-driven approaches that follow (multi-) linear and technical solutions. When a problem has been narrowed down and found to be specific, it may facilitate action; however, links and impacts to other issues may not be being considered;
- ▶ For transformative approaches toward restorative action and sustainability, looking to represent (as possible) the complex behavior of the system is needed, in order to understand possible leverage points and prioritize accordingly;
- ▶ Urban land degradation is a complex situation; it is not static nor the same for different scales and contexts, therefore requires systemic approaches.

Student questions and exercises

Exercise 1:

Stocktaking at the Emscher case study. Use the information provided by the Emscher case study, Germany.

Exercise 2:

Action prioritization exercise. – Simplified version of an impact analysis method. Recognizing relationships between variables. Finding leverage points.

See Case Study in Section 5.10 Ruhr Valley: post-industrial landscape restoration, Emscher park, Germany

References/resources

Cubie, D. and T. Natoli. (2022). *Coherence, Alignment and Integration: Understanding the Legal Relationship Between Sustainable Development, Climate Change Adaptation and Disaster Risk Reduction*. In: Yairen, S.F., Columbié, J., Le Tissier, M. & B. O'Dwyer (Eds) *Creating Resilient Futures Integrating Disaster Risk Reduction, Sustainable Development Goals and Climate Change Adaptation Agendas* (P. 23-44) (P. 45-64)

Le Tissier, M. and H. Whyte. (2022). *Why Does Making Connections Through Resilience Indicators Matter?* In: Yairen, S.F., Columbié, J., Le Tissier, M. & B. O'Dwyer (Eds) *Creating Resilient Futures Integrating Disaster Risk Reduction, Sustainable Development Goals and Climate Change Adaptation Agendas* (P. 23-44)

Reed, B. (2007). Shifting from 'sustainability' to regeneration. *Building, Research and Information*, 35(6), 674-680. <https://doi.org/10.1080/09613210701475753>

Snowden, D. and M. Boone. (2007). *A Leader's Framework for Decision Making. Decision making and problem solving. Harvard business review.* <https://hbr.org/2007/11/a-leaders-framework-for-decision-making>

Snowden, D. and A. Rancati. (2021). *Managing complexity (and chaos) in times of crisis. A field guide for decision makers inspired by the Cynefin framework*, Publications Office of the European Union, Luxembourg, 2021, ISBN 978-92-76-28843-5, JRC123629. <https://publications.jrc.ec.europa.eu/repository/handle/JRC123629>

3.10 Assess the status quo—identify actors, their involvement and resources needed (Exercise)

Developed by

Alejandra Ramos (UNU-EHS)

Simone Sandholz (UNU-EHS)

Learning objectives

- Identify key stakeholders, determine required resources, and develop strategic restoration priorities;
- Engage in a practical evaluation of urban restoration needs and opportunities.

Summary

Part 1. Identifying urban land degradation—Starting point—identification in Submodule 1.10

Select an urban area experiencing land degradation (e.g., a brownfield site, a flood-prone neighborhood, or an abandoned industrial district). Identify the primary causes of degradation (e.g., pollution, urban sprawl and land take, deforestation, poor land use policies). Vulnerable populations? (e.g., low-income communities, informal settlements, marginalized groups). Assess the direct and

indirect consequences, including environmental, economic, and social impacts. Document and summarize key findings.

Part 2. Mapping city sectors, actors, and institutions (Stakeholder mapping)

Identify key actors involved in future urban land restoration for the selected degraded area, including: Public sector: local governments, planning departments, environmental agencies. Private sector: developers, real estate firms, and investors. Civil society: local communities, advocacy groups. Discuss their level of involvement, interests, and possible conflicts. Create a stakeholder matrix, categorizing stakeholders by typology, power, and interest. See 4-field matrix in submodule 3.3.

Part 3. Identifying resources needed for a SULR project

Classify key resources required for urban land restoration. Suggested categories: 1) Knowledge and expertise: urban planners, ecologists, engineers, community representatives. 2) Material resources: Land access, construction materials, and techniques. 3) Regulatory support: land use policies, environmental laws, and municipal incentives. 4) Funding: public, private investments, grants, and loans. Discuss barriers to securing these resources and potential solutions. Develop a brief resource map outlining necessary inputs for a restoration project.

Part 4. Developing a theory of change and prioritizing actions—or adapting the one developed in submodule 2.8

Develop a Theory of Change (ToC) for your restoration project: Define the long-term goal for the selected area. Identify key interventions (e.g., removing pollutants, reintroducing native vegetation). Establish short- and medium-term outcomes (e.g., improved biodiversity, enhanced climate resilience). Create a flowchart or diagram illustrating the theory of change for your urban land restoration project. Choose a prioritization method to guide action: a) Impact vs. Feasibility Matrix: Rank restoration actions based on environmental impact and ease of implementation. b) Cost-Benefit Analysis: Evaluate the financial viability of different interventions. c) Stakeholder Consensus Approach: Identify actions with strong multi-sectoral support. Link the prioritization process to financing options, including government funding, private-public partnerships, or international grants. You can use the theory of change developed in submodule 2.8. What would you change and why?

Lessons learned

- ▶ Successful urban land restoration relies on collaboration among public institutions, private investors, and community organizations. Identifying power dynamics and aligning interests is key to ensuring project implementation and success;
- ▶ Securing financial, technical, and material resources is often a barrier to implementing restoration projects. Linking restoration efforts to economic, social, and environmental benefits can help attract investment and funding;
- ▶ A defined theory of change helps structure urban restoration efforts by clarifying objectives, expected outcomes, and required interventions. Prioritizing actions based on feasibility and impact ensures efficient use of resources and maximizes positive change.

Teaching notes

- ▶ Students can carry out the exercise for a specific urban polygon or for the city scale depending on time and resources available.

References/resources

Doyle, L. (2021). A theory of change for complex systems. *Change & complexity: vector theory of change*. The Cynefin Centre. <https://thecynefin.co/library/change-complexity-vector-theory-of-change/>



Module 4. Operationalizing Sustainable Urban Land Restoration

Description

This module covers state-of-the-art tools for assessing, planning, decision-making, implementing, and monitoring SULR, including land degradation assessment tools, citizen science, and decision-making frameworks, while addressing key data and ethical considerations. An in-depth exploration of various tools and frameworks related to data governance, degradation assessment, planning and design and monitoring will equip students with the knowledge to critically analyze and apply the SULR planning cycle. By integrating these concepts, students will be able to enhance restoration outcomes and contribute to sustainable urban land management.

Core competencies

Academic knowledge:

- ▶ Data literacy and data governance;
- ▶ Urban Land Degradation Assessment and Planning frameworks, tools and data;
- ▶ Decision Design Tools, Methods, and Technologies (GIS, Remote Sensing, Drones, Sensors etc.);
- ▶ Citizen science;
- ▶ Monitoring and Evaluation.

Skills:

- ▶ Gain practical skills on data sourcing, management, analysis and interpretation;
- ▶ Understand and apply the knowledge of assessment, decision design, and monitoring tools.

Attitudes:

- ▶ Importance of data literacy;
- ▶ Concern for suitability and applicability of tools and technologies.

Authors

Module coordination: THK

Developers:

(THK)

Prof. Dr. Lars Ribbe

Dr. Nazmul Huq

Sadiya Afrin Mazumder

Arsalan Saghir

(CU)

Arch. Fabiana Torres Espinoza

Cristina Peralta Peñaloza (MSc)

Dr. Mónica González Llanos

Dr. Ximena Salazar Guamán

(UGM)

Dr. Dyah Rahmawati Hizbaron

Dr. rer. nat. Arry Retnowati

Utia Suarma (MSc)

(UNU-EHS)

Dr. Simone Sandholz

Lionel Muñoz Rosas (MSc)

Alejandra Ramos Gálvez (MSc)

(SALAR International)

Helena Ohlsson (MSc)

4.1 Tools for planning, implementing and monitoring Sustainable Urban Land Restoration

Developed by

Lionel Muñoz Rosas (UNU-EHS)

Alejandra Ramos Gálvez (UNU-EHS)

Learning objectives

- ▶ Identify and select digital and non-digital tools for planning, implementation, and monitoring across various urban contexts and stakeholders;
- ▶ Understand capacities, limitations and the existence of biases in digital and non-digital tools;
- ▶ To evaluate tools in order to choose the appropriate ones based on specific potential and application for urban land restoration.

Summary

Part 1: Digital tools, capabilities, implementation and limitations

Part 1 focuses on digital tools, their general application in urban planning, alongside a critical view on their indiscriminate use. This section also presents a list of key digital tools, as well as their implementation within the project cycle, potential limitations, their prerequisites and the related necessary considerations. Participants should understand the centrality of data, as well as its limitations in a digital tools' use, to finally be able to assess when and how a corresponding tool might be implemented.

Part 2: Non-digital tools, capabilities, implementation and limitations

Part 2 delivers non-digital tools with potential for use in SULR. As with digital tools, this part describes different key tools, their use in the implementation cycle, potential limitations, considerations, and prerequisites. Alongside concrete definitions, the section also provides points of general relevance for ULR and exercises for participants.

Part 3: Examples

The final part presents examples of digital and non-digital tool applications for SULR and SULR-alike cases. The objective of the section is to present students with a short evaluation of use cases and encourage a critical perspective on tools they might use.

Lessons learned

- ▶ Digital and non-digital tools are known, and their implementation context is understood;
- ▶ Digital tools are demystified, and their implementation rationalized;
- ▶ Students view digital and non-digital tools as such, supporting decision-making and not taking it over;
- ▶ Contextual characteristics determine the usability of tools;
- ▶ Engaging with citizens, through the tools, can support trust, legitimacy, and restoration efforts.

Student questions and exercises

- ▶ How are informed decision-making, stakeholder engagement and monitoring connected to SULR?
- ▶ While implementing digital tools, why is it important to understand the time horizon technologies work best in and are originally conceived for?
- ▶ Say, you work for a city government with a tight budget. What technologies would you choose to prioritize SULR and why?
- ▶ While implementing referenced non-digital tools, why is it important that the implementation involve local communities?

- ▶ When designing a SULR intervention, which non-digital tools would you use? Follow these steps:
 - o Think about a general order of implementation
 - o Describe a SULR intervention where such an approach might make sense
 - o Discuss why implementation order might have consequences for results.

References/resources

AI4SoilHealth (n. d.). About us. Available online at About - [AI4SoilHealth](#)

Beuschel, W. (2008). *Dashboards for management*. In: Adam, F., Humphreys, P. (eds.) *Encyclopedia of Decision Making and Decision Support Technologies*, pp. 116-123. Information Science Reference, Hershey, PA (2008). [doi:10.4018/978-1-59904-843-7.ch014](#).

Kerins, P., Guzder-Williams, B., Mackres, E., Rashif, T. and Pietraszkiewicz (2021). *Mapping Urban Land Use in India and Mexico using Remote Sensing and Machine Learning*. World Resources Institute: Technical Note. <https://doi.org/10.46830/writn.20.00048>.

Kontgis, C., Kerins, P., and E. Mackres. (2021). *How Land Use Paints a Clearer Picture of Urban Life*. World Resources Institute: Technical Perspective. [How Land Use Paints a Clearer Picture of Urban Life | World Resources Institute \(wri.org\)](#)

World Economic Forum. (2022). *Using Digital Technology for a green and just recovery in cities*. Global Future Council on Cities of Tomorrow. [WEF_C4IR_GFC_on_Cities_Digital_Technology_2022.pdf \(weforum.org\)](#)

4.2 Data, Data Literacy, and Data Governance for Urban Land Restoration

Developed by

Nazmul Huq (TH Köln, Germany)

Lars Ribbe (TH Köln, Germany)

Learning objectives

- ▶ Develop critical thinking about data, including how and why data is generated, collected, and handled;
- ▶ Understand the needs and principles of data governance regarding urban land restoration;
- ▶ Reflect on data protection and ethics and learn to handle data responsibly regarding urban land restoration.

Summary

Part 1: Data and data literacy

- ▶ Data literacy is crucial in today's data-driven world, as it enables individuals and organizations to harness the power of data to drive growth, innovation, and success;
- ▶ Understanding the significance of data and how it can be utilized to solve problems, inform decision-making, and generate knowledge and wisdom is crucial.

Part 2: Sources and types of data

- ▶ Understanding various data types, formats, and sources. This includes knowing where to find relevant data, how it is structured, and how it can be used to address specific business questions;

- ▶ Data-literate individuals approach data with a critical mindset, questioning assumptions, verifying data quality, and considering potential biases. Critical thinking involves asking questions, evaluating data, and making informed decisions based on evidence.

Part 3: Understanding data governance for SULR

- ▶ Emerging technologies (e.g., Internet of Things, Geographic Information Systems, Big Data) are critical for SULR for data and data processing but require robust governance to ensure ethical use, transparency, and accountability;
- ▶ Global frameworks (e.g. General Data Protection Regulation - GDPR) provide guidelines for data and technology governance, but challenges like data privacy, interoperability, and equity remain.

Part 4: Building a data governance structure

- ▶ Data governance ensures high-quality, accessible, and secure data for informed decision-making in SULR;
- ▶ Frameworks help standardize data governance practices;
- ▶ Examples include urban planning projects that leverage data governance to optimize land use and resource allocation;
- ▶ Creating data governance frameworks for cities requires substantial effort in planning, resource management, following best practices and standards, and involving stakeholders;
- ▶ Spatial data governance is more than just data, but how data will create value in urban decision-making while complying to national and international regulations, standards and practices such as data ethics and trust.

Lessons learned

- ▶ Data-literate individuals and organizations are more likely to embrace data-driven innovation;
- ▶ Data collection and handling require planning and care to ensure the quality of analyses and the protection of participants;

- ▶ Data comes from various sources and exists in many forms;
- ▶ Data governance is foundational for sustainable urban land restoration. Without proper governance, data can be misused, inaccessible, or of poor quality, undermining decision-making;
- ▶ Organizations should strive for a higher level of data governance maturity to enable proper handling of data and enable data-driven decision;
- ▶ Developing a data governance structure at the organizational level is a step-by-step iterative process.

Student questions and exercises

Exercise 1:

Think about your city. Your city office is collecting personal health information from citizens to assess the impacts of nearby coal mines. What types of data should the city office collect? How do you think officials treat the personal data collected from the citizens? Do you think it is OK city officials make the responses available to the mayor with the residents names

Exercise 2:

In pairs or small groups, discuss: Who are the different data actors and what are their roles? What rights do you feel data providers (e.g., citizens, research participants) should have in relation to their data, and why? What duties do you feel data collectors and users (e.g., municipalities, businesses, researchers) should have in relation to data and data providers, and why?

References/resources

De Martino, S. A., Stanley, V., and R. Tonchovska. (2012). *Spatial data infrastructure and INSPIRE (English)*. Europe and Central Asia Knowledge Brief (Issue No. 55). World Bank Group. <http://documents.worldbank.org/curated/en/900421468249889891/Spatial-data-infrastructure-and-inspire>

Herweg, K. G., and K. Steiner. (2002). Impact monitoring and assessment. Volume 1: Procedure: Instruments for use in rural development projects with a focus on sustainable land management. Centre for Development and Environment. <https://boris.unibe.ch/85037/1/imavol1en.pdf>

Herweg, K. G., and K. Steiner. (2002). *Impact monitoring and assessment. Volume 2: Toolbox: Instruments for use in rural development projects with a focus on sustainable land management.* Centre for Development and Environment. <https://boris.unibe.ch/85036/1/imavol2en.pdf>

World Bank. (2021). *CPL municipal spatial data infrastructure (MSDI) manual.* World Bank Group. https://collaboration.worldbank.org/content/sites/collaboration-for-development/en/groups/city-planning-labs/documents.entry.html/2021/03/18/_manual_cpl_municipalspatialdatainfrastructure-2Rxa.html

4.3 Data requirements for Urban Land Restoration

Developed by

Nazmul Huq (TH-Köln, Germany)

Arsalan Saghir (TH-Köln, Germany)

Lars Ribbe (TH-Köln, Germany)

Learning objectives

- ▶ A comprehensive understanding of the role of data across different stages of SULR planning and implementation;
- ▶ Various data and data sources used for ULD assessment;
- ▶ Data characteristics, challenges in data acquisition, and strategies for obtaining the right data for effective SULR planning.

Summary

Part 1. A System View of Land Restoration and Data

- ▶ Land degradation occurs through complex interaction between different components of urban systems, *i.e.*, social and ecological systems;
- ▶ Data is critical to understanding the degradation process and status of restoration decision-making and planning;
- ▶ Different types of data are utilized throughout the various stages of the restoration planning cycle, which includes assessment, planning, design, and implementation.

Part 2. Data types and characteristics for Urban Land Restoration Planning Including Assessment

- ▶ Understanding of both spatial and socio-economic data, examples, and application in different stages of ULR planning;
- ▶ Both spatial and socio-economic data have temporal dimensions, such as seasonal variations and urban growth trends, changes of ecosystem services over time;
- ▶ Importance of data quality for SULR and sourcing of good quality data.

Part 3. Challenges for data sourcing and quality

- ▶ Overview of challenges related to data, data quality, and sourcing, including data uncertainty, missing and unavailable data, and limited access to tools or confidential government-declared data;
- ▶ Challenges in data sourcing include limited accessibility, varying quality standards, and high acquisition costs;
- ▶ Ensure data accuracy by using cross-verified datasets, recent updates, advanced tools like AI, stakeholder collaboration, statistical gap analysis, and periodic updates for long-term reliability.

Part 4: Case Study - Application of ITT smartSense for data collection

- ▶ Real-time environmental parameter monitoring from local sites, including Data Collection parameters and sensors, Application of sensors for measuring land degradation Results of Smartsense weather station.

Lessons learned

- ▶ Effective urban land restoration relies on integrating data from societal, environmental, and economic factors to develop comprehensive solutions;
- ▶ High-quality data, including biophysical and socio-economic data with their temporal dimensions, is essential for informed decision-making and strategic planning;
- ▶ Temporal data helps track restoration progress, while spatial data supports zoning and land-use planning;
- ▶ The quality of data is particularly important for informed decision-making, and it requires careful planning to acquire high-quality data;
- ▶ Challenges in data sourcing include limited accessibility, varying quality standards, and high acquisition costs.

Student questions and exercises

- ▶ Students will identify the different data necessary to restore an open solid waste disposal site within a municipality
 - o Students to write down max. 5 types of data they think are necessary
 - o Students will further categorize into key data types
 - o Lecturer will discuss the results with students and enters the lecture contents

References/resources

- Cen, X., Wu, C., Xing, X., Fang, M., Garang, Z., and Y. Wu. (2015). Coupling Intensive Land Use and Landscape Ecological Security for Urban Sustainability: An Integrated Socioeconomic Data and Spatial Metrics Analysis in Hangzhou City. *Sustainability*, 7(2), 1459–1482. <https://doi.org/10.3390/su7021459>
- FAO. (2022). *Global indicators for monitoring ecosystem restoration – A contribution to the UN Decade on Ecosystem Restoration*. Food and Agriculture Organization. <https://openknowledge.fao.org/server/api/core/bitstreams/8c813ba8-377a-4c30-965f-42a62a9dd492/content>
- IPBES. (2018). *The IPBES assessment report on land degradation and restoration*. Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. <https://doi.org/10.5281/zenodo.3237392>
- IUCN. (2014). *A guide to the Restoration Opportunities Assessment Methodology (ROAM): Assessing forest landscape restoration opportunities at the national or sub-national level* (Working Paper – Road-test edition). International Union for Conservation of Nature. <https://portals.iucn.org/library/sites/library/files/documents/2014-030.pdf>
- Kakani, M., Singh, R., Buckingham, K., Faruqi, S., Calmon, M., Ding, H., Duraisami, M., DeWitt, S., Warman, J., Batista, A., Cervera, M., Shelar, K., Soares, D. S., and P. P. F. Costa (2024). *The Restoration Launchpad: A Step-by-Step Guide for Restoration Planners and Practitioners*. World Resources Institute. <https://doi.org/10.46830/wrigb.21.00045>

4.4 Overview of tools and methods for assessing Urban Land Degradation

Developed by

Nazmul Huq (TH-Köln, Germany)

Arsalan Saghir (TH-Köln, Germany)

Lars Ribbe (TH-Köln, Germany)

Learning objectives

- ▶ Understanding the basic concepts and processes of land degradation assessment
- ▶ Understanding and evaluation of major tools of land degradation assessments and visualization Overview of session

Summary

This sub-module introduces key concepts, tools, and methods for evaluating land degradation in urban areas. It begins with an overview of land degradation, covering types such as climatic, soil, and human factors, along with qualitative and quantitative assessment methods. The sub-module then focuses on major tools such as: Global Assessment of Soil Degradation (GLASOD), Global Land Degradation Assessment (GLADA), Land Degradation Assessment in Drylands (LADA), and indices such as NDVI, exploring their application at local scales for urban degradation assessment. It also covers site-scale models like SWAT and USLE for evaluating soil erosion and water management. The sub-module emphasizes the importance of using these tools and methods to guide effective urban land restoration and planning.

Part 1: Introduction to and Overview of Urban Land Degradation Assessment

- ▶ What is Land Degradation Assessment? (LDA):
Introduction to LDA, explaining how it is used to measure and understand the extent of land degradation in urban areas, with real-world examples illustrating its significance;
- ▶ Types of Land Degradation Assessment: Introducing the main types of LDA:
 - i. Land Take and Sealing
 - ii. Soil Erosion and Degradation
 - iii. Pollution and Contamination
 - iv. Physical Degradation
 - v. Urban Vegetation Loss
- ▶ Objective of Land Degradation Assessment: Discussing the critical role LDA plays in informing restoration decisions, highlighting how it helps prioritize actions and allocate resources for effective land management through informed decision-making;
- ▶ Methods of Land Degradation Assessment: Explanation of both qualitative (field surveys, expert judgment) and quantitative (remote sensing, GIS, modeling) methods for assessing urban land degradation. Also mixed-method approaches;
- ▶ Scales of Land Degradation Assessment: Understanding how LDA can be applied at different levels, including global (large-scale trends), regional (intermediate zones), national (policymaking), and local (site-specific) assessments.

Part 2: Introduction to Major Urban Land Degradation Assessment Tools

- ▶ Key Methods for Urban Land Degradation Assessment: Overview of key tools used in urban land degradation assessment, including GLASOD (Global Assessment of Soil Degradation), GLADA (Global Land Degradation Assessment), and LADA (Land Degradation Assessment in Drylands);

- i. Comparative Analysis of Tools: A detailed comparison of these tools in relation to urban land degradation, exploring their advantages, drawbacks, scale, application for ULR and Data Sourcing;
- ▶ Common Steps to Perform LADA Tools for Local Scale Degradation Assessment: Step-by-step guidance on using LADA tools for localized degradation assessment, including data collection, analysis, and actionable recommendations to support urban restoration initiatives;
- ▶ Key Indices for ULD: NDVI (Normalized Difference Vegetation Index).

Part 3: Site-Scale Degradation Assessment Using SWAT/USLE

- ▶ Importance of Site-Scale Frameworks: Aid in mitigation and restoration planning. Helps in making informed decisions regarding mitigation and restoration;
- ▶ Comparative Analysis– SWAT vs. USLE: Detailed comparative analysis between two with respect to model type, application, complexity, output and use case;
- ▶ Introduction to SWAT and USLE: Overview of the SWAT (Soil and Water Assessment Tool) and USLE (Universal Soil Loss Equation) frameworks, both of which are used for assessing soil erosion and degradation at the local site scale;
- i. Application of SWAT/USLE for Site Scale Assessment: Detailed explanation of how these frameworks can be applied at local scales to evaluate degradation, focusing on key components and steps.

Lessons Learned

- ▶ Local-scale assessments help address specific land degradation problems effectively;
- ▶ SWAT is good for detailed studies, while USLE is simple and quick to use;
- ▶ Accurate data is essential for reliable results in both tools;
- ▶ Combining multiple factors like climate, soil, and human impact gives a complete understanding of degradation;

- ▶ Real-world case studies show how these tools work in practice;
- ▶ These tools support better decisions for managing erosion and restoring degraded land.

Student questions and exercises

1. First set of questions:

- a. What are some common causes of urban land degradation? (open ended)
- b. What is the most commonly used technology for large-scale land degradation assessment? (RS and GIS)
- c. Which of the following indicators best represents soil degradation? (Reduced vegetation cover and soil erosion)
- d. Which policy tool is most commonly used to reduce urban land degradation? (Zoning laws and green infrastructure policies (such as green roofs, permeable surfaces, and urban forests))

2. Open discussion with students:

- a. How would you assess the land degradation in a data poor technology poor region. Given that you do not have available data on restoration parameters and you do not have access to technology to measure.
- b. Which approach would you follow?

References/ Resources

- Bagegnehu, B., Alemayehu, M., and W. Nigatu (2019). Geographic Information System (GIS) based soil loss estimation using Universal Soil Loss Equation Model (USLE) for soil conservation planning in Karesa Watershed, Dawuro Zone, South West Ethiopia. *International Journal of Water Resources and Environmental Engineering*, 11(8), 143–158. <https://doi.org/10.5897/IJWREE2018.0820>
- Brabant, P. (2010). *A land degradation assessment and mapping method*. Institut de Recherche pour le Développement. https://horizon.documentation.ird.fr/exl-doc/pleins_textes/divers12-04/010054639.pdf

FAO. (2016). *Land degradation assessment in drylands: Manual for local level assessment of land degradation and sustainable land management. Part 1: Planning and methodological approach, analysis and reporting*. Food and Agriculture Organization of the United Nations. <https://openknowledge.fao.org/server/api/core/bitstreams/2aad7976-f67d-4e14-bdb5-55b71b605a34/content>

IPBES. (2018). *The IPBES assessment report on land degradation and restoration*. Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. <https://doi.org/10.5281/zenodo.3237392>

4.5 Mapping Urban Land Degradation

Developed by

Mónica González Llanos (UC, Territorium Research Group, Ecuador)

Ximena Salazar Guamán (UC, Territorium Research Group, Ecuador)

Fabiana Torres Espinoza (UC, Territorium Research Group, Ecuador)

Cristina Peralta Peñaloza (UC, Territorium Research Group, Ecuador)

Learning objectives

- ▶ Understand how GIS and remote sensing technologies work as complementary tools for identifying, analyzing and monitoring urban land degradation processes;
- ▶ Learn the key stages of urban land degradation mapping as a foundation for restoration planning, including the application of emerging geospatial technologies and GIS-based decision support systems;
- ▶ Explore practical applications of urban land degradation mapping tools through case studies like the Google Earth Engine example from Ecuador, demonstrating real-world implementation of these technologies.

Summary

Part 1. Mapping degraded land

A brief overview of degraded land mapping and its main applications. This section explains how mapping technologies help identify, analyze, and monitor land degradation across various ecosystems, providing essential data for environmental management, urban planning, and restoration efforts.

Part 2. GIS and Remote Sensing for mapping Urban Land Degradation

This part examines the role of Geographic Information Systems (GIS) and remote sensing in degraded land mapping, both as primary monitoring tools and as complementary technologies. It covers the fundamental aspects of GIS and remote sensing, including definitions, characteristics, and operating principles. Additionally, this section provides resources and links for those interested in further training and education in GIS and remote sensing technologies.

Part 3. Mapping Urban Land Degradation processes

Building on the foundational knowledge of GIS and remote sensing, this section outlines the key stages of urban land degradation mapping as an essential preliminary step for urban land restoration. It explores emerging technologies that offer new approaches to geographic mapping and discusses GIS-based support systems as tools for managing urban land degradation and planning future restoration efforts.

Part 4. Urban Land Degradation mapping example

This section presents a practical example of using Google Earth Engine as an emerging tool for mapping one stage of the degradation process in Ecuador. The case study demonstrates the application of concepts covered in previous sections and includes additional features and information on new technologies relevant to degraded land mapping.

Lessons learned

- ▶ Understanding remote sensing and GIS principles provides essential tools for analyzing and visualizing urban land degradation data effectively;
- ▶ Knowledge of emerging technologies and their integration with remote sensing and GIS significantly enhances our ability to study and address urban land degradation;
- ▶ The interrelationship between traditional mapping tools and new technologies creates powerful systems for monitoring degradation processes and planning restoration efforts in urban environments.

Student questions and exercises

Example of using Google Earth Engine as an emerging tool for mapping one stage of the process of mapping degraded land. This exercise contains additional features and useful information about new technologies for mapping. We recommend that you have some basic knowledge of programming concepts.

References/resources

Annappa N.N., Bhavya N., Govinda K., Uday Kumar S.N. and Krishna Murthy R. (2023). Eco-Rehabilitation through Satellite Eyes: Mapping Degraded Land with Remote Sensing and GIS. In *Advanced Innovative Technologies in Agricultural Engineering for Sustainable Agriculture* (Vol. 8, pp. 73-98). AkiNik Publications <https://doi.org/10.22271/ed.book.2464>

Bojórquez-Tapia, L.A., Cruz-Bello, G.M. and L. Luna-Gonzalez. (2013) Connotative Land Degradation Mapping: A Knowledge-Based Approach to Land Degradation Assessment. *Environmental Modelling and Software*, 40, 51-64. <https://doi.org/10.1016/j.envsoft.2012.07.009>

Bosque Sendra, J. (2000) *Sistemas de Información Geográfica* Ed. Rialp, Madrid, 451 pp.

Taboada González, J.A. and J.M. Cotos Yáñez. (2005) *Sistemas de información medioambiental* Ed. Netbiblo

4.6 Urban planning and design tools

Developed by

Simone Sandholz (UNU-EHS)

With contributions from SALAR International

Learning objectives

- ▶ Students will learn about available planning tools and toolboxes aiming at sustainable urban planning, or particular aspects, including in particular nature-based solutions; community-based and participatory approaches that engage urban citizens in planning;
- ▶ They will know which tools/ tool boxes can be used for which planning step(s);
- ▶ They will receive needed information to incorporate considerations of sustainable land restoration/land use into given tools and toolboxes;
- ▶ Students will be able to evaluate planning and design tools regarding the applicability for sustainable land restoration/land use.

Summary

Part 1. Tools and toolboxes for sustainable urban planning and design

Part 1 of the module will start with introducing the wealth of available planning and design tools from many different organizations that can support better and more sustainable urban development. It will structure which are the planning stages covered, what are key goals and underlying frameworks, and how these tools and toolboxes could align with SULR, while there are (still) no tools available on SULR only. Lastly, the introductory part will make a connection to different user groups addressed.

Part 2. Presentation of selected toolboxes (open source) and relating them to SULR

Part 2 will present selected toolboxes to give an exemplary overview of how they can be applied and how and where SULR can be incorporated. This part will focus on open source materials from leading international organisations only, to allow for maximum uptake. 3-5 tool boxes are described, more are presented as a summary, reflecting on the growing number of available materials and that new ones keep emerging. The SymbioCity process is presented in more detail, as the approach aligns with SULR design principles.

Part 3. Thinking out of the box – incorporating participatory and creative tools

The concluding part broadens the scope beyond “classic” urban planning but introduces tools that come from different disciplines, with a focus on participatory and creative methods from arts and culture. The aim is to go beyond technology-driven approaches and tools but focus on context-specific ones. Finally, the sub-module concludes with a summary on key features of planning and design tools to support SULR.

Lessons learned

- ▶ There are multiple tools and toolboxes that aim to support urban planners and decision-makers in implementing the best urban planning / design possible;
- ▶ Urban planning and design tools can include sustainable land restoration land use considerations, for any planning stage;
- ▶ Such tools can also help assessing and mitigating potential negative implications from land restoration;
- ▶ The number of tools and toolboxes keeps growing, there is a benefit in staying informed to find the best-suited tool or toolbox for a given context.

Student questions / exercises

The submodule provides a broader overview on existing planning tools and toolboxes. While the overview is not comprehensive, some open source materials

are presented, based on which students can see how to apply them and how to look for other tools as needed.

It includes a few reflection questions after presenting toolboxes in Part 2, namely:

- ▶ Having heard about tools and toolboxes, where do you see links to SULR?
- ▶ How could they benefit SULR?

Furthermore, there is a 2nd set of reflection questions in Part 3 on creative tools:

- ▶ What could be comparable benefits of creative tools?
- ▶ For which purpose would you use them?

References/resources

Barashkov, J. (2024). Re-commoning Urban Spaces from the Bottom-Up: Empowering Urban Communities: A Digital Toolbox for Bottom-Up Intervention in Kottbusser Tor, Berlin. *Proceedings of the International Conference on Computer-Aided Architectural Design Research in Asia*, 2:241 – 250. <http://dx.doi.org/10.52842/conf.caadria.2024.2.241>

Geropanta, V. and E.M. Cornelio-Mari. (2022). Inclusiveness and Participation in the Design of Public Spaces: Her City and the Challenge of the Post-Pandemic Scenario. *International Journal of E-Planning Research (IJEPR)*, 11(1), 1-15. <https://doi.org/10.4018/IJEPR.309380>

Mueller, J., Asada, S., and L. Tomarchio. (2020). Engaging the Crowd: Lessons for Outreach and Tool Design From a Creative Online Participatory Study. *International Journal of E-Planning Research (IJEPR)*, 9(2), 66-79. <https://doi.org/10.4018/IJEPR.2020040101.0a>

Ramalho, T.M.T., and M.O. Guarneri. (2022). *Leave No Agenda Behind: Participatory Approaches to Supporting Local Governments in Africa to Implement the New Urban Agenda and the SDGs*. In: Croese, S., Parnell, S. (eds) *Localizing the SDGs in African Cities*. Sustainable Development Goals Series. Springer, Cham. https://doi.org/10.1007/978-3-030-95979-1_13

Schuchert, C.L., Schwehr, P., and B. Gabriel. (2022). Urban qualities for dense mixed-use spaces. Theses, case studies and a toolbox for integrated tangible urban planning. *IOP Conference Series: Earth and Environmental Science*, 1078 012086. [Urban qualities for dense mixed-use spaces. Theses, case studies and a toolbox for integrated tangible urban planning.](#) - IOPscience

Links to toolkits:

Linking doughnut economics to urban planning - manual: <https://doughnutecconomics.org/tools/doughnut-for-urban-development-manual-and-tools>

Related canvas by C40: https://www.c40knowledgehub.org/s/article/The-City-Portrait-Canvas-Guide?language=en_US

Our City Plans: An incremental and participatory toolbox for urban planning | UN-Habitat

<https://symbiocity.org/wp-content/uploads/2021/06/SymbioCity-Tools-and-Methods-spreads.pdf>

https://symbiocity.org/wp-content/uploads/2020/01/5403-SKL-Planning-our-Towns-Toolkit-Nov-2019_WEB-spreads-kopia.pdf

<https://unhabitat.org/planning-for-climate-change-toolkit>

4.7 Sustainable Urban Land Restoration decision design tools

Developed by

Nazmul Huq (TH-Köln, Germany)

Lars Ribbe (TH-Köln, Germany)

Sadiya Afrin Mazumder (TH-Köln, Germany)

Learning objectives

- ▶ Understand the importance of decision design in SULR;
- ▶ Evaluate and compare commonly used decision-design tools for urban restoration planning and interventions;
- ▶ Apply tools, such as Multi-Criteria Decision Making (MCDM), to implement sustainable urban land restoration strategies, specifically within the context of cities in developing countries.

Summary

Part 1. Overview of decision-design tools for SULR

Decision design reduces the complexity in the decision-making process in SULR by helping the planners to frame decisions in a clear and informed way by considering multiple economic, societal, and environmental factors and stakeholder involvement.

Part 2. A comparative assessment of different decision-design tools in SULR

Decision-design tools used in SULR include Cost-Benefit Analysis (CBA), Social Cost-Benefit Analysis (SCBA), MCDM, and Participatory Impact Assessment (PIA). Each technique has advantages and disadvantages, such as CBA's inability to address distributional effects, SCBA's difficulty measuring social and environmental implications, MCDM's lack of prescriptive solutions, and PIA's potential lack of general acceptance. Supporting techniques such as GIS methodologies, Social Return on Investment (SROI), etc. give additional metrics for evaluating feasibility, sustainability, and stakeholder engagement in SULR initiatives.

Part 3: Application of decision-design tools for SULR

The example from Lilongwe City, Malawi, shows how to apply MCDM with a detailed workflow, including the process of defining criteria and evaluating different alternatives. It focuses on how the MCDM method can help decision makers select the most suitable and priority locations to implement urban green infrastructure to solve the existing land degradation problems in the context of urban land restoration.

Lessons learned

- ▶ Decision design balances social, economic, and environmental aspects and is vital for effective urban landscape rehabilitation;
- ▶ Objective evaluation and comparison of restoration solutions are made easier by tools such as CBA, MCDM, SCBA, and PIA;
- ▶ MCDM provides a quick and sustainable framework for assessing SULR methods using sustainability indices, which helps decision-makers choose the best solutions for urbanized areas, especially in developing nations.

Student questions and exercises

Exercise 1: Lecturers will ask the questions after the first session of the lecture:

- ▶ Can different decision-design tools be applied to a single decision-making context (yes/no)?
- ▶ Is it necessary to include participatory approaches in every decision-making scenario (yes/no)?

Exercise 2: Group exercise: Imagine a city experiencing significant green space loss, leading to the Urban Heat Island (UHI) effect. The potential alternatives to solve the problem are: Implementation of (i) Urban green infrastructure and reforestation, (ii) reflective & cool surfaces, and (iii) sustainable water management.

Task: Now discuss and identify some key sustainability criteria that could be considered to evaluate and compare multiple alternatives to mitigate the issue.

References/resources

Bakolo, C., Kayitete, L., De Dieu Tuyizere, J., Tomlinson, J., Fawcett, J., and R. F Alfaro. (2024). Identification of optimal locations for green space initiatives through GIS-based multi-criteria analysis and the analytical hierarchy process. *Environmental Systems Research*, 13(1), 46. <https://doi.org/10.1186/s40068-024-00377-0>

Campos, P. C. D. O., Paz, T. D. S. R., Lenz, L., Qiu, Y., Alves, C. N., Simoni, A. P. R., Amorim, J. C. C., Lima, G. B. A., Rangel, M. P., and I. Paz. (2020). Multi-Criteria Decision Method for Sustainable Watercourse Management in Urban Areas. *Sustainability*, 12(16), 6493. <https://doi.org/10.3390/su12166493>

IPBES. (2018). *The IPBES assessment report on land degradation and restoration*. (p. 744 pages). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. <https://www.ipbes.net/assessment-reports/ldr>

OECD. (2018). *Cost-Benefit Analysis and the Environment: Further Developments and Policy Use*. Organisation for Economic Co-operation and Development. <https://doi.org/10.1787/9789264085169-en>

Van Der Jagt, A. P. N., Buijs, A., Dobbs, C., Van Lierop, M., Pauleit, S., Randrup, T. B., and T. Wild. (2023). An action framework for the participatory assessment of nature-based solutions in cities. *Ambio*, 52(1), 54–67. <https://doi.org/10.1007/s13280-022-01772-6>

4.8 Citizen Science and Local Empowerment for Sustainable Urban Land Restoration

Developed by

Dyah Rahmawati Hizbaron (UGM, Indonesia)

Arry Retnowati (UGM, Indonesia)

Utia Suarma (UGM, Indonesia)

Learning objectives

- ▶ Understand the concept of Citizen Science (CS) for SULR;
- ▶ Engage the fundamental concepts of CS and local empowerment into SULR programs;

- ▶ Analyze the importance of CS and local empowerment and participation using drones, alarms, sensors, etc. for SULR programs;
- ▶ Evaluate the use of CS drones, CS alarms, CS sensors, CS non-digital data, etc. for digital twins of cities.

Summary

Part 1: Introduction to CS for SULR

CS plays a pivotal role in SULR, offering innovative solutions to bridge data gaps and actively engage the community in long-term restoration efforts. In the planning phase there are classical processes which can be done repetitively, such as assessment, planning, implementation and operationalization. The cycle of planning goes traditionally from identifying community problems, collecting information, analyzing information, forming community goals, and developing alternative plans, comparing alternative plans and selecting preferred plans to be implemented and monitored. Here, CS is an alternative mechanism which offers collecting information, analyzing information and developing alternative plans from the collected data by creating engagement from community level. Detailed historical data often are missing from the records, (e.g. detailed effects of past flood events). Other types of data such as perception and level of understanding on flood hazard are also often absent. Filling this gap is the domain of CS.

Topics covered, include:

- ▶ CS fundamentals;
- ▶ The importance of CS for SULR;
- ▶ Problem identification leading to CS for SULR;
- ▶ Mapping data (data availability and gaps);
- ▶ Example of data gaps: time and scale of observation.

Part 2: Roles of Citizen Science for drone monitoring of SULR

CS can also be applied to innovation technology, which can be fully implemented and monitored through local stakeholder involvement. CS

generated data can support situation analyses of land degradation or land restoration programs in urban areas. Drones can be applied as a reconnaissance tool through CS to assess flood hazards in small areas, however stakeholder involvement may be limited.

Topics covered, include:

- ▶ CS drones programs (*i.e.*, problem analysis, stakeholders, tools, timeline, data analytic processes, data storage mechanisms, identifying solutions);
- ▶ The use of CS drones (e.g., mapping SULR, Monitoring and Evaluation of future urban conditions)

Part 3 : Digital Twins of Cities

Citizen-centric digital twins (CCDT) place citizens as the central element of the system and integrate all other technologies within it.

- ▶ The use of big data from CS drones, CS sensors, CS alarms and other devices for future modelling;

Part 4 : Digital Twins of Cities: Example from Indonesia

Lessons learned

- ▶ There are broad definitions of CS depending on the context, however it always includes generation of scientific data, enlists volunteers, and undertakes a politically relevant issue, often filling data gaps at local and detailed scales;
- ▶ Pursuing urban land restoration requires many options of methods and techniques. Among those, there are numerous gaps and challenges which can be tackled through CS. As a way of closing data gaps, CS offers long term solutions and leverages networks for greater citizen engagement in SULR programs;
- ▶ Problem identification is a crucial initial step in CS for SULR, where community members actively participate in recognizing and addressing environmental issues in urban areas;

- ▶ There are a variety of methods on doing CS in SULR and methods are varied according to the existing initiatives, scale of problems and availability of resources;
- ▶ Digital twins are an innovative step in undertaking CS for SULR.

Student questions and exercises

1. Explain the understanding of citizen science in the context of urban scale.
2. Which or what scale do you think local empowerment can be applied in urban restoration?
3. Find an example of digital twin implementation and the correlation with the SULR.

See Case Study in Section 5.9 New Zealand/ Aotearo: Indigenous Knowledge for Sustainable Urban Land Restoration

References/resources

Abdeen, F. N., Shirowzhan, S., and S. M. E. Sepasgozar. (2023). Citizen-centric digital twin development with machine learning and interfaces for maintaining urban infrastructure. *Telematics and Informatics*, 84 (102302), 2023. <https://doi.org/10.1016/j.tele.2023.102302>

Dodman D., Diep L., and S. Colenbrander. (2017). *Resilience and Resource Efficiency in Cities*. UNEP. <https://www.unep.org/resources/report/resilience-and-resource-efficiency-cities>

Marfai, M. A., and D.R. Hizbaron. (2011). Community's adaptive capacity due to coastal flooding in Semarang coastal city, Indonesia. *International Journal of Seria Geografie. Annals of the University of Oradea*, 21(2), 2011. https://istgeorelint.uoradea.ro/Reviste/Anale/Art/2011-2/6_536_AUOG_Marfai.pdf

Saputra, A., Sigit, A. A., Priyana, Y., Abror, A. M., Lia Sari, A. N., and O. Nursetiyani. (2022). A Low-Cost Drone Mapping and Simple Participatory GIS to Support the Urban Flood Modelling. *Geographia Technica*, 17(2). 10.21163/GT_2022.172.04

Vohland, K., Land-Zandstra, A., Ceccaroni, L., Lemmens, R., Perelló, J., Ponti, M., ... and K. Wagenknecht. (2021). *The science of citizen science*. Springer Nature. <https://doi.org/10.1007/978-3-030-58278-4>

4.9 Monitoring and Evaluation in Sustainable Urban Land Restoration

Developed by

Nazmul Huq (TH-Köln, Germany)

Sadiya Afrin Mazumder (TH-Köln, Germany)

Learning objectives

- ▶ Understanding the key concepts of monitoring and evaluation (M&E) for SULR
- ▶ Understanding the application of tools and data for SULR intervention monitoring

Summary

Part 1: Introduction to M&E and Key Concepts

- ▶ M&E is critical for SULR, ensuring that projects achieve their goals. They enable adaptive management and evidence-based decision-making in urban restoration;
- ▶ Key concepts of M&E: Successful monitoring relies on fundamental concepts such as baseline assessment, which establishes initial conditions for comparison, and indicators and metrics, which provide measurable parameters for tracking progress;

- ▶ Different scales of monitoring—spatial and temporal—define how data is collected and analyzed over time. Theory of Change (ToC) plays a crucial role by mapping out the expected sequence of events in restoration efforts, linking interventions to measurable outcomes and long-term impact.

Part 2: Key steps of M&E

- ▶ Presents key steps of M&E, including tools, techniques, and sources: (1) Defining priority areas of interest, (2) Setting measurable indicators, (3) Establishing baseline conditions, (4) Implementing data collection and analysis methods, and (5) Reporting findings for decision-making.

Part 3: Examples of key global urban monitoring frameworks, indicators, and data

- ▶ Introduction to major global urban monitoring frameworks, e.g., UN-Habitat's City Prosperity Index, World Bank's Urban Sustainability Framework, SDG framework, and city biodiversity index, including indicators and data sources.
- ▶ SDG urban monitoring framework
- ▶ UN-Habitat's City Prosperity Index
- ▶ IUCN Urban Nature Index
- ▶ City biodiversity index
- ▶ Examples of global urban monitoring indicators

Lessons learned

- ▶ M&E enables systematic tracking of restoration projects to measure progress, effectiveness, and impact;
- ▶ ToC helps define monitoring needs at different stages;
- ▶ Global frameworks, indicators, and data sets enable urban managers to perform monitoring at different spatial and temporal scales.

Student questions and exercises

- ▶ Develop a project monitoring indicator framework for a local-level restoration project. Develop indicators for impact, outcome, output, and activities level based on provided ToC

References/resources

Chan, L., Hillel, O., Werner, P., Holman, N., Coetzee, I., Galt, R., and T. Elmqvist. (2021). *Handbook on the Singapore Index on Cities' Biodiversity (also known as the City Biodiversity Index)*. Secretariat of the Convention on Biological Diversity & National Parks Board, Singapore. <https://www.cbd.int/doc/publications/cbd-ts-98-en.pdf>

FAO and WRI. (2019). *The road to restoration: A guide to identifying priorities and indicators for monitoring forest and landscape restoration*. Food and Agriculture Organization of the United Nations & World Resources Institute. <https://openknowledge.fao.org/server/api/core/bitstreams/fda6a9d2-73c0-4bb8-a4b5-fd1e7cc6cc07/content>

GIZ. (2023). *Tools and applications for ecosystem restoration monitoring*. Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH. <https://www.giz.de/de/downloads/giz2023-en-Tools-and-Applications-for-Ecosystem-Restoration-Monitoring.pdf>

IUCN. (2014). *A guide to the restoration opportunities assessment methodology (ROAM): Assessing forest landscape restoration opportunities at the national or sub-national level (Working Paper – Road-test edition)*. International Union for Conservation of Nature. <https://portals.iucn.org/library/sites/library/files/documents/2014-030.pdf>

UN-Habitat. (2024). *Implementation of resolutions adopted by the United Nations Habitat Assembly: Report of the Executive Director*. United Nations Human Settlements Programme. https://unhabitat.org/sites/default/files/2024/04/english_6.pdf

4.10 Assess, implement and monitor a Sustainable Urban Land Restoration intervention (Exercise)

Developed by

Alejandra Ramos (UNU-EHS)

Leticia Domingos Vellozo (UNU-EHS)

Simone Sandholz (UNU-EHS)

Learning objectives

- ▶ Identify main tools and methods needed to execute a SULR intervention;
- ▶ Apply theoretical knowledge to localized urban land degradation and restoration challenges.

Summary

Part 1 Select tools for assessment and develop a planning strategy

(1) Choose an assessment approach by identifying urban land degradation issues and (2) Select tools such as GIS mapping. Then (3) Identify key stakeholders, including the public and private sectors, and civil society. Next, (4) Determine financial resources needed, considering possible funding sources. The spatial and environmental challenges must be (5) Analyzed through mapping land degradation hotspots and examining land-use conflicts. With this information, (6) Draft a general diagnostic report, and according to it (7) Develop a SULR strategy by defining objectives, SULR specific actions, time frames, and phases based on SULR principles. Define the vision.

Part 2 Define how to implement, monitor and consider changes

(1) Define an execution plan by outlining requirements and resources needed, governance frameworks necessary for implementation, followed

by establishing (2) A monitoring and adaptive management system. This involves setting performance indicators (e.g., vegetation cover, water quality, community engagement) and (3) Tracking progress using methods like periodic assessments, citizen science initiatives, and sensor-based monitoring. Consider contingency plans to address potential disruptions such as policy shifts, funding limitations, or climate-related challenges.

Part 3 Adapt the theory of change

(1) Evaluate initial assumptions by comparing the Theory of Change developed in 2.8 and already adapted in 3.10 with new data and observations, then (2) Modify the SULR intervention plan by adjusting project phases, priorities, and tools based on evolving challenges. Finally, (3) Create a revised Theory of Change flowchart to illustrate updated goals, interventions, and expected outcomes, ensuring alignment with the evolving urban restoration project.

Part 4 Create a visualization of expected results

(1) Develop a drawing or collage that visually represents expected restoration results, using images, maps, blueprints, and concept sketches to highlight before-and-after comparisons of degraded versus restored urban spaces. (2) Reflect on desired outcomes, identifying key improvements such as enhanced public spaces, increased biodiversity, and social equity advancements, ultimately demonstrating how the intervention contributes to long-term urban resilience and sustainability.

Lessons learned

- ▶ Effective restoration planning integrates environmental, social, economic, and governance factors;
- ▶ Collaboration among urban planners, landscape architects, ecologists and policymakers is essential;
- ▶ Monitoring and adaptability are key to success. Restoration is not a linear process—continuous assessment and flexibility in implementation are crucial. Real-time data collection and community feedback help in adjusting the strategy when needed;

- ▶ Stakeholder engagement and financial resources define feasibility. Without involvement and secured funding, urban restoration projects risk failure. Ensuring participation from diverse actors and securing multiple funding sources strengthens project viability;
- ▶ Visualization and strategic planning enhance restoration outcomes. A well-defined theory of change and visual representation of goals help align stakeholders, guide implementation and measure progress toward SULR.
- ▶ The purpose is to assess urban land degradation, plan and implement a restoration strategy, and adapt it based on proposed ongoing monitoring. Integrate selected tools, financial planning, and visualization of expected outcomes.

References/resources

Doyle, L. (2021). *A theory of change for complex systems. Change & complexity: vector theory of change*. The Cynefin Centre. <https://thecynefin.co/library/change-complexity-vector-theory-of-change/>

Teaching notes

- ▶ Students can carry out the exercise for a specific urban polygon or for the city scale depending on time and resources available;



Module 5. Catalyzing Sustainable Urban Land Restoration through Innovation

Description

This module explores innovation as a driver of SULR, focusing on creating, modifying, or introducing improved initiatives or processes aligned with urban sustainability targets. Students will learn about ecopreneurship and other forms of innovation, and how to develop related mindsets. They will be introduced to the SULR Innovation Model Canvas, (inspired by the NbS Business Model Canvas), a 10-step process that integrates economic, environmental, social, and cultural dimensions into SULR strategies. Key steps include identifying value propositions, beneficiaries, activities, and resources, as well as cost structures and value capture strategies.

The module also emphasizes the relevance of effective governance and communication for operationalizing SULR innovation, with cross-cutting governance facilitating enabling environments and strategic communication ensuring stakeholder engagement. The module concludes by illustrating opportunities for scaling SULR innovations and presenting good practices, emphasizing partnerships among public, private, and locally led initiatives.

Core competencies

Academic knowledge:

- ▶ Conceptualization of innovation, introducing different types and criteria;
- ▶ Ecopreneurship principles and the need for mindset shifts towards more sustainable urban futures;
- ▶ The SULR Innovation Model Canvas aims to brainstorm, expand, and refine ideas, moving beyond business as usual mindsets;
- ▶ Value proposition aspects, creation, criterion and capture, and the main concepts in the model.

Skills:

- ▶ Understand the role of innovation, ecopreneurship and start-up mindsets in SULR;
- ▶ Analyze good SULR practices and what are key success factors;
- ▶ Mainstream and expand SULR initiatives through collaboration;
- ▶ Apply the 10-steps SULR Innovation Model Canvas to create sustainable and scalable urban land restoration solutions;
- ▶ Developing and tailoring communication formats for different target audiences, create a compelling elevator pitch and design outreach strategies to communicate SULR innovation ideas;

Attitudes:

- ▶ Integrity and authenticity: Commitment and ability to act with sincerity, honesty and integrity when using the SULR Innovation Model Canvas;
- ▶ Co-creation: Motivation to build, develop and facilitate collaborative relationships to accelerate SULR.

Authors

Module coordination: UNU-EHS

Developers:

(UNU-EHS)

Dr. Andrea Ramírez Agudelo
Alejandra Ramos Gálvez (MSc)
Leticia Domingos Vellozo (MSc)

(ASU)

Dr. Aya Elsisy
Prof. Dr. Marwa Khalifa
(FAO Green Cities Initiative)
Federica di Cagno (MSc)
Simone Borelli (MSc)

(KU)

Christine Majale (MSc)
Dr. Peter K. Kamau
(SALAR International)
Helena Ohlsson (MSc)

5.1 Innovation in Sustainable Urban Land Restoration

Developed by

Andrea Ramírez-Agudelo (UNU-EHS)

Alejandra Ramos Gálvez (UNU-EHS)

Learning objectives

- ▶ Define innovation in the context SULR;
- ▶ Recognize the different types and criteria of innovations related to SULR;
- ▶ Identify the relevance of the innovation process and key methodologies.

Summary

Part 1. Innovation in SULR

This first part, ‘Innovation in SULR,’ introduces the concept of innovation in the context of SULR. It begins by contextualizing innovation as a critical tool for addressing urban land degradation caused by global and local pressures such as urban sprawl, land resource depletion, and ecosystem fragmentation. Participants explore the definition of innovation as the creation, modification, or introduction of new or improved products, processes, or services that align with urban sustainability targets. This part examines the types of innovations relevant to SULR, including incremental, disruptive, radical, and architectural innovations. It also includes the dimensions of innovations, such as social, ecological, and technological. Lastly, it outlines the criteria for innovations relevant to SULR, focusing on adaptability, community-centered, purpose-driven solutions, sustainability, and scalability. By understanding these fundamentals, participants gain a foundation for identifying innovative approaches tailored to urban restoration challenges.

Part 2. Innovation Ecosystems and Collaboration

This second part, ‘Innovation Ecosystems and Collaboration,’ emphasizes the relevance of multi-stakeholder dynamics and collaboration in fostering enabling environments for successful SULR innovations. Participants learn about the principles of SULR innovation, including value propositions aligned with urban sustainability aims and systemic solutions that generate synergetic benefits for cities and communities. This section highlights the role of urban innovation ecosystems in enabling tailored solutions that address context-specific needs while fostering the engagement of diverse stakeholders such as the public sector, private sector, local communities, research and development and academic institutions, and Non-Governmental Organizations. SULR innovations present essential practices, such as financing mechanisms, collaborative approaches, knowledge sharing, and feedback routines for an iterative development process and continuous improvement.

Part 3. Ecopreneurship and Start-up Mindsets

The third part, ‘Ecopreneurship and Start-up Mindsets,’ introduces participants to ecopreneurship principles and their application in developing sustainable urban land restoration initiatives. Participants learn about the SULR start-up mindset, which emphasizes the SULR principles of adaptability, community-centered (co-creating with stakeholders), and purpose-driven innovation to address complex urban challenges. It explores business-driven and value-driven approaches that balance economic viability with environmental sustainability. This section also highlights the importance of a mindset shift beyond business-as-usual approaches to foster transformative solutions that align with long-term sustainability goals.

Part 4. Ideation for SULR Innovation

This fourth part, ‘Ideation for SULR Innovation,’ provides practical guidance to participants through the initial ideation process for developing SULR innovative solutions. It focuses on creative thinking techniques and structured approaches to generate and refine ideas that can help address urban land restoration challenges. Participants learn the role of framing specific problems and applying ideation techniques, such as brainstorming and biomimicry, and develop concepts integrating social, ecological, and technological aspects. The process includes evaluating ideas using criteria like feasibility and impact,

refining concepts based on feedback, and preparing the basic drafts for compelling pitches for stakeholders and potential funders.

Lessons learned

- ▶ Innovation as a Strategic Approach: SULR requires innovative approaches that integrate social, ecological, and technological dimensions to address urban land degradation effectively. This approach involves moving beyond traditional solutions to create systemic, synergistic solutions for value-driven changes that address land degradation and enhance urban sustainability;
- ▶ Collaboration and Contextualization: For an idea to become a successful SULR innovation, it must result from multi-stakeholder collaborations within urban innovation ecosystems. Tailoring solutions to specific urban challenges while considering local beneficiaries' needs ensures these initiatives' effectiveness and long-term success;
- ▶ Adaptive Mindset and Continuous Improvement: Adopting an 'ecopreneurial' mindset, characterized by adaptability, co-creation of SULR ideas for continuous learning and iteration helps refine ideas and ensure relevance and effectiveness in dynamic urban contexts.

Student questions and exercises

SULR Innovation Concept (And, But, Therefore Framework - ABT)

Exercise Instructions:

- ▶ Read the ABT framework template carefully;
- ▶ Fill in the blanks with your ideas, ensuring you address all components listed, including Innovation Definition, Innovation Type, SULR Criteria, Problem Framing, Proposed Solution, Potential Impact, Balance, and Target Audience.

References/resources

Cowie, A. L., Orr, B. J., Castillo Sanchez, V. M., Chasek, P., Crossman, N. D., Erlewein, A., ... and S. Welton. (2018). *Scientific conceptual framework for land degradation neutrality: A report of the Science-Policy Interface*. <https://doi.org/10.1016/j.envsci.2017.10.011>

FAO. (2024). *Restoration of degraded lands through agroforestry and innovative approaches*. <https://www.fao.org/3/cd1211en/online/src/html/restoration-degraded-lands-agroforestry.html>

International Resource Panel (IRP). (2019). *Land Restoration for Achieving the Sustainable Development Goals: An International Resource Panel Think Piece*. Herrick, J.E., Abrahamse, T., Abhilash, P.C., Ali, S.H., Alvarez-Torres, P., Barau, A.S., Branquinho, C., Chhatre, A., Chotte, J.L., Cowie, A.L., Davis, K.F., Edrisi, S.A., Fennessy, M.S., Fletcher, S., Flores-Díaz, A.C., Franco, I.B., Ganguli, A.C., Speranza, C.I., Kamar, M.J., Kaudia, A.A., Kimiti, D.W., Luz, A.C., Matos, P., Metternicht, G., Neff, J., Nunes, A., Olaniyi, A.O., Pinho, P., Primmer, E., Quandt, A., Sarkar, P., Scherr, S.J., Singh, A., Sudoi, V., von Maltitz, G.P., Wertz, L., and G. Zeleke. A think piece of the International Resource Panel. United Nations Environment Programme, Nairobi, Kenya <https://www.resourcepanel.org/reports/land-restoration-achieving-sustainable-development-goals>

UNU-EHS. (2024). *5 highlights of sustainable urban land restoration*. <https://unu.edu/ehs/article/5-highlights-sustainable-urban-land-restoration>

Urban Innovative Actions (UIA). (n.d.). *Sustainable use of land and nature-based solutions: A new approach to urban challenges*. <https://www.uia-initiative.eu/en/sustainable-use-land-nature-based-solutions>

5.2 Developing your Sustainable Urban Land Restoration Innovation Model Canvas

Developed by

Andrea Ramírez-Agudelo (UNU-EHS)

Learning objectives

- Understand a comprehensive SULR Innovation Model Canvas following a 10-step process;
- Identify the potential of an innovative idea, applying the SULR Innovation Model Canvas to ensure a solid foundation for successful implementation.

Summary

Part 1. SULR Innovation Model Canvas (S1)

This first part of the SULR Innovation Model Canvas introduces the model for developing innovation models, facilitating SULR implementation with a 10-step process. This part covers Step 1, Value Propositions. This step highlights the need to identify the beneficiaries' needs (context-specific), generate value for urban inhabitants (people), and ensure the environmental quality of the cities (planet). Submodule 5.3 presents a more detailed explanation of this component.

Part 2. SULR Innovation Model Canvas (S2-6)

This second part of the SULR Innovation Model Canvas covers Step 2, Key Activities; Step 3, Key Resources; Step 4, Key Partners; Step 5, Key Beneficiaries; and Step 6, Cross-cutting Governance. These steps detail the elements necessary to successfully operationalize SULR, including effective project management for developing, prototyping, testing, and evaluating SULR innovation ideas to address the identified beneficiaries' needs while supporting community development. In addition, cross-cutting governance facilitates an

enabling environment for innovation management. Submodule 5.4 presents a more detailed explanation of these components.

Part 3. SULR Innovation Model (S7-9)

This third part of the SULR Innovation Model Canvas complements the model by explaining Step 7, Cost Structure; Step 8, Cost Reduction; and Step 9, Capturing Value. This step emphasizes the feasibility of SULR by avoiding, reducing, and reversing degraded urban landscapes, identifying strategies for cost reduction and capturing value from SULR initiatives to contribute to LDN. Submodule 5.5 presents a more detailed explanation of these components.

Part 4. SULR Innovation Model (S10)

This fourth part of the SULR Innovation Model Canvas introduces Step 10: Communication, which emphasizes the role of strategic communication for meaningful engagement with SULR initiatives and feedback. Effective communication is essential to engage key partners, convey benefits, and mobilize resources for impactful relationships, visibility, trust and credibility. Step 10 facilitates the implementation and impact of SULR innovations for avoiding, reducing, and reversing degraded urban land by implementing feedback mechanisms that ensure iterative development and continuous improvement for long-term viability. Submodule 5.6 presents a more detailed explanation of this component.

Lessons learned

- Comprehensive perspective: The SULR Innovation Model Canvas integrates economic, social, and cultural dimensions to align social-ecological-technological innovation goals;
- Iterative Development: By examining the various elements, participants learned about the SULR innovation model to develop and evaluate their SULR innovation ideas for benefits beyond a business-as-usual mindset;
- Strategic Communication: Effective communication is essential for engaging stakeholders and ensuring the long-term success of SULR initiatives.

Student questions and exercises

Exercise Instructions:

1. Mark “Y” for each component in the “Basic Idea Present?” column.
2. Count the total number of “Yes” entries.
3. Record the totals at the bottom of the table.

Readiness Assessment:

0-3 Yes: Early innovation idea stage.

- ▶ You’ve identified some initial components of your SULR innovation. Keep brainstorming!

4-6 Yes: Growing innovation idea potential.

- ▶ You’ve identified some basic elements of your SULR innovation. Keep expanding!

7-10 Yes: Good innovation idea progress.

- ▶ You’ve identified the core elements of your SULR innovation. Keep refining!

References/resources

Bocken, N. (forthcoming). *Business models for sustainability*. Oxford Research Encyclopedia of Environmental Science. Retrieved from SSRN: <https://ssrn.com/abstract=4731205> or <http://dx.doi.org/10.2139/ssrn.4731205>

Connecting Nature. (n.d.). *Nature-based economy policy*. <https://connectingnature.eu/sites/default/files/images/inline/Nature%20based%20Economy%20policy%20doc.pdf>

Innovate Now. (2025). *10-step innovation process*. <https://innovatenow.app/10-step-innovation-process/>

Sustainable Business Canvas. (n.d.). *Sustainable business canvas: A tool for designing sustainable business models*. <https://www.sustainablebusinesscanvas.org/>

Threebility. (n.d.). *The sustainable business model canvas: A common language for sustainable innovation*. <https://www.threebility.com/post/the-sustainable-business-model-canvas-a-common-language-for-sustainable-innovation>

5.3 Sustainable Urban Land Restoration Value Proposition

Developed by

Marwa Khalifa (ASU, Egypt)

Aya Elsisy (ASU, Egypt)

Learning objectives

- ▶ Understand the relation between the different aspects of value creation, criteria, and capture;
- ▶ Analyze potential sources for value creation in SULR projects;
- ▶ Evaluate the trade-offs between different value propositions in SULR projects.

Summary

Introduction: Understanding Value

This part explains the difference between human moral values (the self-conceptions of righteousness) and attribute values (the worthiness or favorable qualities in things). It reflects on how we value things (attach worthiness to them) through our own values. The value of things transcends their monetary worth, as some of them are attached to self-actualization, self-preservation, and/or self-worth. This understanding aligns with the dual nature of NbS values that differentiates between monetary values (like direct revenue generation) and non-monetary ones (like community cohesion)

Part 1: The Relation between SULR Value 3Cs (Creation, Criterion, and Capture)

This part explains the relationship between the rationale of value creation in SULR projects, its evaluation criteria and mobilization of its capture instruments. This part also discusses the difference between essential values (that have independent existence on human existence) versus constructed values (that have dependent existence on human existence); tangible values (which could be measured with quantitative methods) versus intangible values (which could be measured with qualitative methods); as well as monetary versus non-monetary values.

Part 2: SULR Innovation Value Propositions

This part focuses on the different types of value propositions (e.g. environmental, social, and economic). It reflects on how the SULR Innovation Model Canvas (inspired by the NbS Business Model Canvas) has broadened the spectrum of value propositions, leading to adopting a value case rationale instead of the commonly known business-case. This part concludes with the trade-offs between different value propositions and the key questions for developing ethical value propositions for SULR projects.

Part Three: Application

This part allows students to analyze a real case of land degradation and estimate different values to be persevered, captured, and/or created by an SULR project.

Lessons learned

- ▶ The rationale of SULR value creation, criterion, and capture;
- ▶ The trade-offs between different value propositions;
- ▶ Context-based value proposition for developing SULR projects.

Teaching notes/Students questions and exercises

Part 1: The Relation between SULR Value 3Cs (Creation, Criterion, and Capture)

Student Discussion (1) : Is the land value essential or constructed?

The instructor could guide the students to understanding that there are different components within land value. Some of them are essential and some are constructed, depending on the community perception and dominating discourses within a specific context. For example, the proximity to natural resources could be considered as an essential value in a specific land plot as it is challenging to recreate these natural resources while the proximity to efficient markets could be considered as a constructed value because those activities could be recreated.

Student Discussion (2): How to evaluate the increase/decrease of social cohesion without monetizing this social value?

The goal of this discussion is to understand how intangible components of land value could be identified and assessed. The instructor could guide the students towards different means of qualitative methods that could be used to evaluate non-quantifiable costs and benefits. The instructor could choose any other intangible social value for this exercise.

Part 2: SULR Innovation Value Propositions

Student Activity: What are the trade-offs between different value propositions?

This exercise discusses the trade-offs between different value propositions. This activity reflects on the ethical considerations for understanding how value propositions should be framed, what are the key questions and steps for this process, and who is responsible for addressing each of these concerns to ensure transparent and inclusive SULR.

Steps:

- ▶ Ask students to identify 3 potential environmental value propositions for increasing the tree cover in a densely populated low-income urban area;
- ▶ Ask them to list 3 potential negative impacts on each of the social and economic value propositions for increasing the tree cover in a certain area;
- ▶ Reflect on the similar and different themes in their work.

Part Three: Application

Student Activity: Application Case Study – El-Warraq Island, Egypt

The students are invited to reflect on a real-life case of ULD. Instructors could choose a case well-known within their context or use the case of El-Warraq Island (in Cairo, Egypt). The goal of this activity is to allow students to analyze contextual potentials and challenges, and identify possible value propositions for an SULR project.

Steps:

- ▶ Explain the case study for the students and then separate them into working groups.
- ▶ Ask them in stages to do the following:
 1. Identify some of the essential & constructed values on the island.
 2. Identify potential value propositions for an SULR project on this Island.
 3. Identify potential tradeoffs between different value propositions.
- ▶ Reflect on the students' work, stressing on the key aspects of developing a sustainable value proposition for an SULR project.

References/resources

Elder-Vass, D. (2022). *Inventing value: The social construction of monetary worth*. Cambridge University Press.

Elsisy, A. M. H. (2024). *Understanding Land Value through Investigating Redevelopment of Informal Areas in Egypt* (Doctoral dissertation, University of Sheffield). <https://etheses.whiterose.ac.uk/id/eprint/35364/>

Faivre, N., Fritz, M., Freitas, T., De Boissezon, B., and S. Vandewoestijne. (2017). Nature-Based Solutions in the EU: Innovating with nature to address social, economic and environmental challenges. *Environmental research*, 159, 509-518. <https://doi.org/10.1016/j.envres.2017.08.032>

Konijnendijk, C., Di Cagno, F., Borelli, S., and T. Wild. (2024). *Capturing the Values and Making the Business Case for Nature-Based Solutions – A step-by-step Guide*. Deliverable 5.3, H2020 CONEXUS project. <https://networknature.eu/product/31880>

McQuaid, S. (2019) *Nature-Based Solutions Business Model Canvas Guidebook*, Dublin: Trinity College <https://connectingnature.eu/sites/default/files/downloads/NBC-BMC-Booklet-Final-%28for-circulation%29.pdf>

5.4 Operationalizing Sustainable Urban Land Restoration value proposition

Developed by

Federica Di Cagno & Simone Borelli (FAO Green Cities Initiative)

Learning objectives

- ▶ Identify key activities and necessary resources for implementing nature-based SULR projects;
- ▶ Define the roles of key partners and beneficiaries, including co-beneficiaries and potential trade-offs;
- ▶ Explore governance models for SULR projects and collaborative frameworks;

- Problem-based learning through scenario-based exercise.

Summary

Part 1. Identifying and optimizing key activities and resources for SULR projects

This section introduces the SULR Innovation Model Canvas focusing on the essential activities and resources needed for SULR projects so that they create and deliver economic and non-economic values. Accessing and mobilizing resources is vital for SULR projects, as they determine the feasibility and scalability of NBS interventions. A key aspect of resource management is the ability to leverage existing assets while integrating new ones to optimize project outcomes. The urban green loop serves as a nature-inspired model for optimizing resources by repurposing by-products from one component of the project as an input for another. By linking key activities and resources within a circular system, SULR nature-based projects can maximize their multifunctionality and, therefore, land use efficiency.

Part 2. Key partners, (direct and indirect) beneficiaries and trade-offs

SULR projects usually involve multiple stakeholders, which can be categorized into key partners, beneficiaries, co-beneficiaries, and those who may experience unintended negative consequences (trade-off groups). Identifying potential trade-offs and engaging stakeholders in decision-making is crucial for minimizing these negative impacts and ensuring the ULR initiatives deliver the broadest possible set of benefits. In fact, understanding the distribution of benefits is essential, especially when conflicts over land use occur.

Part 3. Governance

Governance structures define how decisions over SULR projects are made, who holds responsibility, and how resources are allocated. This section presents different governance models and instruments, exploring their role in sustainability and adaptability of SULR projects. Quito's example highlights the challenges and opportunities in governance for SULR projects, illustrating the potential misalignment between formal and informal governance.

Part 4. Exercise

A scenario-based exercise is proposed to apply concepts learned in the previous parts. The exercise concludes with group presentations and different ideas/solutions are discussed.

Lessons learned

- Identifying and optimizing activities and resources is essential for the effective implementation and economic sustainability of the SULR/NBS project/initiative;
- Partnerships with key stakeholders and a clear understanding of beneficiaries and potential trade-offs ensures the full realization of NBS values;
- Governance plays a crucial role in SULR project sustainability, influencing decision-making and resource allocation;
- A combination of regulatory, economic and awareness raising instruments can support project sustainability.

Student questions and exercises

Part 3. Case study - Formal and informal governance duality in peri-urban Quito, Ecuador

The case highlights that informal systems (community engagement, participatory design) foster innovation and community participation but need formal policy and regulatory alignment for impact

Questions for students:

1. How could the formal barriers (funding gap, outdated tree-planting bans) have been anticipated or mitigated?

- ▶ Pre-project agreements with the municipality (e.g. Memorandum of understanding), early legal/policy analysis during project planning phase, alternative funding sources (e.g. NGOs, private donors, crowdfunding, etc.).

2. How can local governments better support adaptive policies for SULR?

- ▶ Institutionalize community feedback in policymaking and update planning regulations for climate/ecological needs
- ▶ Introduce experimental zoning or pilot-friendly frameworks for SULR projects

Part 4: Exercise: Identifying partners and beneficiaries for a rain garden in a community park

Invite students to work in groups to identify key partners, beneficiaries (and proposed governance models) needed for implementing a rain garden as mandated by the local council or imagine a similar scenario in your own city. Building on lessons from Quito, where governance barriers impacted project implementation, consider how governance and partnerships can strengthen this initiative.

1. Identifying partners: (i) Which organizations, businesses, or institutions could provide funding (where municipal funding gap exist), technical expertise, or resources? (ii) Could local schools, universities, or environmental NGOs contribute to research or maintenance?

2. Identifying beneficiaries: (i) Who will benefit from improved stormwater management and park usability? (ii) Are there indirect beneficiaries, such as businesses near the park that might experience increased foot traffic?

3. Governance considerations: (i) Should maintenance be community-led, municipal-managed, or a hybrid approach? (ii) How could different governance models (public-led, co-governance, community-led) impact long-term sustainability? (iii) What trade-offs exist between formal governance structures (institutional oversight) and informal governance (community participation)?

References/resources

Egusquiza, S., Cortese, F., and M. Perfido. (2019). Polycentric Governance Models for Urban Sustainability. *Urban Studies Review. IOP Conf. Series: Earth*

and Environmental Science 323 (2019) 012081 IOP Publishing doi:10.1088/1755-1315/323/1/012081

Konijnendijk, C., Di Cagno, F., Borelli, S., Wild, T. (2024). *Capturing the Values and Making the Business Case for Nature-Based Solutions – A step-by-step Guide*. Deliverable 5.3, H2020 CONEXUS project. pp.49-56 <https://zenodo.org/records/14368705>

Porreca, R., & Janoschka, M. (2024). *Pathways of Co-Production: Negotiations and contextual insights into Quito's peripheral urbanisation*. Habitat International, 146, 103043: <https://doi.org/10.1016/j.habitatint.2024.103043>

UNEP Copenhagen Climate Centre (UNEP-CCC). (2024). *Business Models for Financing Nature-based Solutions in Urban Climate Action: A Knowledge Resource for Cities in Developing Countries*. Copenhagen, Denmark. <https://unepccc.org/wp-content/uploads/2025/02/24442-unepccc-business-models-v09-au-pages-hq.pdf>

5.5 Sustainable Urban Land Restoration feasibility

Developed by

Federica Di Cagno & Simone Borelli (FAO Green Cities Initiative)

Learning objectives

- ▶ Key aspects of Cost Structure and Value Capture in SULR;
- ▶ Analyze cost structure for NBS, including planning, implementation, and maintenance;
- ▶ Investigate methods to reduce costs and enhance efficiency in NBS/SULR projects;
- ▶ Differentiate between direct and indirect monetary, as well as non-monetary value capture mechanisms;

- Use tools to estimate and capture values such as revenues, cost savings, and non-monetary benefits.

Summary

Part 1. Cost structure

This section explores the financial components of the SULR Innovation Model Canvas, which includes: cost structure, cost reduction, and value capture.

Cost structures refers to all financial investments necessary to plan, implement, and maintain a project over its lifecycle. These costs can be categorized into: capital costs, planning costs and operational costs. A key consideration in cost management is in which phase are the highest costs incurred and why. Identifying the right financing mechanisms that can support the different phases and ensuring cost-efficiency throughout all project phases is crucial for the financial sustainability of any SULR project. A cost breakdown exercise for an urban rain garden provides practical insights into financial planning and stakeholder involvement.

Part 2. Cost reduction

This section introduces strategies for optimizing costs in SULR projects without compromising the effectiveness and long-term sustainability of the SULR intervention. It also discusses cost comparison which serves both as a communication tool and an economic estimation method. This comparison highlights how NBS not only provide direct cost savings but can also generate indirect benefits and/or co-benefits for indirect beneficiaries; reinforcing the argument for ULR activities development and investment.

Part 3. Value capture

This section explains how SULR projects can generate both monetary and non-monetary values (benefits). It introduces valuation tools to measure economic, social, environmental and health benefits which can provide quantitative evidence to support NBS investment decisions and help secure funding while ensuring long-term ULR project sustainability.

Part 4. Case study

This case study illustrates how non-monetary benefits of SULR projects can be measured and translated into monetary terms through tools such as Health Impact Assessments (HIA) and how increased green cover can lead to better public health outcomes, which in turn generate cost savings for healthcare systems and enhance overall community well-being.

Lessons learned

- Financial planning and cost breakdown are essential for building a strong business case for NbS;
- Cost reduction strategies can make NBS feasible even with limited budgets;
- Capturing value involves showcasing both direct financial benefits and broader social/environmental co-benefits;
- Quantitative valuation tools strengthen business proposals and increase the chance to attract investments;
- Startups and community initiatives can leverage both direct and indirect revenue streams for gaining financial and community support.

Student questions and exercise

Exercise Part 1: Cost structure

Exercise: In this exercise, students will analyze the potential planning, capital, and operational costs associated with the urban rain garden project presented in Submodule 5.4. Based on given Key activities, Key Resources and building on the Key Partners, Beneficiaries and Governance structure identified in the previous submodule 5.4., students will break down the costs and evaluate the feasibility and long-term financial sustainability of the intervention. The exercise should highlight how financial planning affects project success with a focus on three main reflections: (i) How do these different cost categories interact over time? For example, can higher upfront investments reduce long-term operational

costs?; (ii) Who should be responsible for each cost category (e.g. municipality, donors, etc.); (iii) Can certain costs be optimized through partnerships, volunteer engagement or resource sharing across municipal departments? The latter also lays the foundation for the following section 2. Cost reduction

Exercise Part 4: Case study - Barcelona Health Impact Assessment

Use the case study to reflect on the broader concept of value capture in SULR projects.

Questions for students:

1. How can the methodology used in this case study support a broader value capture strategy for SULR projects?
 - o Can HIA results help attract public or private investment? Can they inform long-term maintenance or municipal budgeting? Can they inform policy/regulatory reforms?
2. Which type of value (monetary or non-monetary) do you believe is (or should be) more powerful in motivating investment in SULR?
 - o Consider different stakeholders (policymakers, communities, the private sector and more)
3. Why is it important to move beyond a purely monetary-based mindset when formulating projects whose objective is to produce public goods?

See Case study in Section 5.1 Barcelona Metropolitan Area, Besòs River: Sustainable Urban Land Restoration

Also see our documentary film, “Green Revival: From the River to the Heart of the City, Barcelona, Spain”: <https://youtu.be/Uml0NY8P9ac?si=d0MLbyzVUoYNkkrR>

References/resources

Central Park Conservancy. (2015). *The Central Park Effect: Assessing the Value of Central Park's Contribution to New York City's Economy*. Central Park Conservancy. https://assets.centralparknyc.org/pdfs/about/The_Central_Park_Effect.pdf

Guerry, A., Lonsdorf, E., Nootenboom, C., Remme, R., Griffin, R., Waters, H., Polasky, S., Han, B., Wu, T. Janke, B., Meacham, M., Hamel, P., and X. Wang. (2023). Mapping, measuring, and valuing the benefits of nature-based solutions in cities. 259-293 In: McPhearson, T., Kabisch, N. and N. Frantzeskaki (eds.) *Nature-based Solutions for Cities*. <https://doi.org/10.4337/9781800376762.00023>

Guerry, A., et al. (2023). *Integrated Valuation of Ecosystem Services and Tradeoffs (InVEST): Applications for Urban NBS*.

Konijnendijk, C., Di Cagno, F., Borelli, S., and T. Wild. (2024). *Capturing the Values and Making the Business Case for Nature-Based Solutions – A step-by-step Guide*. Deliverable 5.3, H2020 CONEXUS project. <https://zenodo.org/records/14368705>

Panduro, T., Nainggolan, D., Taylor, T., Zandersen, M., Zhang, Y. and Y. Wang. (2021). *Cost-Effectiveness of NBS: A Guide for Practitioners*. ReGreen Project. <https://www.regreen-project.eu/wp-content/uploads/REGREEN-D2.3-Cost-effectiveness-of-NBS.pdf>

5.6 Communication: Scaling Sustainable Urban Land Restoration Innovations

Developed by

Andrea Ramírez-Agudelo (UNU-EHS)

Learning objectives

- ▶ Apply the understanding of opportunities for scaling SULR innovations;
- ▶ Analyze good practices and SULR's scaling factors;
- ▶ Understand how strategic communication tools enhance the visibility, engagement, and mainstreaming of SULR.

Summary

Part 1. Opportunities for Scaling SULR Innovations and Good Practices

This first part, 'Opportunities for Scaling SULR Innovations and Good Practices' focuses on the social, ecological and technological enablers that facilitate the scaling of SULR initiatives. Key topics include scaling through partnerships, which include public sector leadership, R&D and academic institutions, private sector, locally-led initiatives, and NGO's. By recognizing the potential avenues for scaling SULR initiatives, participants will learn how diverse partnerships can unlock significant opportunities for mainstreaming and impact.

Part 2. SULR: Scaling Factors and Good Practices

The second part highlights SULR as a social, ecological and technological innovation, detailing how Value Propositions (VPs) can enable scaling. It examines the interplay between cost structures, business models, and value creation, providing participants with insights into how to design and implement scalable SULR initiatives. By analyzing good practices, participants will critically examine the scaling factors.

Part 3. Communication: Strategies for Scaling SULR

The third part, 'Communication: Strategies for Scaling SULR,' focuses on the essential role of communication in expanding the reach and impact of SULR innovations. Effective communication enhances visibility, raises awareness, and fosters engagement among target audiences within the context of SULR practices. This alignment strengthens community ties and improves the overall effectiveness and sustainability of urban planning efforts. Students will explore how different tools can strategically promote SULR services and products. They will learn how compelling narratives and targeted messaging facilitate SULR implementation and mainstreaming, including good practices such as accountability and monitoring.

Lessons learned

- ▶ Partnerships and Collaborations: Effectively scaling SULR initiatives relies on efforts across multiple actors and diverse sectors. By combining their expertise and resources, different actors can create synergies that maximize

impact and expand the reach of SULR initiatives, fostering sustainable urban transformations;

- ▶ Identifying and Evaluating Opportunities for Scaling: Value propositions that address beneficiaries' needs are essential for SULR efforts. For this purpose, SULR innovations can identify and evaluate opportunities for scaling their initiatives through social, ecological, and technological value propositions (VPs). VPs and aligned cost structures are key to designing scalable SULR initiatives that meet community needs and market demands for project effectiveness and adaptation to evolving urban challenges;
- ▶ Tailored Communication Strategies: Effective scaling benefits from adapting communication approaches to different target audiences, ensuring that the value and impact of SULR initiatives are conveyed in the terms they understand their needs and expectations.

Student questions and exercises

Exercise Instructions

1. Choose Your SULR Initiative

- ▶ Select your SULR initiative.

2. Craft a Post

- ▶ Write a brief that encapsulates your SULR initiative, engaging social media posts (e.g., for Twitter, Instagram, or LinkedIn).

3. Reflect:

- ▶ Reflect on its primary message and how your headline conveys the core value of your initiative.
- ▶ Consider whether it resonates with your target audience.

References/resources

Brandtner, C. (2023). *Can cities be the source of scalable innovations?* *Stanford Social Innovation Review*. https://ssir.org/articles/entry/can_cities_be_the_source_of_scalable_innovations <https://doi.org/10.48558/pwbd-nn15>

Calliari, E., Castellari, S., Davis, M., Linnerooth-Bayer, J., Martin, J., Mysiak, J., Pastor, T., Ramieri, E., Scolobig, A., Sterk, M., Veerkamp, C., Wendling, L., and M. Zandersen. (2022). Building climate resilience through nature-based solutions in Europe: A review of enabling knowledge, finance and governance frameworks. *Climate Risk Management*, 37, 100450. <https://doi.org/10.1016/j.crm.2022.100450>

NetworkNature. (2023). Scaling nature-based solutions for climate resilience and nature restoration. <https://networknature.eu/nbs-resource/30641>

UNCCD. (2017). *Scaling up sustainable land management and restoration of degraded land*. <https://www.unccd.int/resources/publications/scaling-sustainable-land-management-and-restoration-degraded-land>

van der Heijden, J. (2023). Towards a science of scaling for urban climate action and governance. *European Journal of Risk Regulation*, 14(3), 513-525. <https://doi.org/10.1017/err.2022.13>

5.7 Good Practice I - Locally-led and Grassroot Initiatives

Developed by

Peter K. Kamau (KU, Kenya)

Christine Majale (KU, Kenya)

Learning objectives

Upon completion of this submodule, the participants are able to:

- ▶ Recognize the opportunities for locally led grassroots initiatives in SULR;

- ▶ Explore strategies for enhancing value and benefits in locally led SULR;
- ▶ Understand potential for collaborations and partnerships in locally-led grassroots initiatives for SULR.

Summary

Part 1: Locally led grassroots Initiatives in SULR

Determining how local communities would want to engage in SULR by looking at the drivers and triggers. What are the key triggers? and what are key resources and activities?

- ▶ SULR Scaling through locally led initiatives;
- ▶ Trigger of locally led initiatives for SULR;
- ▶ Examples of land restoration initiatives in urban areas: examples Nairobi and Dresden;
- ▶ Locally led initiatives and key activities;
- ▶ Interventions, roles and responsibilities among communities;
- ▶ Locally led key resources;
- ▶ Local expertise and innovations: non-monetary resources mobilization and local community networks.

Part 2: Locally led initiatives as social enterprise initiatives: How, What and When?

Examining the different value propositions 1. Environment (e.g. tree planting, reclaiming streets from cars) ; 2. Economic (e.g. job creation, urban farming); 3. Social (e.g. monitoring impact of the SULR on community health, creating social interactions). Technologies may include using social media platforms.

HOW: Locally led initiatives– Scaling strategies

- ▶ Social capital creation and return on investments: Local -cooperatives, Community-Based Organizations (CBOs), or Non-Governmental Organizations (NGOs), resources mobilization strategies, case studies of successful CBOs undertaking SULR.

WHAT: Locally led market-driven approaches for SULR

- ▶ Innovations for financing locally led initiatives for SULR.

WHEN: Emerging pathways for resilience in locally led initiatives in SULR.

- ▶ M&E of locally led initiatives for SULR. Tracing the effectiveness of locally led initiatives, emerging technologies, reporting systems and feedback systems and dissemination systems.

Part 3: Lessons learned

- ▶ Review of the different strategies of mobilising resources including; Self-financing, Community groups; social networks; non governmental organisations, grants from charitable/endowment funds among others.
- ▶ Locally led initiatives in SULR: the inherent potential for local community action and the necessary environment to support them
- ▶ Rethinking Locally led initiatives in SULR as social enterprise engagements that generate specific and quantifiable urban communal benefits.
- ▶ Collaborations and partnerships and their role in ensuring long term benefits realization in SULR.

Student questions and exercises

1. What are the locally led initiatives in your locality that provide solutions to land degradation?
2. How do these local formations mobilise resources?
3. What evaluation mechanisms do they use in their operations (for monitoring; reporting..etc)?
4. Do the local communities involved partner with other actors?
5. Who are these actors and in what way do they partner?

See Case Study in Section 5.8 Nairobi: Ecological and Social Recovery of Nairobi River- Michuki Park, Kenya

References/resources

Deffner, J., Joost, J.-M., Weber, M., and I. Stiess. (2021). Bottom-Up Strategies for Shared Mobility and Practices in Urban Housing to Improve Sustainable Planning. *Sustainability*, 13(5), 2897. <https://www.mdpi.com/1024990>

Macedo, P., Huertas, A., Bottone, C., del Río, J., Hillary, N., Brazzini, T., Wittmayer, J.M. and G.I Penha-Lopes. (2020). *Learnings from Local Collaborative Transformations: Setting a Basis for a Sustainability Framework*. *Sustainability* 2020, 12(3), 795; <https://www.mdpi.com/623264>

Mell, I., Clement, S., and F. O'Sullivan. (2023). Mainstreaming Nature-Based Solutions in City Planning: Examining Scale, Focus, and Visibility as Drivers of Intervention Success in Liverpool, UK. *Land*, 12(7). <https://www.mdpi.com/2073-445X/12/7/1371>

Pavesi, R., Orsi, L., and L. Zanderighi. (2025). Enhancing Circularity in Urban Waste Management: A Case Study on Biochar from Urban Pruning. *Environments*, 12(1), 5. <https://doi.org/10.3390/environments12010005>

Zapata Campos, M. J., Carenzo, S., Charles, G., Gutberlet, J., Kain, J.-H., Oloko, M. O., Reynosa, J. P., and P. Zapata. (2022). Grassroots innovations in 'extreme' urban environments. The inclusive recycling movement. *Environment and Planning C*, 41(2), 351-374. <https://doi.org/10.1177/23996544221118191>

5.8 Good Practices II: Private Sector

Developed by

Andrea Ramírez-Agudelo (UNU-EHS)

Leticia Domingos Vellozo (UNU-EHS)

Learning objectives

- ▶ Recognize how the private sector facilitates and leads SULR innovations through good practices, such as Public-Private Partnerships (PPP);

- ▶ Examine key private sector activities, resources, and strategies that support SULR initiatives with a beyond-business-as-usual mindset;
- ▶ Identify the private sector's good practices, including emerging trends, such as financial mechanisms, accountability frameworks and mandatory reporting.

Summary

Part 1. SULR Private Sector: Opportunities and Good Practices

This first part, 'SULR Private Sector: Opportunities and Good Practices,' focuses on the private sector's role in facilitating the scaling of SULR initiatives. It emphasizes the importance of partnerships—led by the private sector—recognizing potential opportunities for mainstreaming and impact. This part examines opportunities for SULR Scaling through new markets, key activities such as Corporate Sustainable Responsibility (CSR) initiatives, and key resources for advanced technologies.

Part 2. Private Sector: How, What, and When

This second part, 'Private Sector: How, What, and When,' highlights the strategies and mechanisms through which the private sector contributes to addressing urban land degradation and scaling SULR. This part highlights mechanisms such as PPP, market-driven approaches, and emerging trends, like accountability frameworks and technological convergence, that define when and how businesses invest in SULR.

Part 3. Private Sector Lessons Learned

This third part, 'Private Sector Lessons Learned,' examines private sector leadership in SULR initiatives, providing insights into the challenges and opportunities related to leadership in circular economy practices, collaboration models, financial mechanisms, and technology adoption.

Lessons learned

- ▶ **Private Sector's Leadership in SULR:** Private businesses can lead SULR initiatives by adopting creative implementation models and resource-efficient practices, such as creating value through circular economy practices and technological innovation.
- ▶ **Market-driven approaches and resource efficiency:** Market-driven approaches can promote circular economy models and collaboration between business sectors, *i.e.*, industrial symbiosis, which can enhance resource efficiency, turning waste into valuable materials and reducing environmental impact.
- ▶ **Financial Mechanisms and Market Dynamics:** By promoting market-driven approaches integrating circular economy principles, the private sector can tap into new markets, technologies, products, and services, contributing to innovation development and SULR's long-term business growth. Innovative financing tools like green bonds and impact investments can help to scale SULR initiatives.
- ▶ **Accountability and Standardization:** Adopting standardized metrics and mandatory reporting frameworks ensures monitoring, which entails measuring and enhancing the impact of private sector efforts in SULR via continuous improvement.

Student questions and exercises

Exercise: Creating a Headline for the Private Sector's Role in Scaling SULR

Exercise Instructions: Develop a headline that communicates the private sector's key role in scaling SULR initiatives.

1. Review the private sector's role in scaling SULR.
2. Craft a concise and engaging headline (max 10 words).
3. Use the checklist to refine your headline.

References/resources

Ellen MacArthur Foundation. (n.d.). *Circular economy*. <https://ellenmacarthurfoundation.org/topics/circular-economy/overview>

FAO. (2021). *FAO strategy for private sector engagement, 2021-2025*. Rome, Italy. <https://doi.org/10.4060/cb3352en>

Selvar. (n.d.). *Integramos biodiversidad y resiliencia en el desarrollo y planificación territorial*. <https://www.selvar.co/>

UNCCD. (n.d.). *Business for Land*. <https://www.unccd.int/our-work/business4land>

UNEP-CCC. (2024). *Business models for financing nature-based solutions in urban climate action: A knowledge resource for cities in developing countries*. Copenhagen, Denmark. <https://unepccc.org/publications/business-models-for-financing-nature-based-solutions-in-urban-climate-action/>

5.9 Good Practices III: Public Sector

Developed by

Andrea Ramírez-Agudelo (UNU-EHS)

Leticia Domingos Vellozo (UNU-EHS)

With contributions from SALAR International

Learning objectives

- ▶ Recognize how the public sector facilitates and leads Sustainable Urban Land Restoration (SULR) innovations through good practices, such as regulatory frameworks and sustainability incentives;
- ▶ Examine key public sector activities, resources, and strategies that support SULR initiatives with a not-for-profit revenue;

- ▶ Analyze successful public sector-led initiatives, identifying the role of policy coherence, adaptive governance models, and strategic urban planning in creating enabling environments for SULRs.

Summary

Part 1. SULR Public Sector: Opportunities and Good Practices

This first part, ‘SULR Public Sector: Opportunities and Good Practices,’ focuses on the public sector’s role in facilitating the scaling of SULR initiatives. It emphasizes the importance of collaborations—led by the public sector—facilitating multi-stakeholder engagement for addressing land degradation while enabling an environment for mainstreaming and impactful SULR initiatives. This part examines opportunities for SULR Scaling through strategic regulations and policies, adaptive governance models, and partnerships; key activities, such as regulatory support; funding and incentives; facilitating access to public urban land; and Key Resources, such as regulatory frameworks; public land access; financial support; data and reports.

Part 2. Public Sector: How, What, and When

This second part, ‘Public Sector: How The What and When,’ highlights the strategies and mechanisms through which the public sector contributes to creating a participatory and enabling environment for addressing urban land degradation and scaling SULR innovations. This part highlights mechanisms such as urban land management and finance instruments, value-driven approaches, and emerging trends, like standardized metrics and impact criteria for mainstreaming urban NbSs; climate finance and blended funding; regenerative and resilient Urban Land Management (Circular Economy Integration) that define when and how the public sector leads in SULR initiatives.

Part 3. Public Sector Lessons Learned

This third part, ‘Public Sector Lessons Learned,’ examines public sector leadership in SULR initiatives, providing insights into the challenges and opportunities related to enabling environment for SULR innovations’ developments and mainstreaming; SULR’s value-driven propositions and urban sustainability alignment (policy coherence and coordinated actions); and their defining role for strategic urban planning and management practices.

Lessons learned

- ▶ Essential contributions of the public sector: Public sector leadership creates enabling environments through regulatory frameworks, such as strategic policies, regulations, urban land management and financial mechanisms to ensure that SULR initiatives address land degradation challenges while aligning with broader sustainability and development goals;
- ▶ Value-driven propositions and urban sustainability alignment: The public sector can align contextual needs and community priorities with SULR initiatives as synergistic efforts in cities. The Public Sector can foster participatory processes by integrating SULR efforts into urban planning frameworks to ensure multiple benefits for long-term viability. Strategic public leadership creates an environment for SULR initiatives to benefit diverse stakeholders, in which value-driven initiatives can promote investments and incentives, encouraging partnerships and supporting inclusive urban development;
- ▶ Urban land management and financing instruments: By promoting value-driven approaches integrating sustainable urban development principles, the public sector can create new opportunities for policy coherence and coordinated actions, facilitating the impact of strategic urban planning and management practices while contributing to an enabling environment for SULR's long-term implementation and affordability.

Student questions and exercises

Exercise: Creating a Headline for the Public Sector's Role in Scaling SULR

Exercise Instructions: Develop a headline that communicates the public sector's key role in scaling SULR initiatives.

- 1) Review the public sector's role in scaling SULR.
- 2) Craft a concise and engaging headline (max 10 words).
- 3) Use the checklist to refine your headline.

See Case Study in Section 5.7 Kathmandu: Schoolyard kitchen gardens for urban land restoration and education, Nepal

References/resources

Climate Alliance. (2022). *Urban ecosystem restoration & nature-based solutions*. Retrieved from https://www.climatealliance.org/fileadmin/Inhalte/4_Activities/Projects/Peer_Learning_Note_Interlace_Project_102022.pdf

ESRI StoryMaps. (n.d.). *Nature-based solutions for urban resilience*. <https://storymaps.arcgis.com/stories/7afa6040cd4e46b48720e280b7238434>

Hodge, I., & Adams, W. M. (2016). Short-term projects versus adaptive governance: Conflicting demands in the management of ecological restoration. *Land*, 5(4), 39. <https://doi.org/10.3390/land5040039>

International Resource Panel. (2020). *Land restoration for achieving the Sustainable Development Goals*. United Nations Environment Programme. <https://www.resourcepanel.org/reports/land-restoration-achieving-sustainable-development-goals>

UNCCD. (2021). *Restored land, healthy people, green recovery*. https://www.unccd.int/sites/default/files/inline-files/UNCCD_Solution%20Brief.pdf

5.10 Pitching your Sustainable Urban Land Restoration Innovation (Final assignment)

Developed by

Andrea Ramírez-Agudelo (UNU-EHS)

Karen Sudmeier-Rieux (UNCCD-G20 Global Land Initiative)

This module was developed to provide instructors with ideas for a final summative assignment or as additional exercises at any point in the course. The entire course has laid the foundation for understanding SULR. Now, let's bring these elements together through some optional exercises.

Learning objectives

- ▶ Promote creativity, critical thinking, and co-creation by bringing together various course elements in the final summative assignments;
- ▶ Develop communications and negotiation skills in pitching your SULR innovation ideas;
- ▶ Create a compelling innovation story for your SULR ecopreneurship or start-up via the SULR Innovation Model Canvas that best communicates the SULR innovation.

Summary

Part 1. Role-Play: Steps for an Effective SULR Elevator Pitch.

Part 1 invites students to create a compelling elevator pitch for a SULR ecopreneurship or start-up, which involves targeting an audience, defining an intent and value proposition, negotiating the outcomes, and delivering your message through a 30-40 minute role play.

The scenario is the following: a SULR project has been developed to improve the liveability of a neighborhood in fictitious SULR City, where a central neighborhood suffers from heat, traffic noise and degradation. The project has divided the neighborhood into those for and against, as it would entail converting it into a pedestrian area and diverting traffic. Local businesses fear losing customers, neighbors fear higher rents, but proponents highlight improved quality of life, new spaces for parks and playgrounds, innovation, and small businesses (e.g., green urban landscaping, community gardens and vegetable markets, neighborhood cafés). The role play simulates a participatory process, voting on the proposed plans, and weighing the pros and cons.

Part 2. Crafting Your SULR Innovation Story (Based on the 10-Step SULR Innovation Model Canvas)

Part 2 proposes exercises highlighting the capacities for effective communication in promoting SULR innovations, enabling urban planners, landscape architects, NGOs, ecopreneurs, and start-ups to be more competent at crafting a message that helps them connect with their audiences to drive initiatives forward.

Lessons learned

- ▶ Role-playing activities help to communicate complex concepts to diverse audiences, inviting participants to brainstorm, co-create, and practice sharing their SULR ideas in realistic scenarios.
- ▶ The integrative approach of crafting an innovation story allows participants to develop problem-solving skills by applying concepts from earlier modules using tools such as the SULR Innovation Model Canvas.

Student questions and exercises

Part 1. Role Play (40-60 minutes): Practical Steps for an Effective Elevator Pitch by a SULR landscape architecture firm:

A. Assign stakeholder roles (5 minutes).

Stakeholders may include, but are not limited to:

- ▶ SULR landscape architecture firm, which pitches a rendition, a budget, and a vision for the proposed SULR transformation of a neighborhood
- ▶ The SULR City Mayor
- ▶ City council members
- ▶ Local business owners
- ▶ Local residents
- ▶ Local environmental and housing NGOs
- ▶ Media representatives

B. The Pitch (5 minutes)

The SULR landscape architecture firm representative gives the Pitch.

For more information on the pitch, refer to Submodule 5.6.

Pitch steps include:

Step 1. Audience Understanding

Problem introduction

- a. Set the stage - Identify the key partners of your SULR project proposal
- b. Detail the Challenge
- c. Create a sense of urgency

Step 2. Intent and Value Proposition

Value proposition of the SULR proposal

- a. Present the solution
- b. Highlight the benefits and potential impact

Step 3. Message Delivery & Feedback Refining

- a. Tell your SULR “Transformational Story”
- b. Finish with your “Call-to-Action”

C. Town Hall Meeting (15-20 minutes)

Each representative is given 3-5 minutes to present their position on the pitch.

D. Town Hall wrap-up (10 minutes)

The mayor calls for a vote, and all the participants can decide to fund the project through the city’s participatory budgeting. Then, the mayor and the public sector representatives can offer some conclusions on this pilot project for the neighborhood (3-5 minutes).

E. Debriefing with the students (10 minutes)

Reflect on the Pros and Cons of the SULR project proposal.

How could the “Elevator Pitch” be improved?

Ask students, especially those playing specific roles, to reflect on the following:

- ▶ How easy or difficult was it to express your message and be heard? Why?

- ▶ Who was missing from the discussion? Should other actors/players be involved? Who?

- ▶ Who had the greater advantage in the discussions? Why?

Part 2. Crafting Your SULR Innovation Story

(Based on the 10-Step SULR Innovation Model Canvas)

Students are given three options for their final assignment: **to develop a SULR innovation project for any city or neighborhood of choice.**

1. A public sector proposal (e.g., a municipality develops a call for proposals) to convert a polluted neighborhood into a SULR project;
2. A private sector business plan (e.g., a private business proposal) for a SULR innovation business or a social enterprise;
3. A public-private proposal (e.g., a landscape architecture firm responds to a municipal bid) to convert a degraded neighborhood into a SULR project;

Instructions:

Students draft their proposal based on the SULR Innovation Model Canvas, outlining their innovative idea for a SULR project (3-5 pages). Each proposal should refer to and explain upon which SULR principle the proposal will act: (AVOID - REDUCE - REVERSE - RENEW)

This exercise combines concepts and tools learned throughout Modules 1-5.

Part 3. Course wrap-up

Part 3 wraps up our course with a few summary slides and a course overview:

Our course journey included:

Module 1: The drivers and challenges facing urban areas;

Module 2: Approaches to SULR;

Module 3: SULR enabling factors & trade-offs;

Module 4: Operationalizing SULR;

Module 5. Catalyzing SULR through innovation;

Ultimately, our course was about understanding the connections between ecological and social processes in urban areas, specifically between land and the built-up environment, and how you can contribute as a future agent of positive change in cities.

We hope this course has been a valuable resource for your students' learning journey.

References/resources

G20 GLI/UNCCD. (2025). *Communicators retreat to review and re-ignite land restoration campaigns*. <https://g20land.org/communicators-retreat-to-review-and-re-ignite-land-restoration-campaigns/>

Smart City Expo World Congress. (2013). *Elevator pitch session EP 3 - Sustainable built environment and city* [Video]. <https://www.youtube.com/watch?v=-EWtLR1Vf1k>

Stael, D. (2022). *Here's the elevator pitch for four new sustainable ideas*. LinkedIn. <https://www.linkedin.com/pulse/heres-elevator-pitch-4-new-sustainable-ideas-dirk-saves-dirk-stael>

UNEP. (2024). *Key principles for effective communications on ecosystem restoration*. <https://www.decadeonrestoration.org/publications/key-principles-effective-communications-ecosystem-restoration>

UNU-EHS. (2024). *5 highlights of sustainable urban land restoration*. <https://unu.edu/ehs/article/5-highlights-sustainable-urban-land-restoration>

5. Case studies descriptions

The ten case studies in table X below were developed by our course developers with the objective of providing a variety of ideas and real-life examples of a host of “Innovations in SULR”. The case studies are each approximately 15–20 minutes in length and can replace or supplement various lectures in the course modules.

City	Country	Case study title	Scale	Authors	Suggested course sub-modules
1. Barcelona Metropolitan Area	Spain	Besòs River: Sustainable Urban Land Restoration	Meso	United Nations University, Institute for Environment and Human Security	1.1, 5.5
2. Beirut	Lebanon	Miyawaki forests for urban renewal and peace building	Micro	TheOtherDada consultancy	1.2, 2.5
3. Buenos Aires	Argentina	Sustainable urban land restoration opportunities in informal settlements	Micro	Instituto Internacional de Medio Ambiente y Desarrollo – America Latina	1.4
4. Cairo	Egypt	Al Fustat Restoration Plan	Meso	Ain Shams University	3.3
5. Cuenca	Ecuador	Shinshin creek renovation	Micro	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) Sirio & Persea, Laboratorio de proyectos de paisaje, arquitectura y diseño.	2.4, 3.5
6. Jakarta	Indonesia	Evaluating Mangrove Sustainability in the context of Sustainable Urban Land Restoration	Micro	Universitas Gadjah Mada	3.2, 3.8
7. Kathmandu	Nepal	Schoolyard kitchen gardens for urban land restoration and education	Nano	Wildlife Conservation Nepal	3.4, 5.9
8. Nairobi	Kenya	Ecological and Social Recovery of Nairobi River- Michuki Park	Micro	Kenyatta University	5.7
9. Multiple cities	New Zealand	Aotearoa Indigenous Knowledge for Sustainable Urban Land Restoration	Meso	Global Land Initiative/ UNCCD	4.8, 5.1
10. Ruhr Valley	Germany	Emscher Landscape Park Restoration	Macro	TH-Köln	3.9

5.1 Barcelona Metropolitan Area, Besòs River: Sustainable Urban Land Restoration

Location	Barcelona Metropolitan Area (BMA), Catalonia, Spain
Spatial scale	Meso scale - Metropolitan area
City size (population)	1.6 million - 5 million (BMA)
Climate zone	<u>Mediterranean Climate Zone</u>
Type of restoration	Urban land, Wetlands, Riparian
Social benefits	Recreation, Human health, Flood management
Author	Andrea Ramírez Agudelo (United Nations University, Institute for Environment and Human Security)

Description

The Besòs River restoration in the Barcelona Metropolitan Area is a long-term example of a SULR addressing urban environmental and social challenges. At the end of the 20th Century, once heavily polluted by industrial-related waste and urban runoff, the river symbolized the high risks of flooding, poor water quality, and environmental degradation in Catalonia and Europe. Local governments, supported by European funds, initiated a comprehensive river restoration project to transform the Besòs River into a hybrid infrastructure of a riverside park and constructed wetlands (1996-2006). This initiative aimed to enhance the riverbed’s environmental conditions, including its natural hydrology and ecosystems, improve water quality through advanced constructed wetland systems, creating floodplains to manage the torrential rains and urban flooding risks. Alongside ecological restoration, the project prioritized recreational use along its riverbanks (5-km-long public use area), creating accessible and multifunctional green spaces for the densely populated Barcelona metropolitan area, which lacked sufficient natural and open areas for recreation and community activities, affecting the quality of life for Barcelona’s most socially vulnerable residents.

Nowadays, the Besòs Axes is a recognized habitat for urban wildlife and biodiversity after its monitoring by an academic initiative (Barcelonarius), which

complements its role as a strategic metropolitan green and blue infrastructure for renaturalization, with recreational and socio-cultural benefits from its parks, walking paths, and bike lanes along its river banks. This metropolitan intervention highlights the transformative power of multi-scale and multi-level implementation of hybrid infrastructural approaches by integrating environmental degradation challenges into nature-based processes. Under a coordinated governance, such as the leadership of the Consorci Besòs and its comprehensive agenda (Agenda Besòs), this restoration aimed for more sustainable urban development, showcasing how revitalized natural systems can enhance local and metropolitan approaches.

Learning outcomes

- ▶ Understand the importance of urban land restoration as a comprehensive process of urban change;
- ▶ Learn how such an NbS implementation has been developed as a local process with coordinated governance;
- ▶ Learn which factors have ensured its sustainability and recognition as a Metropolitan Green Infrastructure.

Watch our case study documentary: “Green Revival: From the River to the Heart of the City, Barcelona, Spain” <https://youtu.be/Uml0NY8P9ac?si=fGiekY-DijrpwJENe>

Figure. 17. Besòs River in Barcelona Metropolitan Area, Catalonia, Spain. Photo: © G20 Global Land Initiative /UNCCD (2024)



5.2 Beirut: Reclaiming urban green spaces and restoring hope, Lebanon

Location	Beirut, Lebanon
Spatial scale	Micro / Miyawaki forests and urban green spaces
City size (population)	2,5 million
Climate zone	<u>Mediterranean arid</u>
Type of restoration	Miyawaki forests, greening of urban spaces
Social benefits	Health, social interaction, biodiversity
Author	Adib Dada, Mohammad Kabakibi (theOtherDada consultancy, Lebanon)/ Karen Sudmeier-Rieux (G20 Global Land Initiative/UNCCD)

Description

TheOtherDada [tOD] Regenerative Consultancy & Architecture Practice, is a Lebanon-based company with a mission to enable mindset shifts which reestablish humans as stewards and healers of the planet. Since 2019, the firm has set out to rewild Beirut, a city scarred by water pollution, waste management, urban heat, conflicts and a lack of public spaces for social interaction. Beirut's RiverLESS forest project works with communities to reclaim public space by planting native Miyawaki forests in urban landfills and along Beirut River's sewage-filled, concrete riverbanks, through yearly afforestation campaigns. Miyawaki forests are native pocket forests with a high number of species, significantly enhancing urban biodiversity. The project has been supported by SUGi, who have organised and funded 230+ SUGi Pocket Forests in 54 cities on 6 continents since 2019.

tOD collaborates with municipalities of Greater Beirut, and undertakes restoration primarily with schools or private lands. Once the plot is identified, native species are planted in collaboration with local volunteers, who then continue to weed, water and care for the plots. Challenges include holding

municipalities accountable for the restoration works, usually requiring the restoration to be completely privately funded and maintained.

To date, tOD has planted over 13 forests across Lebanon, reclaiming more than 4,280 square meters of degraded land, 16,340 native trees and shrubs with 33 different native species thanks to the commitment of 1,700 volunteers over the years. Low-maintenance and highly effective, Miyawaki forests offer a sustainable solution for land restoration, with forests becoming self-sufficient in 3 years time, while providing public spaces for social interactions and restoring hope to the people of Lebanon, as well as minimizing flooding, and eventually helping lower air pollution levels.

Learning outcomes

- ▶ Learn about Miyawaki forest principles and how they were established in degraded areas of Beirut;
- ▶ Understand the benefits of these green spaces to local residents for social cohesion and well-being;
- ▶ Learn about challenges and how these were overcome.

Figure 18. Left: Before Miyawaki forest restoration (2019); Right: Post-restoration (2021), Beirut, Lebanon. Photos: © SUGi



5.3 Buenos Aires - sustainable urban land restoration opportunities in informal settlements, Argentina

Location	Buenos Aires, Argentina
Spatial scale	Micro-Local / 2 neighbourhoods in Buenos Aires City
City size (population)	15 million
Climate zone	Humid Subtropical
Type of restoration	Decontamination, greening / urban heat island reduction
Social benefits	Adequate housing, Health, Public space, Community empowerment
Authors	Jorgelina Hardoy, Florencia Almansi, Instituto Internacional de Medio Ambiente y Desarrollo - America Latina (IIED-AL)

Description

The case study compares two neighborhoods in Buenos Aires, Argentina: Villa 20 and Rodrigo Bueno. Villa 20 is located in the poorest commune of the city and has a mix of low-income settlements, social housing projects, and parks. Villa 20 began to be occupied in the 1940s, coincident with the industrialization period in Argentina and rural–urban migration, during which many “villas de emergencia” (emergency towns) were built by newcomers to the city. Rodrigo Bueno is located next to a wealthy neighborhood and a natural reserve, and became a settlement in the 1980s due to land occupation during one of many economic crises and land access restrictions during the military government in the country. Both sites face land degradation due to pollution. Organized community efforts have led to better living conditions and adequate housing.

Negotiations between stakeholders have been conducted to manage eviction risks and remove and manage pollution. Environmental and housing improvements have been made in both consolidated informal occupations, involving law and regulation modifications, new housing, street openings,

paving replacements, and tree planting. Since 2021, the Transformative Urban Coalitions Project (TUC), implemented by an international consortium with funds from the International Climate Initiative (IKI), has integrated climate action into Villa 20’s reurbanization process. The Urban Lab (UL) is a participatory space where actors come together to find solutions to complex problems like climate change. The TUC project, led by an Argentine NGO, has shown that there is room for nature in Villa 20 despite existing land and legal restrictions.

Learning outcomes

- Understand the characteristics of the places where urban land degradation is present; recognize them as part of connected social processes in the history of cities;
- Recognize the vulnerability of informal settlements to land degradation conditions, compromising its inhabitants health and well-being;
- Identify local characteristics that can challenge responses to degradation;
- Relate the relevance of community-based efforts to successful changes toward improved urban spaces and housing.

Figure 19. Left: Before urban renewal project; Right: After urban renewal project implemented by the community: flower beds, use of native plants, benches and pergolas to add shade, irrigation system and a ramp to improve access to a school in Villa 20, Argentina. Photos: © IIED-AL.



5.4 Cairo: Al Fustat Restoration Plan, Egypt

Location	Cairo, Egypt
Spatial scale	Meso/Micro – Al Fustat Area
City size (population)	25 million (Greater Cairo Region- GCR)
Climate zone	<u>Desert climate</u>
Type of restoration	Regeneration & Regreening plan
Social benefits	Well-being, Heritage Preservation, Recreation, and Economic Development
Authors	Marwa Khalifa, Aya Elsisy, Sandra Sameh (Ain Shams University, Egypt)

Description

Al-Fustat has a unique and significant status; It was founded in 641 AD after the Muslim conquest of Egypt and was the country's first capital under Islamic rule. Al-Fustat area has experienced significant land degradation due to several factors, as follows: The increase in groundwater levels, which has impacted the unique excavation site, Ain El Seira Lake contamination which was characterized by its therapeutic properties and healing sulphuric waters, sprawl of informal settlements including two unsafe areas with deteriorated physical and human conditions, and conversion of large area of land to a construction dump site. These issues jeopardize the area's historical and heritage significance.

To address these challenges and restore the area's cultural and environmental value, the Cairo Urban Development Fund has an ambitious development plan to transform the area into a global touristic destination and a green lung for Cairo, with five key objectives: preserving cultural heritage, creating green spaces, promoting tourism and the local economy, modernizing infrastructure, and connecting the area to surrounding historical sites. These measures aim to restore the land of Al-Fustat and emphasize its cultural and heritage significance through land restoration processes.

Learning outcomes

- Understand the complex nature of urban land restoration in dense urban areas with historical and heritage significance;
- Analyze the environmental, social, and economic impacts of the proposed urban development plan;
- Address the importance of NbS and heritage preservation in the Urban Land Restoration Process;

Figure 20. Left: Work in progress: Urban greening and planting of restored land in Al-Fustat site, Cairo, Egypt. Photo: © ASU/Khalifa (2024); Right: Work in progress: Transforming the construction dump site into green hills “Telal Al-Fustat” Park, Cairo, Egypt. Photo: © UDF (2024)



5.5 Cuenca: Shinshin creek renovation, Ecuador

Location	Cuenca Municipality, Ecuador
Spatial scale	Micro / Shinshin Creek micro-watershed
City size (population)	600,000
Climate zone	<u>Temperate climate</u>
Type of restoration	Recovery, Regreening and Environmental Management Plan
Social benefits	Climate adaptation Habitability, productivity and well-being Environmental connectivity
Authors	Daniel Tello Enríquez, Cristina Peralta Peñaloza, John Guachichulca Mejía, Lina Flórez Díaz, Bastiaan Stoop, María Elissa Torres, Priscila Carpio Cordero, Carolina Auquilla Guamán, Danilo Sánchez León. Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) Sirio & Persea, Laboratorio de proyectos de paisaje, arquitectura y diseño, Ecuador

Description

Cuenca, Santa Ana de los 4 Ríos has a rich water system characterized by an extensive network of urban streams. However, due to the lack of conservation and restoration strategies, these areas are highly vulnerable. As part of GIZ Ecuador's Sustainable Intermediate Cities II program, a Pilot Management Plan was developed for the Shinshin creek in the municipality of Baños. It is based on the collection of photogrammetric information, a historical approach and the creation of a stakeholder map. The social, geophysical and landscape analysis identifies contamination of the riverbed and banks, land degradation and a high vulnerability to disasters. Strategic goals based on productivity, adaptability, livability, and ecological connectivity are developed through intergenerational community workshops.

With these guidelines, a master plan is proposed that redefines the creek as an integrating axis of biodiversity. Thus, a series of facilities are proposed to promote urban and ecosystem services that affirm its biological functionality and justify the balanced coexistence of communities of people, flora and fauna. The management model is structured in a catalog of management measures and lines of action for the protection and sustainable use of the riverbed. The results will be used to develop the Urban Creek Management Manual for Cuenca.

Learning outcomes

- ▶ Consider the functionality of creeks as biodiversity corridors for the balanced coexistence of species and the provision of ecosystem and urban services;
- ▶ Recognize the need to integrate different data sources and tools (photogrammetric and historical cartography and social perceptions) to produce a more robust and socially acceptable plan;
- ▶ Analyze the possibilities of a standardized management model with clear guidelines for urban watercourse restoration.

Figure 21. Left: Shinshin creek before renovation Photo: © Territorium Research Group, UC; Right: Shinshin creek, Cuenca, Ecuador post-renovation rendition. Image: © GIZ Ecuador/Sirio y Persea, (2022)



5.6 Jakarta: Evaluating Mangrove Sustainability in the context of Sustainable Urban Land Restoration, Indonesia

Location	North Jakarta, Indonesia
Spatial scale	Micro (Penjaringan Sub District)
City size (population)	10 million
Climate zone	Tropical climate
Type of restoration	Greening, Social Laboratories, Mangrove Conservation Area
Social benefits	Recreation/Tourism, Green Space, Community Engagement
Authors	Dyah Rahmawati Hizbaron, Husnul Khotimah Setiacahyandari, Arry Retnowati, Utia Suarma (Universitas Gadjah Mada, Indonesia)

Description

Penjaringan Sub District located on the northern coastline of Indonesia's capital, has long been exposed to tidal floods especially during high tide periods. This area is also experiencing land subsidence of up to 6 cm per year. This threat is a factor that influences the increased risk of tidal flooding, so that it can threaten the sustainability of the environment. Moreover, mangrove ecosystems which are known as a natural barrier with various advantages, cover only 1.97% of the city's total area, leaving them vulnerable to land conversion due to urban development.

This ecosystem consists of two main areas: 1) The protected mangrove areas, which are legally safeguarded and managed by designated institutions. These areas include wildlife reserves, nature tourism parks, production forests, and conservation forests. Each of these areas is governed by specific laws and regulations aimed at preserving and sustainably managing the mangrove ecosystem; 2) The unprotected mangrove areas, which lack formal institutional responsibility for their management. These areas do not benefit from specific

legal protections or regulations, leading to less structured and coordinated preservation and management efforts compared to protected areas.

Learning outcomes

- Understand the background of ecosystem vulnerability in North Jakarta which led to decreasing urban function;
- Evaluate the importance of mangrove sustainability as an SULR measure.

Figure 22. Left: Mangrove Ecosystem in North Jakarta, Indonesia. Photo: © UGM/ HK Setiacahyandar (2023); Right: Unprotected Mangrove Areas in North Jakarta, Indonesia. © UGM/ HK Setiacahyandari (2023)



5.7 Kathmandu: Schoolyard kitchen gardens for urban land restoration and education, Nepal

Location	Kathmandu Municipality, Nepal
Spatial scale	Micro scale - green schoolyards
City size (population)	1 million
Climate zone	Humid subtropical
Type of restoration	Urban land
Social benefits	Education on urban restoration, food and nutrition, public, private, partnership
Authors	Sanjeevani Yonzon Shrestha, Brinda Dewan, Subash Kharel (Wildlife Conservation Nepal) / Karen Sudmeier-Rieux (G20 Global Land Initiative/UNCCD)

Description

Kathmandu, Nepal is one of the world’s oldest, most densely populated and fastest growing cities in the world. Air pollution is a major issue in the Kathmandu Valley, causing thousands of premature deaths annually. Public green spaces are few and far between, and children have little access to nature. To address this issue, since 2023, the Kathmandu Municipality, in collaboration with the non-profit organization, Wildlife Conservation Nepal (WCN) have spearheaded an environmental education program, including “green school yards” and “kitchen gardens”. Book-free Fridays encourage more experiential learning, including preparing garden plots for vegetable growing in school yards across the city.

Abandoned plots in corners of school yards are rehabilitated by WCN staff in collaboration with school teachers and students, often first requiring rubble and garbage removal. School children from “Eco-clubs” help to motivate others to participate in the garden plot preparation, vegetable plantation, weeding, watering and harvesting of the vegetables, which are then prepared and

served for school lunches to the entire school. WCN provides information on the vegetables to be planted and their nutritional values. As of December 2024, 8-10 Kathmandu schools had started kitchen gardens. With an annual budget of around USD\$360,000, the Kathmandu Municipality aims to develop green school yards and kitchen gardens in all of the city’s 90 schools.

Learning outcomes

- ▶ Understand the importance of kitchen gardens as a type of urban land restoration;
- ▶ Learn about how such a program is established and which factors are needed to ensure its sustainability;
- ▶ Learn about challenges and opportunities in replicating the program elsewhere.

Figure 23. Public-private-partnership kitchen garden programs at Kathmandu schools, Nepal. Photos: ©G20 GLI, UNCCD/Sudmeier-Rieux (2024)



Also watch our documentary film: “Kitchen gardens, bringing green to urban schools in Kathmandu” <https://youtu.be/PtR1cxpkLJI>

5.8 Nairobi: Ecological and Social Recovery of Nairobi River- Michuki Park, Kenya

Location	Nairobi City, Kenya
Spatial scale	Meso / River' watershed
City size (population)	4.8 million
Climate zone	Subtropical highland climate
Type of restoration	Ecological, Socio-economic
Social benefits	Livelihood, Security
Authors	Christine Majale, Peter K. Kamau (Kenyatta University, Kenya)

Description

The **Michuki Memorial Conservation Park** formerly known as Mazingira Park is located along the Nairobi River from the Globe Roundabout to Museum Bridge and covers an area of approximately 12.30 ha. The park was established in 2008 through concerted efforts of public, private and community actors geared towards creating a public recreation area at a site previously used as a dumpsite and a criminal hideout which came about as a result of the rehabilitation and restoration of the Nairobi River.

Ecological Restoration: Following the rehabilitation program, a total of 20,000 tons of solid waste was removed. It also involved clearing of illegal structures and leveling of the riparian zone and construction of a pathway along the river to mark a clear boundary of the **riparian land**. A total of 6,357 indigenous trees were planted and construction of a modern tree nursery which stocks indigenous tree seedlings for sale to the people of Nairobi.

Socio-Economic Restoration: Rehabilitation of the walkways was undertaken to allow access to the entire park. The park has a 500-sitter amphitheater, a new guard house and events' grounds. Ensuring the security of a place which was previously attracting illegal activities and dumping of waste.

Learning outcomes

- Understand urban river restoration strategy and its contribution to reversing urban land degradation

Figure 24. Left: Indigenous trees. Photo: Majale/KU-(2025); Right: Tree nursery. Photo: Majale/KU-(2025)



5.9 Aotearo (New Zealand): Indigenous Knowledge for Sustainable Urban Land Restoration

Location	New Zealand, 10 cities +
Spatial scale	Micro and Macro
City size (population)	5,223 Mill
Climate zone	Subtropical
Type of restoration	Biodiversity, Wildlife, Soil
Social benefits	Community engagement, cultural heritage, well-being, human health
Authors	María Mercedes Gavilán Galisteo, Karen Sudmeier-Rieux (G20 Global Land Restoration/ UNCCD)

Description

The New Zealand/Aotearo case study focuses on the integration of Indigenous Knowledge (IK) into urban ecological restoration through various local initiatives. It explores how urban restoration efforts can enhance biodiversity, restore cultural relationships with the land, and support community empowerment by weaving IK into restoration practices. The case study focuses on the decline of local species caused by the increase of exotic species, loss of biodiversity due to the conversion of land for agricultural purposes and the shift towards urbanization. The case study presents public-private initiatives that promote Māori-centric urban restoration. The aim of the initiatives is to restore biodiversity by enhancing native species in urban landscapes, retaining and restoring urban wildlife and regenerating soil biodiversity to create sustainable ecosystems.

This restoration work is undertaken considering cultural revitalization by incorporating IK to guide urban land restoration while linking people with their cultural and ecological heritage. The initiatives prioritize equitable partnerships

with indigenous communities and intergenerational knowledge transfer to ensure inclusive restoration processes. The initiatives not only enhance urban biodiversity but also promote social and environmental justice, healing, and community empowerment. This case study demonstrates how integrating IK in urban restoration enhances biodiversity, promotes cultural well-being, and fosters meaningful partnerships with indigenous communities for enduring project success and sustainable urban ecosystems.

Learning outcomes

- Understand how SULR can and should consider working closely with communities from the outset of a SULR project and incorporate IK.

References:

People, Cities, Nature - Restoring indigenous nature in urban environments (n.d)
<https://www.peoplecitiesnature.co.nz/>

Walker, E. (2022) Research report - Maori values, People, Cities, Nature - Restoring indigenous nature in urban environments, 23 p.
<https://static1.squarespace.com/static/62d879f3f9cbe543dab51e33/t/64ed6226b6f71c3a41b6a480/1693278822022/Maori+Values.pdf>

Figure 25. Left: Maori carving NZ. Photo: Aleza van der Werff on Unsplash
Right: Wellington, NZ. Photo: João Marcelo Martins on Unsplash



5.10 Ruhr Valley: post-industrial landscape restoration, Emscher park, Germany

Location	Ruhr Valley, Germany
Spatial scale	Macro scale - watershed in an industrial district
City size (population)	5.2 million
Climate zone	Humid temperate climate
Type of restoration	Urban watershed
Social benefit	Renaturalization, decontamination, recreation
Author	Johannes Hamhaber (TH-Köln, Cologne University of Applied Sciences, Germany)

Description

The Ruhr Valley is the largest industrial district in Germany, emerging from the mid-19th century from the coal and steel sectors. Based on largely uncontrolled urban growth and intensely polluting industries (besides coal mining, coking, iron and steel milling, also metallurgy and chemical industry) large areas were degraded due to pollution, soil sealing and surface remodeling (e.g. tailings). After World War II, the region still was the economic powerhouse of Germany, but soon underwent massive economic crises (coal mining from 1957, steel from 1975), and coal mining finally was abandoned in 2018. In parallel, environmental regulations were drafted and implemented, finally, the EU water framework prescribed an integrated water management for all European rivers.

Emscher became an icon for the revitalization of the whole region. Linked to “Emscher Park” (an urban and architectural experimental exhibition format), the entire region started promoting its new image as innovative, educative, ecological and urban. The Emscher river restoration project was instrumental, following the Master Plan “Future Emscher”, the river and its tributaries were redesigned and renaturalized, people gained access to new creek and river banks for recreation, the whole sewage system was newly set up underground and the fresh water management for the area includes rain water drainage, seals for retention, groundwater recharge and other feature of a sponge city.

This in turn has allowed the tributaries and the Emscher landscape to return to a largely natural state. Icons of this transition are the Phoenix lake in Dortmund-Hörde, the old Emscher in an industrial heritage park in Duisburg, the water crossing of the Emscher and a shipping canal.

Learning outcomes

- Discover the role of integrated and master planning, as well as new narratives for a turn towards sustainability
- Learn about the role of blue-green infrastructure for the restructuring of old industrial districts and thus highly urbanized zones
- Understand the role of culture and arts in urban land restoration

References:

EGLV (ed. 2022): The new Emscher is coming. Emschergenossenschaft, 32p. https://www.eglv.de/app/uploads/2022/02/20220628-Emscher-Umbau-englisch_.pdf

Figure 26. Renaturation of the Emscher river system: The Emscher River in its new bed near Dortmund-Schönau, Germany. Photo: S. Kunzmann (2012) Wikicommons CC-SA 3.0



Figure 27. Emscher Park, Germany. Photo: © G20 GLI/UNCCD (2024)



Watch our case study documentary: “A Landscape Transformation of Post-mining Germany” <https://www.youtube.com/watch?v=sU46y16-UHA>



g20ico@unccd.int



www.g20land.org



**Initiative Coordination Office UN Convention
to Combat Desertification 1, Platz der
Vereinten Nationen D-53113, Bonn, Germany**



United Nations
Convention to Combat
Desertification

United for land

